

Dual Nature of Matter and Radiation

Question1

A particle having electric charge $3 \times 10^{-19} \text{ C}$ and mass $6 \times 10^{-27} \text{ kg}$ is accelerated by applying an electric potential of 1.21 V .

Wavelength of the matter wave associated with the particle is $\alpha \times 10^{-12} \text{ m}$. The value of α is ____ .

(Take Planck's constant $= 6.6 \times 10^{-34} \text{ J.s}$)

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Answer: 10

Solution:

When a particle of charge q is accelerated through a potential difference V , the work done on the particle is converted into its kinetic energy (K):

$$K = qV$$

Given:

- Charge $q = 3 \times 10^{-19} \text{ C}$
- Potential $V = 1.21 \text{ V}$

$$K = (3 \times 10^{-19}) \times (1.21) = 3.63 \times 10^{-19} \text{ J}$$

The momentum (p) of a particle of mass m is related to its kinetic energy (K) by:

$$K = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mK}$$

Substituting $K = qV$:

$$p = \sqrt{2mqV}$$

The de-Broglie wavelength is given by:

$$\lambda = \frac{h}{p}$$

Substituting the expression for momentum:

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

Substituting the given values:

$$h = 6.6 \times 10^{-34} \text{ J.s, } m = 6 \times 10^{-27} \text{ kg, } q = 3 \times 10^{-19} \text{ C and } V = 1.21 \text{ V}$$

So, the de-Broglie wavelength associated with the particle is,

$$\lambda = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times (6 \times 10^{-27}) \times (3 \times 10^{-19}) \times 1.21}}$$

$$\Rightarrow \lambda = \frac{6.6 \times 10^{-34}}{\sqrt{36 \times 10^{-46} \times 1.21}}$$

$$\Rightarrow \lambda = \frac{6.6 \times 10^{-34}}{6 \times 10^{-23} \times 1.1}$$

$$\Rightarrow \lambda = \frac{6.6 \times 10^{-34}}{6.6 \times 10^{-23}}$$

$$\Rightarrow \lambda = 1 \times 10^{-11} \text{ m} = \alpha \times 10^{-12} \text{ m}$$

$$\Rightarrow \lambda = 10 \times 10^{-12} = \alpha \times 10^{-12} \Rightarrow \alpha = 10$$

Therefore, the value of α is 10 .

Hence, the correct answer is **10**.

Question2

The ratio of de Broglie wavelength of a deuteron with kinetic energy E to that of an alpha particle with kinetic energy $2E$, is $n : 1$. The value of n is ____ .

(Assume mass of proton = mass of neutron) :

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Answer: 2

Solution:

The de Broglie wavelength is given by :

$$\lambda = \frac{h}{p}$$

Since kinetic energy $K = \frac{p^2}{2m}$, we can write momentum as $p = \sqrt{2mK}$.

Therefore:

$$\lambda = \frac{h}{\sqrt{2mK}}$$

Let m_p be the mass of a proton or neutron.

Deuteron (d) consists of 1 proton and 1 neutron. The mass of deuteron is $m_d \approx 2 m_p$

Kinetic Energy $K_d = E$

Alpha particle (α) consists of 2 protons and 2 neutrons. The mass of alpha particle is $m_\alpha \approx 4 m_p$

Kinetic Energy $K_\alpha = 2E$

The ratio of wavelengths is :

$$\frac{\lambda_d}{\lambda_\alpha} = \frac{\frac{h}{\sqrt{2 m_d K_d}}}{\frac{h}{\sqrt{2 m_\alpha K_\alpha}}} = \sqrt{\frac{m_\alpha}{m_d}} \times \sqrt{\frac{K_\alpha}{K_d}}$$

Substituting the values:

$$\frac{\lambda_d}{\lambda_\alpha} = \sqrt{\frac{4 m_p}{2 m_p}} \times \sqrt{\frac{2E}{E}} = 2$$

So, the ratio of de-Broglie wavelengths is $2 : 1 \Rightarrow \frac{n}{1} = \frac{2}{1} \Rightarrow n = 2$

Therefore, the value of n is 2 . **Hence, the correct answer is 2.**

Question3

The de Broglie wavelength for an electron accelerated through the potential difference of V_1 volt is λ_1 . When the potential difference is changed to V_2 volt, the associated de Broglie wavelength is increased by 50%. If $(V_1/V_2) = (9/\alpha)$, then the value of α is ____.

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Answer: 4

Solution:

The de Broglie wavelength λ is given by:

$$\lambda = \frac{h}{p}$$

Where h is Planck's constant and p is the momentum.

The kinetic energy (K) of an electron accelerated through a potential V is $K = eV = \frac{p^2}{2m}$

$$p = \sqrt{2meV}$$

Substituting p into the wavelength formula:

$$\lambda = \frac{h}{\sqrt{2meV}}$$

For a given particle (electron), the wavelength is inversely proportional to the square root of the potential,

$$\lambda \propto \frac{1}{\sqrt{V}}$$

So, the ratio of wavelengths in two cases is,

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{V_1}{V_2}}$$

The initial wavelength is λ_1 .

The final wavelength is $\lambda_2 = \lambda_1 + 50\% \text{ of } \lambda_1 = 1.5\lambda_1 = \frac{3}{2}\lambda_1$.

Substituting these into the ratio :

$$\frac{\frac{3}{2}\lambda_1}{\lambda_1} = \sqrt{\frac{V_1}{V_2}}$$

$$\Rightarrow \frac{3}{2} = \sqrt{\frac{V_1}{V_2}}$$

$$\Rightarrow \left(\frac{3}{2}\right)^2 = \frac{V_1}{V_2}$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{9}{4} = \frac{9}{\alpha} \Rightarrow \alpha = 4$$

Therefore, the value of α is 4 .

Question4

A light wave described by

$E = 60 [\sin (3 \times 10^{15})t + \sin (12 \times 10^{15})t]$ (in SI units) falls on a metal surface of work function 2.8 eV . The maximum kinetic energy of ejected photoelectron is (approximately)

_____ eV. ($h = 6.6 \times 10^{-34}$ J.s. and $e = 1.6 \times 10^{-19}$ C)

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Options:

A.

3.8

B.

7.8

C.

6.0

D.

5.1

Answer: D

Solution:

According to Einstein's photoelectric effect, the energy of an incident photon (E) on a metal of work function (ϕ) then the maximum kinetic energy ($K_{\max}$) to the ejected electron is,

$$K_{\max} = E - \phi$$

The energy of a photon is given by :

$$E = hf = \frac{h\omega}{2\pi}$$

Where :

- h is Planck's constant.
- ω is the angular frequency of the light wave.

The given electric field is $E = 60 [\sin(3 \times 10^{15})t + \sin(12 \times 10^{15})t]$

The given electric field equation contains two sinusoidal terms with angular frequencies are $\omega_1 = 3 \times 10^{15}$ rad/s and $\omega_2 = 12 \times 10^{15}$ rad/s

Since $K_{\max}$ depends on the maximum photon energy, we must use the higher angular frequency, $\omega_{\max} = 12 \times 10^{15}$ rad/s, because higher frequency corresponds to higher energy per photon.

The energy in Joules is $E = \frac{h\omega}{2\pi}$.

$$E(\text{ in eV}) = \frac{h \cdot \omega_{\max}}{2\pi \cdot e}$$

Substituting the given values :

$$E = \frac{(6.6 \times 10^{-34}) \times (12 \times 10^{15})}{2 \times 3.14 \times (1.6 \times 10^{-19})}$$

$$\Rightarrow E = \frac{79.2 \times 10^{-19}}{10.048 \times 10^{-19}}$$

$$\Rightarrow E \approx 7.88 \text{ eV}$$

So, the maximum kinetic energy of ejected electron is,

$$K_{\max} = E - \phi$$

$$\Rightarrow K_{\max} = 7.88 \text{ eV} - 2.8 \text{ eV}$$

$$\Rightarrow K_{\max} = 5.08 \text{ eV}$$

$$\Rightarrow K_{\max} \approx 5.1 \text{ eV}$$

Therefore, the kinetic energy of electron is approximately 5.1 eV.

Hence, the correct option is (D).

Question5

Light is incident on a metallic plate having work function $110 \times 10^{-20} \text{ J}$. If the produced photoelectrons have zero kinetic energy then the angular frequency of the incident light is _____ rad/s. ($h = 6.63 \times 10^{-34} \text{ J.s}$).

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Options:

A.

$$1.04 \times 10^{13}$$

B.

$$1.66 \times 10^{16}$$

C.

$$1.66 \times 10^{15}$$

D.

$$1.04 \times 10^{16}$$

Answer: D

Solution:

For photoelectrons to have zero kinetic energy, the incident light must have the **threshold frequency**.

Using Einstein's photoelectric equation:

$$h\nu = \phi$$

where

$$\phi = 110 \times 10^{-20} \text{ J} = 1.10 \times 10^{-18} \text{ J}$$

and

$$\nu = \frac{\phi}{h} = \frac{1.10 \times 10^{-18}}{6.63 \times 10^{-34}}$$

$$\nu \approx 1.66 \times 10^{15} \text{ Hz}$$

Now angular frequency is:

$$\omega = 2\pi\nu$$

$$\omega = 2\pi \times 1.66 \times 10^{15} \approx 10.43 \times 10^{15} = 1.04 \times 10^{16} \text{ rad/s}$$

So, the correct answer is:

$1.04 \times 10^{16} \text{ rad/s}$

Hence, **Option D** is correct.

Question6

The de Broglie wavelength of an oxygen molecule at 27°C is $x \times 10^{-12}$ m. The value of x is (take Planck's constant $= 6.63 \times 10^{-34}$ J.s, Boltzmann constant $= 1.38 \times 10^{-23}$ J/K, mass of oxygen molecule $= 5.31 \times 10^{-26}$ kg)

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Options:

A.

24

B.

30

C.

20

D.

26

Answer: D

Solution:

For a gas molecule, the average kinetic energy (K) is given by kinetic theory:

$$K = \frac{3}{2} k_B T$$

where k_B is the Boltzmann constant and T is the absolute temperature.

The relationship between momentum (p) and kinetic energy (K) is:

$$p = \sqrt{2mK}$$

Substituting K:

$$p = \sqrt{2m \left(\frac{3}{2} k_B T \right)} = \sqrt{3mk_B T} \dots (i)$$

The de Broglie wavelength is defined as:

$$\lambda = \frac{h}{p} \dots$$

From both the relations,

$$\lambda = \frac{h}{\sqrt{3mk_B T}}$$

The given temperature is 27°C .

$$T = 27 + 273 = 300 \text{ K}$$

The given mass is $5.31 \times 10^{-26} \text{ kg}$.

So, the de Broglie wavelength is,

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{3 \times 5.31 \times 10^{-26}}} = \frac{6.63 \times 10^{-34}}{\sqrt{6595.02 \times 10^{-49}}} = \frac{6.63 \times 10^{-34}}{\sqrt{6.595 \times 10^{-46}}} = \frac{6.63 \times 10^{-34}}{2.568 \times 10^{-23}}$$

$$\Rightarrow \lambda \approx 2.58 \times 10^{-11} \text{ m} = 25.8 \times 10^{-12} \text{ m}$$

$$\Rightarrow x \times 10^{-12} \text{ m} = 25.8 \times 10^{-12} \text{ m}$$

$$\Rightarrow x \approx 26$$

Therefore, the value of x is 26 . Hence, the correct option is (D).

Question7

When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V . If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V . The wavelength of first light is _____ m .

$$(h = 6.63 \times 10^{-34} \text{ J.s}, e = 1.6 \times 10^{-19} \text{ C}, c = 3 \times 10^8 \text{ m/s})$$

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Options:

A.

$$2.9 \times 10^{-8}$$

B.

$$2.5 \times 10^{-7}$$

C.

$$3.1 \times 10^{-7}$$

D.

$$2.2 \times 10^{-8}$$

Answer: B

Solution:

When a light of wavelength (λ) is incident on a metal of work function ϕ then the maximum kinetic energy of the photo-electron is given by Einstein's photoelectric equation:

$$K_{\max} = E - \phi \Rightarrow E = \phi + K_{\max}$$

$$\Rightarrow \frac{hc}{\lambda} = \phi + eV_s$$

where V_s is the stopping potential and ϕ is the work function of the metal.

Case 1 : Wavelength is λ , stopping potential is 3.2 V .

$$\frac{hc}{\lambda} = \phi + 3.2e \dots (i)$$

Case 2 : Wavelength is 2λ , stopping potential is 0.7 V .

$$\frac{hc}{2\lambda} = \phi + 0.7e \dots (ii)$$

Subtracting equation (ii) from equation (i) :

$$\left(\frac{hc}{\lambda} - \frac{hc}{2\lambda} \right) = (3.2e - 0.7e)$$

$$\Rightarrow \frac{hc}{2\lambda} = 2.5e$$

$$\Rightarrow \lambda = \frac{hc}{5e}$$

Substituting the given values :

$$\lambda = \frac{(6.63 \times 10^{-34}) \times (3 \times 10^8)}{5 \times (1.6 \times 10^{-19})}$$

$$\Rightarrow \lambda = \frac{19.89 \times 10^{-26}}{8 \times 10^{-19}} \text{ m} \approx 2.5 \times 10^{-7} \text{ m}$$

Therefore, the initial wavelength of light used was $2.5 \times 10^{-7} \text{ m}$.

Hence, the correct option is (B).

Question8

Number of photons of equal energy emitted per second by a 6 mW laser source operating at 663 nm is _____.
(Given : $h = 6.63 \times 10^{-34} \text{ J.s}$ and $c = 3 \times 10^8 \text{ m/s}$)

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Options:

A.

$$10 \times 10^{15}$$

B.

$$5 \times 10^{16}$$

C.

$$5 \times 10^{15}$$

D.

$$2 \times 10^{16}$$

Answer: D

Solution:

The power (P) of the laser source is, $P = 6 \text{ mW} = 6 \times 10^{-3} \text{ J/s}$

The wavelength (λ) of the emitted photon is $\lambda = 663 \text{ nm} = 663 \times 10^{-9} \text{ m}$

Planck's constant $h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$

Speed of light $c = 3 \times 10^8 \text{ m/s}$

The energy of a single photon is given by :

$$E = \frac{hc}{\lambda}$$

Total Power is the number of photons per second multiplied by the energy of one photon:

$$P = n \times E \Rightarrow n = \frac{P}{E} = \frac{P\lambda}{hc}$$

Substituting the given values into the equation :

$$n = \frac{(6 \times 10^{-3}) \times (663 \times 10^{-9})}{(6.63 \times 10^{-34}) \times (3 \times 10^8)}$$

$$\Rightarrow n = \frac{600}{3} \times \frac{10^{-12}}{10^{-26}}$$

$$\Rightarrow n = 200 \times 10^{14}$$

$$\Rightarrow n = 2 \times 10^{16}$$

Therefore, the laser source emits 2×10^{16} photons every second. Hence, the correct option is (D).

Question9

For a certain metal, when monochromatic light of wavelength λ is incident, the stopping potential for photoelectrons is $3V_0$. When the same metal is illuminated by light of wavelength 2λ , then the stopping potential becomes V_0 . The threshold wavelength for photoelectric emission for the given metal is $\alpha\lambda$. The value of α is _____.

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Options:

A.

1

B.

4

C.

2

D.

3

Answer: B

Solution:

The energy of an incident photon is $E = \frac{hc}{\lambda}$, where h is the Planck's constant, c is the speed of light in vacuum and λ is the wavelength of incident light.

According to Einstein's equation of photoelectric effect;

$$\frac{hc}{\lambda} = \phi + K_{\max}$$

Where ϕ is the work function is the minimum energy required to eject electron from the surface of metal, $\left(\phi = \frac{hc}{\lambda_0}\right)$ and $K_{\max} = eV_s$ (V_s is the stopping potential).

$$\text{So, } \frac{hc}{\lambda} = \frac{hc}{\lambda_0} + eV_s$$

Case 1: Wavelength is λ , stopping potential is $3V_0$.

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + e(3V_0) \dots (i)$$

Case 2: Wavelength is 2λ , stopping potential is V_0 .

$$\frac{hc}{2\lambda} = \frac{hc}{\lambda_0} + eV_0$$

$$\Rightarrow 3\left(\frac{hc}{2\lambda}\right) = 3\left(\frac{hc}{\lambda_0}\right) + 3eV_0 \dots (ii)$$

On subtracting equation (i) from equation (ii):

$$\left(\frac{3hc}{2\lambda} - \frac{hc}{\lambda}\right) = \left(\frac{3hc}{\lambda_0} - \frac{hc}{\lambda_0}\right) + (3eV_0 - 3eV_0)$$

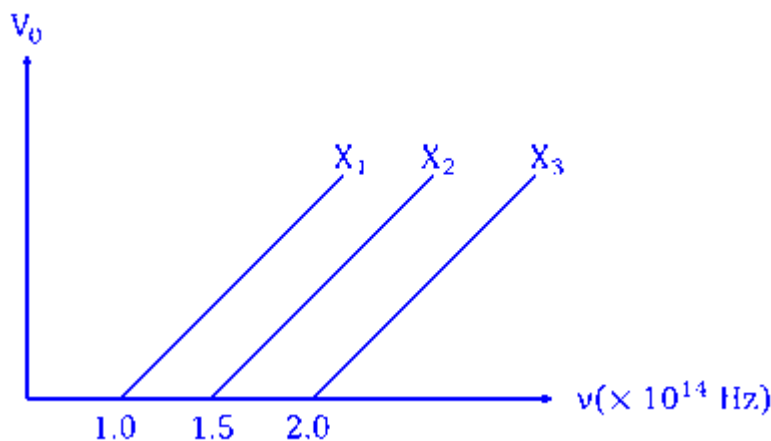
$$\frac{hc}{2\lambda} = \frac{2hc}{\lambda_0} \rightarrow \frac{1}{4\lambda} = \frac{1}{\lambda_0} \rightarrow \lambda_0 = 4\lambda$$

$$\Rightarrow \alpha\lambda = 4\lambda \Rightarrow \alpha = 4$$

Therefore, the value of α is 4. Thus, the correct option is (B).

Question10

The graph shows variation of stopping potential V_0 with the frequency ν of the incident radiation for three photosensitive metals X_1 , X_2 and X_3 . Which metal will give out electrons with greater kinetic energy, for the same wavelength of incident radiation?



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Options:

- A.
 X_1
- B.
 X_2
- C.
 X_3
- D. All the metals will give out photo electrons with same kinetic energies.

Answer: A

Solution:

The Einstein's photoelectric equation states that the energy of incident photons to the work function of the metal and the maximum kinetic energy of the emitted electrons.

$$K_{\max} = h\nu - \Phi_0$$

Where $K_{\max}$ is the maximum kinetic energy of the emitted photoelectrons, $h\nu$ is the energy of the incident photon (where h is Planck's constant and ν is frequency) and Φ_0 is the work function of the metal, which is the minimum energy required to eject an electron.

The work function can also be expressed in terms of the threshold frequency (ν_0), which is the minimum frequency required for photoelectric emission:

$$\Phi_0 = h\nu_0$$

The stopping potential (V_0) is the negative potential applied to stop the most energetic photoelectrons. The work done by this potential equals the maximum kinetic energy:

$$K_{\max} = eV_0$$

Substituting this into the photoelectric equation:

$$eV_0 = h\nu - \phi \Rightarrow V_0 = \left(\frac{h}{e}\right)\nu - \frac{\Phi_0}{e}$$

The equation $V_0 = \left(\frac{h}{e}\right)\nu - \Phi_0$ is in the form of a straight line ($y = mx + c$).

The slope $m = \frac{h}{e} = \text{constant}$ for all metals, which is why the lines are parallel.

X-intercept represents the threshold frequency (ν_0), where $V_0 = 0$. At this frequency, the incident energy exactly equals the work function ($\Phi_0 = h\nu_0$).

From the graph, we can identify the threshold frequencies for each metal:

$$\text{For } X_1 : \nu_0 \approx 1.0 \times 10^{14} \text{ Hz}$$

$$\text{For } X_2 : \nu_0 \approx 1.5 \times 10^{14} \text{ Hz}$$

$$\text{For } X_3 : \nu_0 \approx 2.0 \times 10^{14} \text{ Hz}$$

We can see that $\nu_{01} < \nu_{02} < \nu_{03}$. This means the work function of metal X_1 is the lowest ($\Phi_1 < \Phi_2 < \Phi_3$).

Now, considering incident radiation of the same wavelength (λ).

Since energy $E = \frac{hc}{\lambda}$, the incident energy is the same for all three metals.

Substituting this into the energy equation:

$$K_{\max} = E_{\text{incident}} - \Phi_0$$

Since E_{incident} is constant, the metal with the smallest work function (Φ_0) will result in the greatest kinetic energy ($K_{\max}$).

As metal X_1 has the lowest threshold frequency and thus the lowest work function, it will emit electrons with the greatest kinetic energy.

Hence, the correct option is (A).

Question11

The de Broglie wavelength associated with an electron accelerated through a potential difference V is λ_e and the de Broglie wavelength associated with a proton accelerated through the same potential difference is λ_p . If their corresponding masses are m_e and m_p , respectively, then the ratio of their de Broglie wavelengths $\left(\frac{\lambda_e}{\lambda_p}\right)$ is _____ .

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Options:

A.

$$\sqrt{\frac{m_p}{m_e}}$$

B.

$$\sqrt{\frac{m_e}{m_p}}$$

C.

$$\frac{m_p}{m_e}$$

D.

$$\left(\frac{m_p}{m_e}\right)^2$$

Answer: A

Solution:

According to de Broglie's hypothesis, the wavelength associated with a particle of momentum p is:

$$\lambda = \frac{h}{p}$$

Where h is Planck's constant.

Kinetic energy is given by $K = \frac{1}{2}mv^2$.

In terms of linear momentum, the kinetic energy can be written as

$$K = \frac{m^2v^2}{2m} = \frac{p^2}{2m}$$

Thus, momentum $p = \sqrt{2mK}$.

When a particle of charge q is accelerated through a potential difference V , the work done on it becomes its kinetic energy:

$$K = qV$$

Substituting this into the momentum equation:

$$p = \sqrt{2mqV}$$

So, the de-Broglie's wavelength is $\lambda = \frac{h}{\sqrt{2mqV}}$

The charge on electron is $q_e = e$, so the wavelength for an electron is $\lambda_e = \frac{h}{\sqrt{2 m_e eV}}$

The charge on proton is $q_p = e$, so the wavelength for an proton is $\lambda_p = \frac{h}{\sqrt{2 m_p eV}}$

Taking the ratio $\frac{\lambda_e}{\lambda_p}$:

$$\frac{\lambda_e}{\lambda_p} = \frac{h/\sqrt{2 m_e eV}}{h/\sqrt{2 m_p eV}} = \frac{\sqrt{2 m_p eV}}{\sqrt{2 m_e eV}}$$

$$\Rightarrow \frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$$

Therefore, the ratio of their de Broglie wavelengths is $\sqrt{\frac{m_p}{m_e}}$.

Thus, the correct option is (A).

Question12

Light source having wavelength 331 nm is used to generate photo-electrons whose stopping potential is 0.2 V . The work function of the used metal in the experiment is $\alpha \times 10^{-19}$ J. The value of α is _____ .

($h = 6.62 \times 10^{-34}$ J s, $e = 1.6 \times 10^{-19}$ C and $c = 3 \times 10^8$ m/s)

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Options:

A.

3.68

B.

4.68

C.

5.68

D.

2.68

Answer: C

Solution:

If the energy of an incident photon is E then it is equal to the energy required to overcome the work function (ϕ) of the metal and the maximum kinetic energy ($K_{\max}$) to the emitted electron.

$$E = \phi + K_{\max}$$

Therefore, $\phi = E - K_{\max}$.

Energy is given by $E = \frac{hc}{\lambda}$ where the Planck's constant $h = 6.62 \times 10^{-34}$ J s, the speed of light $c = 3 \times 10^8$ m/s, and the wavelength of light is $\lambda = 331$ nm $= 331 \times 10^{-9}$ m :

Putting the values,

$$E = \frac{(6.62 \times 10^{-34}) \times (3 \times 10^8)}{331 \times 10^{-9}}$$

$$\Rightarrow E = \frac{19.86 \times 10^{-26}}{331 \times 10^{-9}} = \frac{1986 \times 10^{-28}}{331 \times 10^{-9}}$$

$$\Rightarrow E = 6 \times 10^{-19} \text{ J}$$

The maximum kinetic energy is related to the stopping potential (V_0) by:

$$K_{\max} = e V_0$$

The electronic charge is $e = 1.6 \times 10^{-19}$ C and the stopping potential is $V_0 = 0.2$ V :

$$K_{\max} = (1.6 \times 10^{-19}) \times 0.2 = 0.32 \times 10^{-19} \text{ J}$$

So, the work function of metal is,

$$\phi = E - K_{\max}$$

$$\Rightarrow \phi = (6 \times 10^{-19}) - (0.32 \times 10^{-19})$$

$$\Rightarrow \phi = 5.68 \times 10^{-19} \text{ J} = \alpha \times 10^{-19} \text{ J} \Rightarrow \alpha = 5.68$$

Therefore, the correct option is (C).

Question13

An electron of mass m is moving in an electric field

$\vec{E} = -2E_0\hat{i}$ ($E_0 = \text{constant} > 0$), with an initial velocity

$\vec{V} = v_0\hat{i}$ ($v_0 = \text{constant} > 0$). If $\lambda_0 = \frac{h}{4mv_0}$, its de Broglie wavelength at time t is

_____ .

(e = charge of electron)

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Options:

A.

$$\frac{4\lambda_0}{\left[1 - \frac{E_0 e}{2m} \frac{t}{v_0}\right]}$$

B.

$$\frac{4\lambda_0}{\left[1 + \frac{E_0 e}{2m} \frac{t}{v_0}\right]}$$

C.

$$\frac{4\lambda_0}{\left[1 + \frac{2E_0 e}{m} \frac{t}{v_0}\right]}$$

D.

$$\frac{4\lambda_0}{\left[1 - \frac{2E_0 e}{m} \frac{t}{v_0}\right]}$$

Answer: C

Solution:

The force on an electron (charge $-e$) in an electric field $\vec{E}$ is $\vec{F} = -e\vec{E}$.

The electric field is given $\vec{E} = -2E_0\hat{i}$,

So, the force acting on electron is,

$$\vec{F} = (-e) \left(-2E_0\hat{i} \right) = 2eE_0\hat{i}$$

Using Newton's second law of motion, $\vec{F} = m\vec{a}$ the acceleration of electron is;

$$\vec{a} = \frac{\vec{F}}{m} = \frac{2eE_0}{m}\hat{i}$$

Using the first equation of motion ($\vec{v} = \vec{u} + \vec{a}t$), the final velocity of electron at any time t is;

$$\vec{v}(t) = v_0 + \left(\frac{2eE_0}{m} \right)t$$

$$\Rightarrow \vec{v}(t) = v_0 \left(1 + \frac{2eE_0t}{mv_0} \right)$$

The de Broglie wavelength is $\lambda = \frac{h}{p} = \frac{h}{mv}$.

Substituting the velocity:

$$\lambda_t = \frac{h}{mv_0 \left(1 + \frac{2eE_0t}{mv_0} \right)}$$

It is given that $\lambda_0 = \frac{h}{4mv_0} \Rightarrow \frac{h}{mv_0} = 4\lambda_0$.

$$\lambda_t = \frac{4\lambda_0}{1 + \frac{2eE_0t}{mv_0}}$$

Therefore, the correct option is (C).

Question14

An electron is travelling with a velocity v in free space and when it enters a medium, its velocity is reduced by 20%. The de Broglie wavelength of electron in the medium is $\alpha\lambda_0$, where λ_0 is its de Broglie wavelength in free space. The value of α is ____ .

Options:

A.

1.20

B.

1.0

C.

1.25

D.

0.75

Answer: C

Solution:

The de Broglie wavelength (λ) of a particle with mass m and velocity v is given by:

$$\lambda = \frac{h}{mv}$$

Where h is Planck's constant.

For a particle of constant mass (like an electron), the wavelength is inversely proportional to its velocity:

$$\lambda \propto \frac{1}{v}$$

Let the initial velocity in free space be v_0 .

The velocity in the medium (v_m) is reduced by 20% :

$$v_m = v_0 - 20\% \text{ of } v_0 = v_0 - 0.20v_0 = 0.80v_0$$

Let λ_0 be the wavelength in free space and λ_m be the wavelength in the medium.

$$\frac{\lambda_m}{\lambda_0} = \frac{v_0}{v_m}$$

Substituting the value of v_m :

$$\frac{\lambda_m}{\lambda_0} = \frac{v_0}{0.80v_0} = \frac{1}{0.80}$$

$$\Rightarrow \frac{\lambda_m}{\lambda_0} = \frac{100}{80} = 1.25$$

$$\Rightarrow \lambda = 1.25\lambda_0 = \alpha\lambda_0$$

$$\Rightarrow \alpha = 1.25$$

Therefore, the correct option is (C).

Question15

K_1 and K_2 be the maximum kinetic energies of photoelectrons emitted from a surface of a given material for the light of wavelength λ_1 and λ_2 , respectively. If $\lambda_1 = 2\lambda_2$ then the work function of material is given by :

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Options:

A.

$$K_2 + 2K_1$$

B.

$$2K_2 - K_1$$

C.

$$K_1 - 2K_2$$

D.

$$K_2 - 2K_1$$

Answer: D

Solution:

According to the photoelectric effect, when light of a certain frequency or wavelength hits a metal surface, it provides energy to the electrons. Part of this energy is used to overcome the attractive forces of the metal (the work function, ϕ), and the remaining energy becomes the maximum kinetic energy (K) of the emitted photoelectron. The energy of an incident photon is $E = \frac{hc}{\lambda}$. The equation is:

$$E = \phi + K_{\max}$$

$$\Rightarrow \frac{hc}{\lambda} = \phi + K_{\max}$$

$$\Rightarrow K_{\max} = \frac{hc}{\lambda} - \phi$$

We are given two different wavelengths and their corresponding kinetic energies:

For wavelength λ_1 :

$$K_1 = \frac{hc}{\lambda_1} - \phi \dots \quad (i)$$

For wavelength λ_2 :

$$K_2 = \frac{hc}{\lambda_2} - \phi \dots \quad (ii)$$

It is given that $\lambda_1 = 2\lambda_2$.

$$K_1 = \frac{hc}{2\lambda_2} - \phi \dots \quad (iii)$$

From equation (ii), we can find an expression for $\frac{hc}{\lambda_2}$:

$$\frac{hc}{\lambda_2} = K_2 + \phi$$

Now, substituting this into equation (iii) :

$$K_1 = \frac{1}{2}(K_2 + \phi) - \phi$$

$$\Rightarrow 2K_1 = K_2 + \phi - 2\phi$$

$$\Rightarrow 2K_1 = K_2 - \phi$$

$$\Rightarrow \phi = K_2 - 2K_1$$

Therefore, the work function of the material is $K_2 - 2K_1$.

Hence, the correct option is (D).

Question16

A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons /s. The region of an electromagnetic spectrum to which the emitted electromagnetic radiation belongs to ____.

(Take $h = 6.6 \times 10^{-34}$ J.s and $c = 3 \times 10^8$ m/s).

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Options:

A.

Microwave

B.

Infrared

C.

Visible

D.

Ultraviolet

Answer: D

Solution:

The power (P) of a light source is the total energy emitted per unit time. If the source emits n photons per second, and each photon has an energy E , then the power is:

$$P = n \times E$$

The given power of the source is $P = 15 \text{ kW} = 15 \times 10^3 \text{ W}$ and the number of photons per second is $n = 2.5 \times 10^{22} \text{ s}^{-1}$

$$E = \frac{P}{n}$$

$$\Rightarrow E = \frac{15 \times 10^3}{2.5 \times 10^{22}}$$

$$\Rightarrow E = 6 \times 10^{-19} \text{ J}$$

The energy of a photon is related to its wavelength by the formula:

$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$$

Substituting the constants and energy,

The Plank's constant is $h = 6.6 \times 10^{-34} \text{ J} \cdot \text{s}$,

The speed of light is $c = 3 \times 10^8 \text{ m/s}$

$$\lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{6 \times 10^{-19}} \text{ m}$$

$$\Rightarrow \lambda = \frac{19.8 \times 10^{-26}}{6 \times 10^{-19}} \text{ m}$$

$$\Rightarrow \lambda = 3.3 \times 10^{-7} \text{ m}$$

$$\Rightarrow \lambda = 3.3 \times 10^{-7} \times 10^9 \text{ nm} = 330 \text{ nm}$$

Visible light typically ranges from approximately 400 nm to 700 nm . And the ultraviolet (UV) radiation ranges from approximately 10 nm to 400 nm . Since 330 nm is less than 400 nm , the radiation belongs to the ultraviolet region.

Therefore, the emitted radiation belongs to the Ultraviolet region.

Hence, the correct option is (D).

Question17

An electron in the ground state of the hydrogen atom has the orbital radius of 5.3×10^{-11} m while that for the electron in third excited state is 8.48×10^{-10} m. The ratio of the de Broglie wavelengths of electron in the ground state to that in the excited state is

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Options:

- A. 4
- B. 3
- C. 9
- D. 16

Answer: A

Solution:

We know, $\lambda = \frac{h}{mv}$

and $mvr = \frac{nh}{2\pi}$

$$\Rightarrow mv = \frac{nh}{2\pi r}$$

$$\text{So, } \lambda = \frac{h}{nh} 2\pi r = 2\pi \frac{r}{n}$$

$$\Rightarrow \lambda \propto \frac{r}{n}$$

We can write, $\frac{\lambda_g}{\lambda_e} = \left(\frac{r_g}{r_e} \right) \left(\frac{n_e}{n_g} \right)$

$$\Rightarrow \frac{\lambda_g}{\lambda_e} = \left(\frac{5.3 \times 10^{-11}}{84.8 \times 10^{-11}} \right) \left(\frac{4}{1} \right) = \frac{1}{4}$$

$$\Rightarrow \frac{\lambda_e}{\lambda_g} = 4$$

Question18

The work functions of cesium (Cs) and lithium (Li) metals are 1.9 eV and 2.5 eV , respectively. If we incident a light of wavelength 550 nm on these two metal surfaces, then photo-electric effect is possible for the case of

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Options:

- A. Both Cs and Li
- B. Neither Cs nor Li
- C. Li only
- D. Cs only

Answer: D

Solution:

For photo-electric effect, energy of the photon must be greater than the work function of the metal.

We know, $E = \frac{1240}{\lambda} = \frac{1240}{550}$ (as $\lambda = 550$ nm given)

$$\Rightarrow E = 2.25 \text{ eV}$$

here, $E > \phi_{cs}$

So, Cs only.

Question19

A light source of wavelength λ illuminates a metal surface and electrons are ejected with maximum kinetic energy of 2 eV . If the same surface is illuminated by a light source of wavelength $\frac{\lambda}{2}$, then the maximum kinetic energy of ejected electrons will be (The work function of metal is 1 eV)

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Options:

A. 5 eV

B. 3 eV

C. 2 eV

D. 6 eV

Answer: A

Solution:

Let's break down the problem step by step.

The photoelectric equation is given by:

$$K_{\max} = \frac{hc}{\lambda} - \phi$$

where:

$K_{\max}$ is the maximum kinetic energy of the electrons.

$\frac{hc}{\lambda}$ is the energy of the incident photon.

ϕ is the work function of the metal.

For the initial light source of wavelength λ , we know:

$$2 \text{ eV} = \frac{hc}{\lambda} - 1 \text{ eV}$$

Solving for $\frac{hc}{\lambda}$ gives:

$$\frac{hc}{\lambda} = 2 \text{ eV} + 1 \text{ eV} = 3 \text{ eV}$$

Now, if the wavelength is halved to $\frac{\lambda}{2}$, the photon energy becomes:

$$\frac{hc}{\lambda/2} = \frac{2hc}{\lambda} = 2 \times 3 \text{ eV} = 6 \text{ eV}$$

The maximum kinetic energy for the new wavelength is then:

$$K'_{\max} = 6 \text{ eV} - 1 \text{ eV} = 5 \text{ eV}$$

Therefore, the maximum kinetic energy of the ejected electrons when the surface is illuminated by a light source of wavelength $\frac{\lambda}{2}$ is 5 eV.

The correct answer is Option A.

Question20

A sub-atomic particle of mass 10^{-30} kg is moving with a velocity 2.21×10^6 m/s. Under the matter wave consideration, the particle will behave closely like ($h = 6.63 \times 10^{-34}$ J.s)

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Options:

- A. X-rays
- B. Infra-red radiation
- C. Gamma rays
- D. Visible radiation

Answer: A

Solution:

To find the matter wave behavior of the particle, we use the de Broglie wavelength formula:

$$\lambda = \frac{h}{mv}$$

where:

$h = 6.63 \times 10^{-34}$ J · s is Planck's constant,

$m = 10^{-30}$ kg is the mass of the particle, and

$v = 2.21 \times 10^6$ m/s is its velocity.

Follow these steps:

Substitute the values into the formula:

$$\lambda = \frac{6.63 \times 10^{-34}}{(10^{-30})(2.21 \times 10^6)}$$

Calculate the denominator:

$$10^{-30} \times 2.21 \times 10^6 = 2.21 \times 10^{-24}$$

Now compute the wavelength:

$$\lambda \approx \frac{6.63 \times 10^{-34}}{2.21 \times 10^{-24}} \approx 3 \times 10^{-10} \text{ m}$$

This wavelength, roughly 3×10^{-10} m (or 0.3 nm), lies within the X-ray portion of the electromagnetic spectrum, since X-rays typically have wavelengths in the range of about 0.01 nm to 10 nm.

Therefore, under the matter wave consideration, the particle will behave closely like:

Option A: X-rays.

Question21

In photoelectric effect an em-wave is incident on a metal surface and electrons are ejected from the surface. If the work function of the metal is 2.14 eV and stopping potential is 2 V , what is the wavelength of the em-wave?(Given $hc = 1242\text{eVnm}$ where h is the Planck's constant and c is the speed of light in vaccum.)

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Options:

- A. 400 nm
- B. 600 nm
- C. 300 nm
- D. 200 nm

Answer: C

Solution:

Let's break down the problem step by step:

In the photoelectric effect, the energy of a photon is given by:

$$h\nu = \phi + K_e$$

where:

ϕ is the work function of the metal,

K_e is the kinetic energy of the ejected electron.

The kinetic energy of the ejected electrons is provided by the stopping potential:

$$K_e = eV$$

Given that the stopping potential is $V = 2 \text{ V}$, we have:

$$K_e = 2 \text{ eV}$$

Now, we can calculate the photon energy:

$$h\nu = \phi + K_e = 2.14 \text{ eV} + 2 \text{ eV} = 4.14 \text{ eV}$$

The relation between the photon's energy and its wavelength is:

$$h\nu = \frac{hc}{\lambda}$$

Rearranging for λ :

$$\lambda = \frac{hc}{h\nu}$$

Given that $hc = 1242 \text{ eV}\cdot\text{nm}$, substitute the values:

$$\lambda = \frac{1242 \text{ eV}\cdot\text{nm}}{4.14 \text{ eV}}$$

Calculate the wavelength:

$$\lambda \approx \frac{1242}{4.14} \approx 300 \text{ nm}$$

Thus, the wavelength of the incident electromagnetic wave is approximately 300 nm.

The correct answer is Option C: 300 nm.

Question22

An electron of mass ' m ' with an initial velocity $\vec{v} = v_0 \hat{i}$ ($v_0 > 0$) enters an electric field $\vec{E} = -E_0 \hat{k}$. If the initial de Broglie wavelength is λ_0 , the value after time t would be

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Options:

A. $\frac{\lambda_0}{\sqrt{1 - \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$

B. λ_0

C. $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$

$$D. \lambda_0 \sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}$$

Answer: C

Solution:

Derivation of the de Broglie wavelength after time t

The de Broglie wavelength is given by

$$\lambda = \frac{h}{p},$$

where h is Planck's constant and p is the momentum of the particle.

Initial Conditions:

Initially, the electron has a velocity

$$\vec{v} = v_0 \hat{i},$$

so its momentum is

$$p_0 = mv_0.$$

The initial de Broglie wavelength is therefore

$$\lambda_0 = \frac{h}{mv_0}.$$

Effect of the Electric Field:

The electron enters an electric field

$$\vec{E} = -E_0 \hat{k}.$$

For an electron with charge $-e$, the force is given by

$$\vec{F} = -e \vec{E} = -e (-E_0 \hat{k}) = eE_0 \hat{k}.$$

This force causes an acceleration in the $\hat{k}$ direction:

$$a_z = \frac{F_z}{m} = \frac{eE_0}{m}.$$

Momentum Components After Time t :

In the $\hat{i}$ direction, there is no force, so the momentum remains constant:

$$p_x = mv_0.$$

In the $\hat{k}$ direction, starting from rest, the momentum becomes:

$$p_z = mv_z = m(a_z t) = eE_0 t.$$

Total Momentum of the Electron:

The total momentum is the vector sum of the components:

$$p = \sqrt{p_x^2 + p_z^2} = \sqrt{(mv_0)^2 + (eE_0t)^2}.$$

New de Broglie Wavelength:

Substituting the expression for the total momentum into the de Broglie relation gives:

$$\lambda = \frac{h}{\sqrt{m^2v_0^2 + e^2E_0^2t^2}}.$$

Expressing this in terms of the initial de Broglie wavelength $\lambda_0 = \frac{h}{mv_0}$:

$$\lambda = \frac{\lambda_0}{\sqrt{1 + \frac{e^2E_0^2t^2}{m^2v_0^2}}}.$$

This final expression matches the option:

$$\frac{\lambda_0}{\sqrt{1 + \frac{e^2E_0^2t^2}{m^2v_0^2}}}.$$

Question23

In photoelectric effect, the stopping potential (V_0)v/s frequency (ν) curve is plotted.

(h is the Planck's constant and ϕ_0 is work function of metal)

(A) V_0 v/sv is linear.

(B) The slope of V_0 v/sv curve = $\frac{\phi_0}{h}$

(C) h constant is related to the slope of V_0 v/sv line.

(D) The value of electric charge of electron is not required to determine h using the V_0 v/sv curve.

(E) The work function can be estimated without knowing the value of h .

Choose the correct answer from the options given below :

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Options:

- A. (A), (C) and (E) only
- B. (C) and (D) only
- C. (A), (B) and (C) only
- D. (D) and (E) only

Answer: A

Solution:

$$h\nu = \phi + KE_{\max}$$

$$KE_{\max} = eV_0$$

$$V_0 = \frac{h\nu - \phi}{e}$$

(A) V_0 vs ν is linear correct

(B) Slope

$$V_0 = \left(\frac{h}{e}\right)\nu - \frac{\phi}{e} \text{ Wrong}$$

Slope $\frac{h}{e}$

(C) Correct

(D) Incorrect

(E) Correct

Question 24

A proton of mass ' m_P ' has same energy as that of a photon of wavelength ' λ '. If the proton is moving at non-relativistic speed, then ratio of its de Broglie wavelength to the wavelength of photon is.

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Options:

A. $\frac{1}{c} \sqrt{\frac{E}{m_p}}$

B. $\frac{1}{c} \sqrt{\frac{2E}{m_p}}$

C. $\frac{1}{2c} \sqrt{\frac{E}{m_p}}$

D. $\frac{1}{c} \sqrt{\frac{E}{2 m_p}}$

Answer: D

Solution:

To find the ratio of the de Broglie wavelength of a proton to the wavelength of a photon when they have the same energy, we follow these steps:

Energy Equivalence:

Both the proton and the photon have the same energy, denoted by E .

Energy of the Photon:

The energy of a photon with wavelength λ is given by:

$$E = \frac{hc}{\lambda}$$

Kinetic Energy of the Proton:

Since the proton moves at a non-relativistic speed, its kinetic energy can be expressed as:

$$KE_{\text{proton}} = \frac{1}{2} m_P v^2 = E$$

Momentum of the Proton:

The momentum p of the proton is derived from its kinetic energy:

$$p = \sqrt{2m_P E}$$

de Broglie Wavelength of the Proton:

The de Broglie wavelength λ_{proton} is:

$$\lambda_{\text{proton}} = \frac{h}{p} = \frac{h}{\sqrt{2m_P E}}$$

Ratio of Wavelengths:

The ratio of the de Broglie wavelength of the proton to the wavelength of the photon is:

$$\frac{\lambda_{\text{proton}}}{\lambda_{\text{photon}}} = \left(\frac{h}{\sqrt{2m_P E}} \right) \times \left(\frac{E}{hc} \right)$$

Simplifying, we get:

$$\frac{\lambda_{\text{proton}}}{\lambda_{\text{photon}}} = \frac{1}{c} \sqrt{\frac{E}{2m_P}}$$

Thus, the ratio of the de Broglie wavelength of the proton to the wavelength of the photon is $\frac{1}{c} \sqrt{\frac{E}{2m_P}}$.

Question25

Which of the following phenomena cannot be explained by wave theory of light?

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Options:

A.

Refraction of light

B.

Reflection of light

C.

Diffraction of light

D.

Compton effect

Answer: D

Solution:

Among the four listed phenomena:

Reflection of light

Wave theory (Huygens's principle, for example) adequately explains reflection.

Refraction of light

Wave theory also explains refraction via the change of wave speed in different media.

Diffraction of light

Wave theory (again, Huygens–Fresnel principle) explains the bending of waves around obstacles.

Compton effect

This involves *inelastic scattering of photons* by electrons (where the photon loses energy and is scattered with a longer wavelength).

Wave theory alone cannot explain the discrete energy exchange and wavelength shift; it requires the *particle (photon)* concept from quantum theory.

Hence, the phenomenon **not** explained by the (classical) wave theory is the **Compton effect**.

Question 26

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Emission of electrons in photoelectric effect can be suppressed by applying a sufficiently negative electron potential to the photoemissive substance.

Reason (R) : A negative electric potential, which stops the emission of electrons from the surface of a photoemissive substance, varies linearly with frequency of incident radiation.

In the light of the above statements, choose the most appropriate answer from the options given below :

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Options:

A.

Both (A) and (R) are true and (R) is the correct explanation of (A).

B.

Both (A) and (R) are true but (R) is not the correct explanation of (A).

C.

(A) is false but (R) is true

D.

(A) is true but (R) is false

Answer: B

Solution:

Assertion (A) is **true** because a sufficiently negative potential can repel and stop the emitted electrons from leaving the surface.

Reason (R) is **also true** because the stopping potential V_{stop} varies linearly with the frequency ν of the incident light ($V_{\text{stop}} = \frac{h}{e} \nu - \frac{\phi}{e}$).

However, (R) does **not** directly explain **why** a negative potential suppresses electron emission; it only shows **how much** potential is needed for different frequencies.

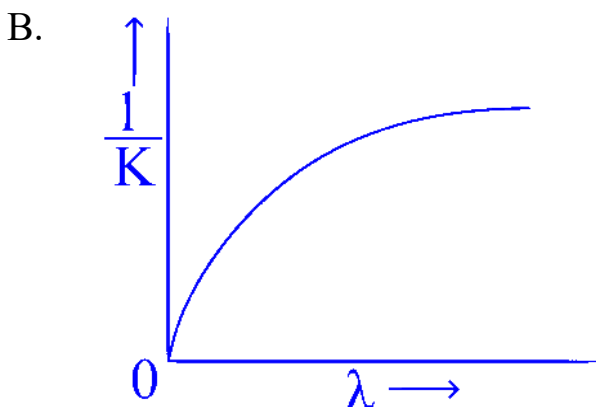
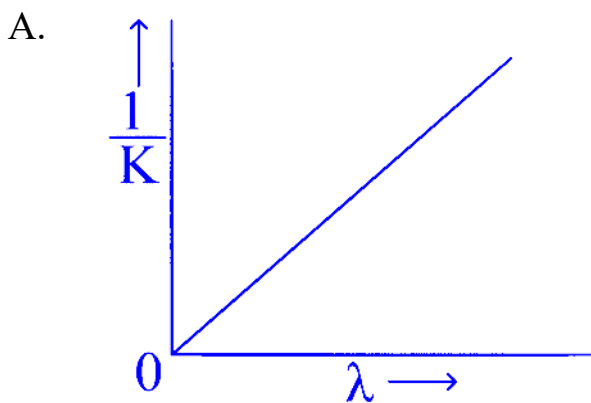
Therefore, **both (A) and (R) are true, but (R) is not the correct explanation of (A).**

Question27

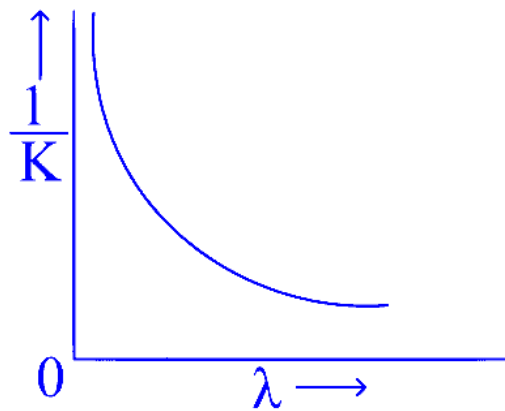
If λ and K are de Broglie wavelength and kinetic energy, respectively, of a particle with constant mass. The correct graphical representation for the particle will be :

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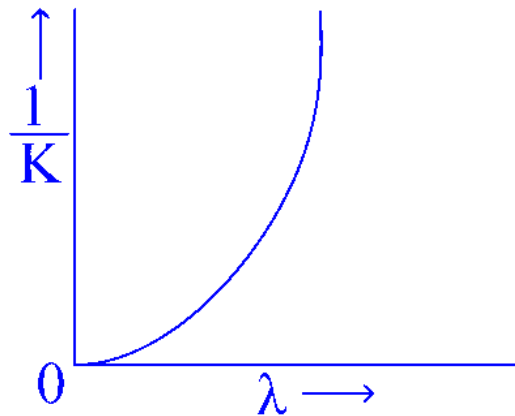
Options:



C.



D.



Answer: D

Solution:

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mK}}$$

$$\lambda^2 = \frac{h^2}{2m} \left(\frac{1}{K} \right)$$

$$Y = cx^2$$

Upward facing parabola passing through origin.

Question28

In an experiment with photoelectric effect, the stopping potential,

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Options:

A.

is $\left(\frac{1}{e}\right)$ times the maximum kinetic energy of the emitted photoelectrons

B.

increases with increase in the intensity of the incident light

C.

decreases with increase in the intensity of the incident light

D.

increases with increase in the wavelength of the incident light

Answer: A

Solution:

We know, $eV_0 = hv - \phi$ (i)

$\Rightarrow eV_0 = KE_{\max}$ (where ϕ = work function of the metal and v_0 = stopping potential)

$\Rightarrow V_0 = \left(\frac{1}{e}\right)KE_{\max}$

Stopping potential depends solely on the frequency of the incident light $+(v)$ & the work function of the metal, not the intensity.

From eq. (i), we can see,

$v \uparrow \Rightarrow v_0 \uparrow$

$\Rightarrow \frac{e}{\lambda \downarrow} \Rightarrow v_0 \uparrow$ (as $v = \frac{c}{\lambda}$)

So, stopping potential increases with decrease in the wavelength of the incident light.

Hence, option 1 is correct.

Question 29

A monochromatic light is incident on a metallic plate having work function ϕ . An electron, emitted normally to the plate from a point A with maximum kinetic energy, enters a constant magnetic field, perpendicular to the initial velocity of electron. The electron passes through a curve and hits back the plate at a point B . The distance between A and B is : (Given : The magnitude of charge of an electron is e and mass is m , h is Planck's constant and c is velocity of light. Take the magnetic field exists throughout the path of electron)

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Options:

A. $\sqrt{2m \left(\frac{hc}{\lambda} - \phi \right)} / eB$

B. $\sqrt{8m \left(\frac{hc}{\lambda} - \phi \right)} / eB$

C. $\sqrt{m(hc/\lambda - \phi)} / eB$

D. $2\sqrt{m(hc/\lambda - \phi)} / eB$

Answer: B

Solution:

To determine the distance between points A and B (where the electron re-enters the metallic plate), we first need to understand the electron's behavior in a magnetic field.

Maximum Kinetic Energy (KE): The energy of the emitted electron can be given by the photoelectric equation:

$$KE_{\max} = \frac{hc}{\lambda} - \phi$$

Here, $\frac{hc}{\lambda}$ is the energy of the incident photons, and ϕ is the work function of the metal.

Momentum (p): The momentum of the electron is related to its kinetic energy by the equation:

$$p = \sqrt{2m \cdot KE_{\max}} = \sqrt{2m \left(\frac{hc}{\lambda} - \phi \right)}$$

Path of Electron in Magnetic Field: When an electron moves perpendicularly through a magnetic field, it follows a circular path. The radius R of this path is given by:

$$R = \frac{p}{eB}$$

where e is the charge of the electron, and B is the magnetic field strength.

Distance Between A and B : Since the electron travels back to the plate, forming a complete semicircle, the distance between points A and B is twice the radius of this circular path:

$$d_{A-B} = 2R = 2 \left(\frac{p}{eB} \right)$$

Substituting the expression for momentum, we find:

$$d_{A-B} = \frac{2\sqrt{2m\left(\frac{hc}{\lambda} - \phi\right)}}{eB} = \frac{\sqrt{8m\left(\frac{hc}{\lambda} - \phi\right)}}{eB}$$

Therefore, the distance between A and B is $\frac{\sqrt{8m\left(\frac{hc}{\lambda} - \phi\right)}}{eB}$.

Question30

An electron with mass ' m ' with an initial velocity

($t = 0$) $\vec{v} = v_0 \hat{i}$ ($v_0 > 0$) enters a magnetic field $\vec{B} = B_0 \hat{j}$. If the initial de-Broglie wavelength at $t = 0$ is λ_0 then its value after time ' t ' would be :

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Options:

A. $\frac{\lambda_0}{\sqrt{1 - \frac{e^2 B_0^2 t^2}{m^2}}}$

B. λ_0

C. $\lambda_0 \sqrt{1 + \frac{e^2 B_0^2 t^2}{m^2}}$

D. $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 B_0^2 t^2}{m^2}}}$

Answer: B

Solution:

Magnetic field does not work

∴ Speed will not change, so De-Broglie wavelength remains same.

Question 31

The radiation pressure exerted by a 450 W light source on a perfectly reflecting surface placed at 2 m away from it, is

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Options:

A. 3×10^{-8} Pascals

B. 0

C. 1.5×10^{-8} Pascals

D. 6×10^{-8} Pascals

Answer: D

Solution:

First, the intensity I at a distance r from a point light source spreading its energy isotropically is given by

$$I = \frac{P}{4\pi r^2},$$

where

$$P = 450 \text{ W}$$

$$r = 2 \text{ m.}$$

So,

$$I = \frac{450}{4\pi(2)^2} = \frac{450}{16\pi} \approx \frac{450}{50.27} \approx 8.96 \text{ W/m}^2.$$

For a perfectly reflecting surface, the radiation pressure p is given by

$$p = \frac{2I}{c},$$

where $c \approx 3 \times 10^8$ m/s is the speed of light.

Substituting the value of I :

$$p = \frac{2 \times 8.96}{3 \times 10^8} \approx \frac{17.92}{3 \times 10^8} \text{ Pa.}$$

Calculating the final value:

$$p \approx 5.97 \times 10^{-8} \text{ Pa},$$

which is approximately $6 \times 10^{-8} \text{ Pa}$.

Thus, the correct answer is:

Option D: $6 \times 10^{-8} \text{ Pascals}$.

Question32

The work function of a metal is 3 eV . The color of the visible light that is required to cause emission of photoelectrons is

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Options:

A. Red

B. Green

C. Blue

D. Yellow

Answer: C

Solution:

To determine which color of visible light will cause the emission of photoelectrons from a metal with a work function of 3 eV, we need to consider the relationship between the photon energy, the work function, and the wavelength of light.

The maximum kinetic energy (KE_{max}) of the emitted photoelectrons is given by:

$$\text{KE}_{\text{max}} = \frac{hc}{\lambda} - \phi$$

For photoemission to occur, the energy of the incident photons must be greater than the work function (ϕ):

$$\frac{hc}{\lambda} > \phi$$

This implies:

$$\lambda < \frac{hc}{\phi}$$

Substitute the values $h = 4.135667696 \times 10^{-15} \text{ eV} \cdot \text{s}$, $c = 3 \times 10^8 \text{ m/s}$, and $\phi = 3 \text{ eV}$:

$$\lambda < \frac{1242}{3} \text{ nm} \approx 414 \text{ nm}$$

Since the wavelength must be less than 414 nm to cause emission, we need light with a shorter wavelength. Among the visible colors, blue light has a wavelength in this range, making it the correct choice. Thus, blue light will cause the emission of photoelectrons with this work function.

Question33

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : In photoelectric effect, on increasing the intensity of incident light the stopping potential increases.

Reason R : Increase in intensity of light increases the rate of photoelectrons emitted, provided the frequency of incident light is greater than threshold frequency.

In the light of the above statements, choose the correct answer from the options given below

JEE Main 2025 (Online) 4th April Morning Shift

Options:

- A. Both **A** and **R** are true and **R** is the correct explanation of **A**
- B. **A** is false but **R** is true
- C. Both **A** and **R** are true but **R** is NOT the correct explanation of **A**
- D. **A** is true but **R** is false

Answer: B

Solution:

In the photoelectric effect, the stopping potential (V_s) is determined by the equation:

$$V_s = \frac{hv - \phi}{e}$$

where h is Planck's constant, v is the frequency of the incident light, ϕ is the work function of the material, and e is the charge of the electron. This shows that the stopping potential depends on the frequency of the

incident light and the work function, not on the intensity of the light.

The intensity of light (I) affects the number of photons hitting the surface per second, given by:

$$I = \frac{\eta h \nu}{A}$$

where η is the efficiency of photoelectron emission and A is the area. Increasing the intensity increases the number of photons per second (n), which in turn increases the number of emitted electrons. However, the stopping potential remains unaffected by changes in intensity; it is solely influenced by the frequency of the incident light relative to the work function of the material.

Question34

A small mirror of mass m is suspended by a massless thread of length l . Then the small angle through which the thread will be deflected when a short pulse of laser of energy E falls normal on the mirror

(c = speed of light in vacuum and g = acceleration due to gravity)

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. $\theta = \frac{E}{mc\sqrt{gl}}$

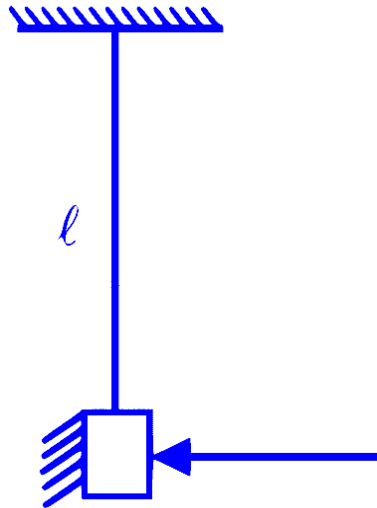
B. $\theta = \frac{E}{2mc\sqrt{gl}}$

C. $\theta = \frac{3E}{4mc\sqrt{gl}}$

D. $\theta = \frac{2E}{mc\sqrt{gl}}$

Answer: D

Solution:

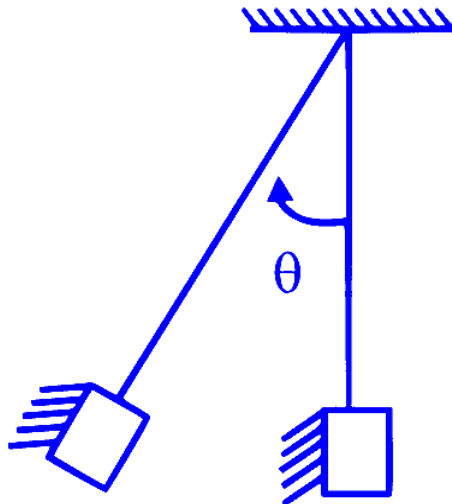


Force due to beam assuming complete reflection $F = \frac{2P}{C} = \frac{2}{C} \frac{dE}{dt}$; P is power

So change in momentum of mirror.

$$m(V - 0) = \int F dt = \frac{2}{C} \int dE = \frac{2E}{C}$$

Now using work energy theorem (1)



$$W_g = \Delta k$$

$$-mg\ell(1 - \cos \theta) = 0 - \frac{1}{2}mv^2$$

$$g\ell \left(2 \sin^2 \frac{\theta}{2} \right) = \frac{v^2}{2}$$

as θ is small

$$g\ell 2 \left(\frac{\theta}{2} \right)^2 = \frac{1}{2} \frac{4E^2}{m^2 c^2} \quad (\text{from eq. (1)})$$

$$g\ell \theta^2 = \frac{4E^2}{m^2 c^2}$$

$$\theta = \frac{2E}{mc\sqrt{g\ell}}$$

Question 35

A photoemissive substance is illuminated with a radiation of wavelength λ_i so that it releases electrons with de-Broglie wavelength λ_e . The longest wavelength of radiation that can emit photoelectron is λ_o . Expression for de-Broglie wavelength is given by:

(m: mass of the electron, h: Planck's constant and c: speed of light)

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Options:

A.

$$\lambda_e = \frac{\sqrt{h\lambda_i}}{\sqrt{2mc}}$$

B.

$$\lambda_e = \frac{h}{\sqrt{2mc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_o}\right)}}$$

C. $\lambda_e = \sqrt{\frac{h}{2mc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_o}\right)}}$

D. $\lambda_e = \sqrt{\frac{h\lambda_0}{2mc}}$

Answer: C

Solution:

Kinetic Energy (K.E.): The kinetic energy of the emitted electron is given by subtracting the work function (W) of the material from the energy of the incident radiation (E).

$$\text{K.E.} = E - W$$

Expressions for Energy:

The energy of the incoming radiation E is calculated using the formula:

$$E = \frac{hc}{\lambda_i}$$

The work function W , which is the minimum energy needed to release an electron, when the radiation wavelength is at its longest λ_0 , is:

$$W = \frac{hc}{\lambda_0}$$

De-Broglie Wavelength of the Electron:

The de-Broglie wavelength λ_e of the emitted electron is given by:

$$\lambda_e = \frac{h}{\sqrt{2m \cdot \text{K.E.}}}$$

Substituting the Values:

The kinetic energy of the emitted electron can be expressed in terms of energies as:

$$\frac{h^2}{2m\lambda_e^2} = \frac{hc}{\lambda_i} - \frac{hc}{\lambda_0}$$

Final Expression for λ_e :

Solving for λ_e , the de-Broglie wavelength of the electron, we have:

$$\lambda_e = \sqrt{\frac{h}{2mc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_0}\right)}}$$

This expression determines the de-Broglie wavelength of the emitted electron based on the wavelengths of the incident and threshold radiation.

Question36

The ratio of the power of a light source S_1 to that the light source S_2 is 2. S_1 is emitting 2×10^{15} photons per second at 600 nm . If the wavelength of the source S_2 is 300 nm , then the number of photons per second emitted by S_2 is _____ $\times 10^{14}$.

JEE Main 2025 (Online) 24th January Evening Shift

Answer: 5

Solution:

Since power emitting by a source is given as

$$= \frac{\text{Total energy emitted}}{\text{time}}$$

$$= \frac{(E_1 \text{ photon}) \times \text{Number of photons (N)}}{t}$$

$$P_1 = (E_1)n$$

$$\frac{P_1}{P_2} = \frac{(E_1)n_1}{(E_2)n_2} = \frac{\left(\frac{hc}{\lambda_1}\right)n_1}{\left(\frac{hc}{\lambda_2}\right)n_2}$$

$$\frac{P_1}{P_2} = \left(\frac{\lambda_2}{\lambda_1}\right) \frac{n_1}{n_2}$$

Substituting the given values

$$2 = \left(\frac{300}{600}\right) \times \frac{2 \times 10^{15}}{n_2}$$

$$n_2 = \frac{1}{2} \times 10^{15} = 5 \times 10^{14} \text{ Photon /sec}$$

Question37

An electron is released from rest near an infinite non-conducting sheet of uniform charge density ' $-\sigma$ '. The rate of change of de-Broglie wave length associated with the electron varies inversely as n^{th} power of time. The numerical value of n is ____.

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Answer: 2

Solution:

Let the momentum of e^- at any time t is p and its de-broglie wavelength is λ .

$$\text{Then, } p = \frac{h}{\lambda}$$

$$\frac{dp}{dt} = \frac{-h}{\lambda^2} \frac{d\lambda}{dt}$$

$$ma = F = -\frac{h}{\lambda} \frac{d\lambda}{dt} \quad [m = \text{mass of } e]$$

Where, -ve sign represents decrease in λ with time

$$ma = \frac{-h}{(h/p)^2} \frac{d\lambda}{dt}$$

$$a = -\frac{p^2}{mh} \frac{d\lambda}{dt}$$

$$a = -\frac{mv^2}{h} \frac{d\lambda}{dt}$$

$$\frac{d\lambda}{dt} = -\frac{ah}{mv^2} \dots (1)$$

$$\text{here, } a = \frac{qE}{m} = \frac{e}{m} \frac{\sigma}{2\varepsilon_0}$$

$$a = \frac{\sigma e}{2 m \varepsilon_0}$$

$$\text{and } v = u + at$$

$$v = at$$

Substituting values of a & v in equation (1)

$$\frac{d\lambda}{dt} = -\frac{2 h \varepsilon_0}{\sigma t^2}$$

$$\Rightarrow \frac{d\lambda}{dt} \propto \frac{1}{t^2}$$

$$\Rightarrow n = 2$$

Question38

**The threshold frequency of a metal with work function 6.63 eV is :
[27-Jan-2024 Shift 2]**

Options:

A. 16×10^{15} Hz

B. 16×10^{12} Hz

C. 1.6×10^{12} Hz

D. 1.6×10^{15} Hz

Answer: D

Solution:

$$\phi_0 = h\nu_0$$

$$6.63 \times 1.6 \times 10^{-19} = 6.63 \times 10^{-34} \nu_0$$

$$\nu_0 = \frac{1.6 \times 10^{-19}}{10^{-34}}$$

$$\nu_0 = 1.6 \times 10^{15} \text{ Hz}$$

Question39

Two sources of light emit with a power of 200W. The ratio of number of photons of visible light emitted by each source having wavelengths 300 nm and 500 nm respectively, will be :
[29-Jan-2024 Shift 2]

Options:

A. 1 : 5

B. 1 : 3

C. 5 : 3

D. 3 : 5

Answer: D

Solution:

$$n_1 \times \frac{hc}{\lambda_1} = 200$$

$$n_2 \times \frac{hc}{\lambda_2} = 200$$

$$\frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2} = \frac{300}{500}$$

$$\frac{n_1}{n_2} = \frac{3}{5}$$

Question40

The work function of a substance is 3.0 eV. The longest wavelength of light that can cause the emission of photoelectrons from this substance is approximately:
[30-Jan-2024 Shift 1]

Options:

A. 215 nm

B. 414 nm

C. 400 nm

D. 200 nm

Answer: B

Solution:

$$\text{For P.E.E. : } \lambda \leq \frac{hc}{W_s}$$

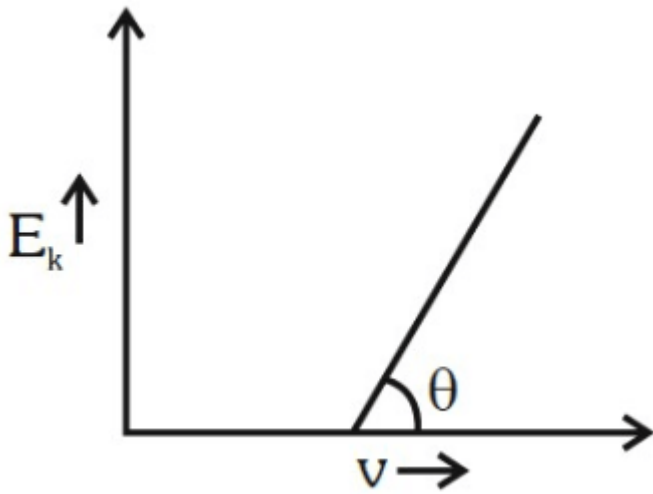
$$\lambda \leq \frac{1240 \text{ nm} \cdot \text{eV}}{3 \text{ eV}}$$

$$\lambda \leq 413.33 \text{ nm}$$

$$\lambda_{\text{max}} \approx 414 \text{ nm for P.E.E.}$$

Question41

For the photoelectric effect, the maximum kinetic energy (E_k) of the photoelectrons is plotted against the frequency (ν) of the incident photons as shown in figure. The slope of the graph gives



[30-Jan-2024 Shift 2]

Options:

- A. Ratio of Planck's constant to electric charge
- B. Work function of the metal
- C. Charge of electron
- D. Planck's constant

Answer: D

Solution:

$$\text{K.E.} = hf - \phi$$

$$\tan \theta = h$$

Question42

When a metal surface is illuminated by light of wavelength λ , the stopping potential is 8V. When the same surface is illuminated by light of wavelength 3λ , stopping potential is 2V. The threshold wavelength for this surface is :

[31-Jan-2024 Shift 1]

Options:

- A. 5λ

B. 3λ

C. 9λ

D. 4.5λ

Answer: C

Solution:

$$E = \phi + K_{\max}$$

$$\phi = \frac{hc}{\lambda_0}$$

$$K_{\max} = eV_0$$

$$8e = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \dots\dots (i)$$

$$2e = \frac{hc}{3\lambda} - \frac{hc}{\lambda_0} \dots\dots (ii)$$

on solving (i) & (ii)

$$\lambda_0 = 9\lambda$$

Question43

If the wavelength of the first member of Lyman series of hydrogen is λ . The wavelength of the second member will be [31-Jan-2024 Shift 1]

Options:

A. $\frac{27}{32}\lambda$

B. $\frac{32}{27}\lambda$

C. $\frac{27}{5}\lambda$

D. $\frac{5}{27}\lambda$

Answer: A

Solution:

$$\frac{1}{\lambda} = \frac{13.6Z^2}{hc} \left[\frac{1}{1^2} - \frac{1}{2^2} \right] \dots\dots (i)$$

$$\frac{1}{\lambda'} = \frac{13.6Z^2}{hc} \left[\frac{1}{1^2} - \frac{1}{3^2} \right] \dots\dots(ii)$$

On dividing (i) & (ii)

$$\lambda' = \frac{27}{32}\lambda$$

Question44

The de Broglie wavelengths of a proton and an α particle are λ and 2λ respectively. The ratio of the velocities of proton and α particle will be :

[1-Feb-2024 Shift 1]

Options:

A. 1 : 8

B. 1 : 2

C. 4 : 1

D. 8 : 1

Answer: D

Solution:

$$\lambda = \frac{h}{p} = \frac{h}{mv} \Rightarrow v = \frac{h}{m\lambda}$$

$$\frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p} \times \frac{\lambda_\alpha}{\lambda_p}$$

$$= 4 \times 2 = 8$$

Question45

Monochromatic light of frequency 6×10^{14} Hz is produced by a laser. The power emitted is 2×10^{-3} W. How many photons per second on an average, are emitted by the source ?

(. Given $h = 6.63 \times 10^{-34}$ Js)

[1-Feb-2024 Shift 2]

Options:

A. 9×10^{18}

B. 6×10^{15}

C. 5×10^{15}

D. 7×10^{16}

Answer: C

Solution:

$$P = nh\nu$$

$$n = \frac{P}{h\nu} = \frac{2 \times 10^{-3}}{6.63 \times 10^{-34} \times 6 \times 10^{14}} \\ = 5 \times 10^{15}$$

Question46

From the photoelectric effect experiment, following observations are made. Identify which of these are correct

A. The stopping potential depends only on the work function of the metal.

B. The saturation current increases as the intensity of incident light increases.

C. The maximum kinetic energy of a photo electron depends on the intensity of the incident light.

**D. Photoelectric effect can be explained using wave theory of light.
Choose the correct answer from the options given below:
[24-Jan-2023 Shift 1]**

Options:

- A. B, C only
- B. A, C, D only
- C. B only
- D. A, B, D only

Answer: C

Solution:

- (A) Stopping potential depends on both frequency of light and work function.
 - (B) Saturation current $\propto$ intensity of light
 - (C) Maximum KE depends on frequency
 - (D) Photoelectric effect is explained using particle theory
-

Question47

**An α -particle, a proton and an electron have the same kinetic energy. Which one of the following is correct in case of their De-Broglie wavelength:
[24-Jan-2023 Shift 2]**

Options:

- A. $\lambda_{\alpha} > \lambda_p > \lambda_e$
- B. $\lambda_{\alpha} < \lambda_p < \lambda_e$
- C. $\lambda_{\alpha} = \lambda_p = \lambda_e$
- D. $\lambda_{\alpha} > \lambda_p < \lambda_e$

Answer: B

Solution:

$$\lambda_D = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

$$\therefore \lambda \propto \frac{1}{\sqrt{m}}$$

$$\because m_\alpha > m_p > m_e$$

$$\lambda_e > \lambda_p > \lambda_\alpha$$

Question48

Electron beam used in an electron microscope, when accelerated by a voltage of 20 kV. has a de-Broglie wavelength of λ_0 . If the voltage is increased to 40 kV. then the de-Broglie wavelength associated with the electron beam would be:

[25-Jan-2023 Shift 1]

Options:

A. $3\lambda_0$

B. $9\lambda_0$

C. $\frac{\lambda_0}{2}$

D. $\frac{\lambda_0}{\sqrt{2}}$

Answer: D

Solution:

When electron is accelerated through potential difference V , then

$$\text{K.E.} = eV$$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2m(\text{KE})}} = \frac{h}{\sqrt{2meV}}$$

$$\therefore \lambda \propto \frac{1}{\sqrt{V}}$$

$$\therefore \frac{\lambda}{\lambda_0} = \sqrt{\frac{20}{40}}$$

$$\therefore \lambda = \frac{\lambda_0}{\sqrt{2}}$$

Question49

Given below are two statements :

Statement I : Stopping potential in photoelectric effect does not depend on the power of the light source.

Statement II : For a given metal, the maximum kinetic energy of the photoelectron depends on the wavelength of the incident light.

In the light of above statements, choose the most appropriate answer from the options given below.

Options :

[25-Jan-2023 Shift 2]

Options:

- A. Statement I is incorrect but statement II is correct
- B. Both Statement I and Statement II are incorrect
- C. Statement I is correct but statement II is incorrect
- D. Both statement I and statement II are correct

Answer: D

Solution:

$$\text{Stopping potential } V_s = \frac{KE_{\max}}{e}$$

$$V_s = \frac{\frac{hC}{\lambda} - \phi}{e}$$

Stopping potential does not depend on intensity or power of light used, it only depends on frequency or wavelength of incident light.

So both statements I and II are correct

Question50

The threshold wavelength for photoelectric emission from a material is $5500\text{\AA}$. Photoelectrons will be emitted, when this material is illuminated with monochromatic radiation from a

- A. 75W infra-red lamp
- B. 10W infra-red lamp
- C. 75W ultra-violet lamp
- D. 10W ultra-violet lamp

Choose the correct answer from the options given below :
[29-Jan-2023 Shift 1]

Options:

- A. B and C only
- B. A and D only
- C. C only
- D. C and D only

Answer: D

Solution:

$$\lambda < 5500\text{\AA} \text{ for photoelectric emission}$$
$$\lambda_{\text{uv}} < 5500\text{\AA}$$

Question51

The ratio of de-Broglie wavelength of an α -particle and a proton accelerated from rest by the same potential is $\frac{1}{\sqrt{m}}$, the value of m is

[29-Jan-2023 Shift 2]

Options:

- A. 4
- B. 16
- C. 8

D. 2

Answer: C

Solution:

$$\frac{\lambda_a}{\lambda_p} = \frac{\frac{h}{\sqrt{2m_a q_a V}}}{\frac{h}{\sqrt{2m_p q_p V}}}$$
$$\frac{\lambda_a}{\lambda_p} = \sqrt{\frac{1}{8}} \quad m = 8$$

Question52

A small object at rest, absorbs a light pulse of power 20 mW and duration 300 ns. Assuming speed of light as $3 \times 10^8 \text{ m/s}$. the momentum of the object becomes equal to :
[30-Jan-2023 Shift 1]

Options:

A. $0.5 \times 10^{-17} \text{ kg m/s}$

B. $2 \times 10^{-17} \text{ kg m/s}$

C. $3 \times 10^{-17} \text{ kg m/s}$

D. $1 \times 10^{-17} \text{ kg m/s}$

Answer: B

Solution:

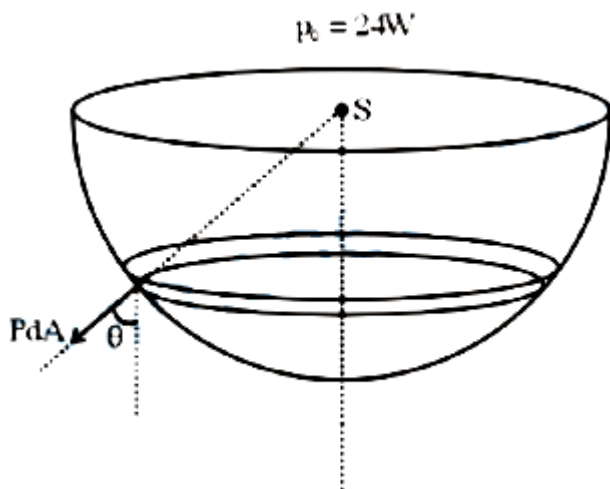
$$\begin{aligned} \text{Momentum} &= \frac{\text{Energy}}{C} \\ &= \frac{\text{Power} \times \text{time}}{C} \\ &= \frac{(20 \times 10^{-3} \text{ W})(300 \times 10^{-9} \text{ s})}{3 \times 10^8 \text{ m/s}} \\ &= 2 \times 10^{-17} \text{ kg-m/s} \end{aligned}$$

Question53

A point source of light is placed at the centre of curvature of a hemispherical surface. The source emits a power of 24W The radius of curvature of hemisphere is 10 cm and the inner surface is completely reflecting. The force on the hemisphere due to the light falling on it is _____ $\times 10^{-8}$ N
[30-Jan-2023 Shift 1]

Answer: 4

Solution:



$$\begin{aligned}\text{Force} &= \int PdA \cos \theta \\ &= \frac{2I}{C} \int dA \cos \theta = \frac{2I}{C} \pi R^2 = 2 \frac{P_0}{4\pi R^2} \cdot \frac{\pi R^2}{C} \\ &= \frac{P_0}{2C} = \frac{24}{2 \times 3 \times 10^8} = 4 \times 10^{-8} \text{ N (Ans)}\end{aligned}$$

Question54

An electron accelerated through a potential difference V_1 has a de-Broglie wavelength of λ . When the potential is changed to V_2 , its de-

Broglie wavelength increases by 50%. The value of $\left(\frac{V_1}{V_2} \right)$ is equal to :
[30-Jan-2023 Shift 2]

Options:

A. 3

B. $\frac{9}{4}$

C. $\frac{3}{2}$

D. 4

Answer: B

Solution:

$$\begin{aligned} KE &= \frac{P^2}{2m}, \quad P = \frac{h}{\lambda} \\ eV_1 &= \frac{\left(\frac{h}{\lambda} \right)^2}{2m} \\ eV_2 &= \frac{\left(\frac{h}{1.5\lambda} \right)^2}{2m} \\ \frac{V_1}{V_2} &= (1.5)^2 = \frac{9}{4} \end{aligned}$$

Question55

If a source of electromagnetic radiation having power 15 kW produces 10^{16} photons per second, the radiation belongs to a part of spectrum is.

(Take Planck constant $h = 6 \times 10^{-34}$ Js)

[31-Jan-2023 Shift 1]

Options:

- A. Micro waves
- B. Ultraviolet rays
- C. Gamma rays
- D. Radio waves

Answer: C

Solution:

$$\text{Energy of one photon} = \frac{\text{Power}}{\text{Photon frequency}}$$

$$E = h\nu = \frac{15 \times 10^3}{10^{16}}$$

$$\nu = \frac{15 \times 10^{-13}}{6 \times 10^{-34}} = 2.5 \times 10^{21}$$

So gamma Rays. Option 3

Question 56

If the two metals A and B are exposed to radiation of wavelength 350 nm. The work functions of metals A and B are 4.8 eV and 2.2 eV. Then choose the correct option [31-Jan-2023 Shift 2]

Options:

- A. Metal B will not emit photo-electrons
- B. Both metals A and B will emit photo-electrons
- C. Both metals A and B will not emit photoelectrons
- D. Metal A will not emit photo-electrons

Answer: D

Solution:

$$\phi = \frac{hc}{\lambda} = \frac{1240}{350} \text{ eV} = 3.54 \text{ eV}$$

∴ Only metal B will emit photoelectron.

Question57

A proton moving with one tenth of velocity of light has a certain de Broglie wavelength of λ . An alpha particle having certain kinetic energy has the same de-Broglie wavelength λ . The ratio of kinetic energy of proton and that of alpha particle is:

[1-Feb-2023 Shift 1]

Options:

A. 2 : 1

B. 4 : 1

C. 1 : 2

D. 1 : 4

Answer: B

Solution:

$$K E = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$$

$$\frac{K E_p}{K E_\alpha} = \frac{m_\alpha}{m_p} = 4 : 1$$

Question58

The threshold frequency of metal is f_0 . When the light of frequency $2f_0$ is incident on the metal plate, the maximum velocity of photoelectron is v_1 . When the frequency of incident radiation is increased to $5f_0$, the maximum velocity of photoelectrons emitted is

v_2 . The ratio of v_1 to v_2 is:

[1-Feb-2023 Shift 2]

Options:

A. $\frac{v_1}{v_2} = \frac{1}{2}$

B. $\frac{v_1}{v_2} = \frac{1}{8}$

C. $\frac{v_1}{v_2} = \frac{1}{16}$

D. $\frac{v_1}{v_2} = \frac{1}{4}$

Answer: A

Solution:

$$K_{\max} = hf - hf_0$$

$$\text{For } f = 2f_0$$

$$\frac{1}{2}mv_1^2 = 2hf_0 - hf_0 = hf_0$$

$$\text{For } f = 5f_0$$

$$\frac{1}{2}mv_2^2 = 5hf_0 - hf_0 = 4hf_0$$

$$\frac{v_1}{v_2} = \frac{1}{2}$$

Question 59

The kinetic energy of an electron, α -particle and a proton are given as $4K$, $2K$ and K respectively. The de-Broglie wavelength associated with electron (λ_e), α -particle (λ_α) and the proton (λ_p) are as follows :

[6-Apr-2023 shift 1]

Options:

A. $\lambda_\alpha > \lambda_p > \lambda_e$

B. $\lambda_\alpha = \lambda_p > \lambda_e$

C. $\lambda_\alpha = \lambda_p < \lambda_e$

D. $\lambda_\alpha < \lambda_p < \lambda_e$

Answer: D

Solution:

According to De-Broglie, Momentum $P = \frac{h}{\lambda}$, where h is plank's constant and λ is wavelength.

Also, relation between Kinetic energy (KE) and momentum(P) is given by: $P = \sqrt{2mKE}$

Now, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mKE}}$

$$\lambda_e = \frac{h}{\sqrt{2m_e KE_e}} = \frac{h}{\sqrt{2m_e \times 4k}} = \frac{h}{\sqrt{8m_e k}}$$

$$\lambda_p = \frac{h}{\sqrt{2m_p KE_p}} = \frac{h}{\sqrt{2m_p k}}$$

$$\lambda_\alpha = \frac{h}{\sqrt{2m_\alpha KE_\alpha}} = \frac{h}{\sqrt{2m_\alpha \cdot 4k}} = \frac{h}{\sqrt{2 \times 2m_p \cdot 2k}} = \frac{h}{\sqrt{8m_p k}}$$

From the above data, $\lambda_\alpha < \lambda_p < \lambda_e$

Question60

The work functions of Aluminium and Gold are 4.1 eV and and 5.1 eV respectively. The ratio of the slope of the stopping potential versus frequency plot for Gold to that of Aluminium is [6-Apr-2023 shift 2]

Options:

A. 1

B. 2

C. 1.24

D. 1.5

Answer: A

Solution:

$$\text{Using } KE_{\text{max}} = eV_s = hf - \phi_0$$

where ϕ_0 is work function, V_s is stopping potential and f is frequency

$$\text{or } V_s = \frac{h}{e}f - \frac{\phi_0}{e}$$

therefore the slope m will be same for all graphs and will be independent of ϕ_0 .

Question61

Proton (P) and electron (e) will have same de-Broglie wavelength when the ratio of their momentum is (assume, $m_p = 1849m_e$):

[8-Apr-2023 shift 1]

Options:

A. 1 : 43

B. 43 : 1

C. 1 : 1849

D. 1 : 1

Answer: D

Solution:

$$\text{Debroglie wavelength } \lambda = \frac{h}{p}$$

$$\lambda_p = \lambda_e$$

$$\frac{h}{p_p} = \frac{h}{p_e} \Rightarrow \frac{p_p}{p_e} = 1$$

Question62

In photoelectric effect

A. The photocurrent is proportional to the intensity of the incident radiation.

B. Maximum Kinetic energy with which photoelectrons are emitted depends on the intensity of incident light.

C. Max K.E with which photoelectrons are emitted depends on the frequency of incident light.

D. The emission of photoelectrons require a minimum threshold intensity of incident radiation.

E. Max. K.E of the photoelectrons is independent of the frequency of the incident light.

Choose the correct answer from the options given below:

[8-Apr-2023 shift 2]

Options:

A. B and C only

B. A and C only

C. A and E only

D. A and B only

Answer: B

Solution:

$$h\nu = \phi + (KE)_{\max}$$

$$(KE)_{\max} = h\nu - \phi$$

Question63

The de Broglie wavelength of a molecule in a gas at room temperature (300K) is λ_1 . If the temperature of the gas is increased to 600K, then the de Broglie wavelength of the same gas molecule becomes

[10-Apr-2023 shift 1]

Options:

A. $2\lambda_1$

B. $\frac{1}{\sqrt{2}}\lambda_1$

C. $\sqrt{2}\lambda_1$

D. $\frac{1}{2}\lambda_1$

Answer: B

Solution:

$$\lambda = \frac{h}{\sqrt{3mK(T)}}$$

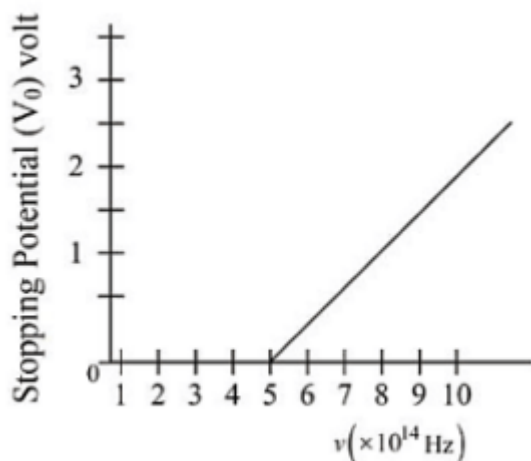
$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}}$$

$$\lambda_2 = \lambda_1 \sqrt{\frac{T_1}{T_2}}$$

$$= \lambda_1 \sqrt{\frac{300}{600}} = \frac{\lambda_1}{\sqrt{2}}$$

Question64

The variation of stopping potential (V_0) as a function of the frequency (ν) of the incident light for a metal is shown in figure. The work function of the surface is



[10-Apr-2023 shift 2]

Options:

A. 2.07 eV

B. 18.6 eV

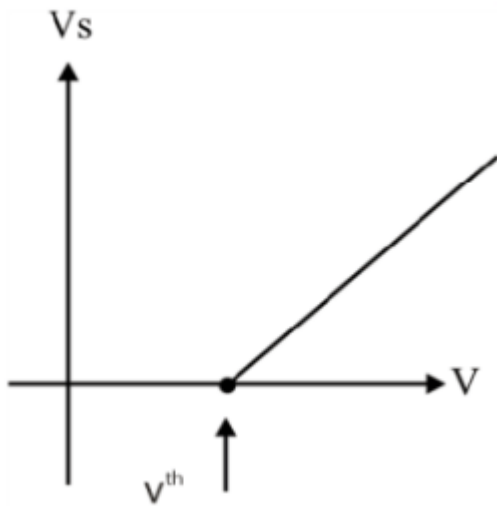
C. 2.98 eV

D. 1.36 eV

Answer: A

Solution:

$$\begin{aligned}\text{Work function } (\phi) &= h\nu^{\text{th}} \\ &= 6.6 \times 10^{-34} \times 5 \times 10^{14} \\ &= 33 \times 10^{-20} \\ \phi &= 3.3 \times 10^{-19} \text{ J} \\ &= \frac{3.3 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} \Rightarrow 2.07\end{aligned}$$



Question65

A metallic surface is illuminated with radiation of wavelength λ , the stopping potential is V_0 . If the same surface is illuminated with radiation of wavelength 2λ , the stopping potential becomes $\frac{V_0}{4}$. The threshold wavelength for this metallic surface will be [11-Apr-2023 shift 1]

Options:

A. $\frac{3}{2}\lambda$

B. 4λ

C. 3λ

D. $\frac{\lambda}{4}$

Answer: C

Solution:

$$E = K \cdot E + \phi_0$$

Now

$$\frac{hc}{\lambda} = eV_0 + \phi_0 \dots (1)$$

$$\text{And } \frac{hc}{2\lambda} = \frac{eV_0}{4} + \phi_0 \dots (2)$$

$$(2) \times 4 \dots (1)$$

$$\frac{2hc}{\lambda} - \frac{hc}{\lambda} = 0 + (4\phi_0 - \phi_0)$$

$$\frac{hc}{\lambda} = 3\phi_0$$

$$\frac{hc}{\lambda} = 3 \frac{hc}{\lambda_0}$$

$$\lambda_0 = 3\lambda$$

Question66

The ratio of the de-Broglie wavelengths of proton and electron having same Kinetic energy: (Assume $m_p = m_e \times 1849$)

[11-Apr-2023 shift 2]

Options:

A. 1 : 62

B. 1 : 30

C. 1 : 43

D. 2 : 43

Answer: C

Solution:

$$\lambda = \frac{h}{P} = \frac{h}{\sqrt{2mK}}$$

$$\frac{\lambda_p}{\lambda_e} = \sqrt{\frac{m_e}{m_p}} = \sqrt{\frac{m_e}{1840 m_e}} = \frac{1}{\sqrt{1840}}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{1}{43} \quad \text{Ans. (3)}$$

Ans. (3)

Question67

A proton and an α -particle are accelerated from rest by 2V and 4V potentials, respectively. The ratio of their de-Broglie wavelength is :
[12-Apr-2023 shift 1]

Options:

A. 2 : 1

B. 4 : 1

C. 8 : 1

D. 16 : 1

Answer: B

Solution:

$$\text{De Broglie wavelength } \lambda = \frac{h}{\sqrt{2mqV}}$$

$$m_\alpha = 4m \rightarrow 4V$$

$$m_p = m \rightarrow 2V$$

$$\lambda = \frac{h}{\sqrt{2mKE}}$$

$$\lambda_p = \frac{h}{\sqrt{2mq(2V)}} \dots (1), \quad \lambda_\alpha = \frac{h}{\sqrt{4mq(4V)}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{h}{\sqrt{2mq(2V)}} \times \frac{\sqrt{4mq(4V)}}{h}$$

$$\frac{\lambda_p}{\lambda_\alpha} = 4 \Rightarrow \frac{\lambda_p}{\lambda_\alpha} = 4 : 1$$

Question68

The difference between threshold wavelengths for two metal surfaces A and B having work function $\phi_A = 9 \text{ eV}$ and $\phi_B = 4.5 \text{ eV}$ in nm is:

{ Given, $hc = 1242 \text{ eV nm}$ }

[13-Apr-2023 shift 1]

Options:

A. 276

B. 264

C. 540

D. 138

Answer: D

Solution:

$$\phi = \frac{hc}{\lambda}$$

$$\lambda_A = \frac{1242}{9} = 138 \text{ nm}$$

$$\lambda_B = \frac{1242}{4.5} = 276 \text{ nm}$$

$$\lambda_B - \lambda_A = 276 - 138 = 138 \text{ nm}$$

Question69

The de Broglie wavelength of an electron having kinetic energy E is λ . If the kinetic energy of electron becomes $\frac{E}{4}$, then its de-Broglie wavelength will be :

[15-Apr-2023 shift 1]

Options:

A. $\frac{\lambda}{\sqrt{2}}$

B. 2λ

C. $\frac{\lambda}{2}$

D. $\sqrt{2}\lambda$

Answer: B

Solution:

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{E_2}{E_1}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

$$\lambda_2 = 2\lambda$$

Question70

When light of frequency twice the threshold frequency is incident on the metal plate, the maximum velocity of emitted electron is v_1 .

When the frequency of incident radiation is increased to five times the threshold value, the maximum velocity of emitted electron becomes v_2 . If $v_2 = xv_1$, the value of x will be

[24-Jun-2022-Shift-1]

Answer: 2

Solution:

Let us say that work function is ϕ

$$\Rightarrow 2\phi = \phi + \frac{1}{2}mv_1^2 \dots\dots (1)$$

$$\text{and } 5\phi = \phi + \frac{1}{2}mv_2^2 \dots\dots (2)$$

From (1) and (2)

$$\frac{v_2^2}{v_1^2} = \frac{4}{1} \text{ or } \frac{v_2}{v_1} = 2$$

Question71

The de Broglie wavelengths for an electron and a photon are λ_e and λ_p respectively. For the same kinetic energy of electron and photon, which of the following presents the correct relation between the de Broglie wavelengths of two ?
[28-Jun-2022-Shift-1]

Options:

A. $\lambda_p \propto \lambda_e^2$

B. $\lambda_p \propto \lambda_e$

C. $\lambda_p \propto \sqrt{\lambda_e}$

D. $\lambda_p \propto \sqrt{\frac{1}{\lambda_e}}$

Answer: A

Solution:

$$\lambda_p = \frac{h}{p} = \frac{hc}{E} \dots\dots (i)$$

$$\lambda_e = \frac{h}{\sqrt{2mE}} \dots\dots (ii)$$

From (i) and (ii)

$$\lambda_p \propto \lambda_e^2$$

Question72

The light of two different frequencies whose photons have energies 3.8 eV and 1.4 eV respectively, illuminate a metallic surface whose

work function is 0.6 eV successively. The ratio of maximum speeds of emitted electrons for the two frequencies respectively will be :
[24-Jun-2022-Shift-2]

Options:

A. 1 : 1

B. 2 : 1

C. 4 : 1

D. 1 : 4

Answer: B

Solution:

$$3.8 = 0.6 + \frac{1}{2}mv_1^2$$

$$1.4 = 0.6 + \frac{1}{2}mv_2^2$$

$$\Rightarrow \frac{v_1^2}{v_2^2} = \frac{3.2}{0.8} = \frac{4}{1}$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{2}{1}$$

Question73

Let K_1 and K_2 be the maximum kinetic energies of photo-electrons emitted when two monochromatic beams of wavelength λ_1 and λ_2 , respectively are incident on a metallic surface. If $\lambda_1 = 3\lambda_2$ then :
[28-Jun-2022-Shift-2]

Options:

A. $K_1 > \frac{K_2}{3}$

B. $K_1 < \frac{K_2}{3}$

C. $K_1 = \frac{K_2}{3}$

D. $K_2 = \frac{K_1}{3}$

Answer: B

Solution:

$$K_1 = \frac{hc}{\lambda_1} - \phi = \frac{hc}{3\lambda_2} - \phi$$

$$\text{and } K_2 = \frac{hc}{\lambda_2} - \phi$$

from (i) and (ii) we can say

$$3K_1 = K_2 - 2\phi$$

$$K_1 < \frac{K_2}{3}$$

Question 74

Given below are two statements :

Statement I : Davisson-Germer experiment establishes the wave nature of electrons.

Statement II : If electrons have wave nature, they can interfere and show diffraction.

In the light of the above statements choose the correct answer from the option given below :

[25-Jun-2022-Shift-1]

Options:

A. Both Statement I and Statement II are true.

B. Both Statement I and Statement II are false.

C. Statement I is true but Statement II is false.

D. Statement I is false but Statement II is true.

Answer: B

Solution:

Davisson-Germer experiment is done and establishes the wave nature of electrons. Interference and diffraction establishes wave nature.

Question 75

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The photoelectric effect does not take place, if the energy of the incident radiation is less than the work function of a metal.

Reason R : Kinetic energy of the photoelectrons is zero, if the energy of the incident radiation is equal to the work function of a metal.

In the light of the above statements, choose the most appropriate answer from the options given below.

[29-Jun-2022-Shift-1]

Options:

- A. Both A and R are correct and R is the correct explanation of A.
- B. Both A and R are correct but R is not the correct explanation of A.
- C. A is correct but R is not correct.
- D. A is not correct but R is correct.

Answer: B

Solution:

To free the electron from metal surface minimum energy required, is equal to the work function of that metal.

So Assertion A, is correct.

$$h\nu = w_0 + K_{\max} \cdot E$$

$$\text{if } h\nu = w_0$$

$$\Rightarrow K_{\max} = 0$$

Hence reason R, is correct, But R is not the correct explanation of A.

When energy of incident radiation is equal to the work function of the metal, then the KE of photoelectrons would be zero. But this statement does not comment on the situation when energy is less than the work function.

Question76

A proton, a neutron, an electron and an α particle have same energy. If λ_p , λ_n , λ_e and λ_a are the de Broglie's wavelengths of proton, neutron, electron and α particle respectively, then choose the correct relation from the following:

[25-Jun-2022-Shift-2]

Options:

A. $\lambda_p = \lambda_n > \lambda_e > \lambda_a$

B. $\lambda_a < \lambda_n < \lambda_p < \lambda_e$

C. $\lambda_e < \lambda_p = \lambda_n > \lambda_a$

D. $\lambda_e = \lambda_p = \lambda_n = \lambda_a$

Answer: D

Solution:

$$\text{de Broglie wavelength } \lambda = \frac{h}{p} \Rightarrow \lambda = \frac{h}{\sqrt{2mK}}$$

Where K : kinetic energy

$$\Rightarrow \text{For same K, } \lambda \propto \frac{1}{\sqrt{m}}$$

$$\text{Since } m_\alpha > m_n > m_p > m_e$$

$$\Rightarrow \lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$$

Question77

The electric field at a point associated with a light wave is given by $E = 200[\sin(6 \times 10^{15})t + \sin(9 \times 10^{15})t]V^{-1}$

Given : $h = 4.14 \times 10^{-15} \text{ eVs}$

If this light falls on a metal surface having a work function of 2.50 eV, the maximum kinetic energy of the photoelectrons will be [29-Jun-2022-Shift-2]

Options:

A. 1.90 eV

B. 3.27 eV

C. 3.60 eV

D. 3.42 eV

Answer: D

Solution:

Frequency of EM waves = $\frac{6}{2\pi} \times 10^{15}$ and $\frac{9}{2\pi} \times 10^{15}$

Energy of one photon of these waves

$$= \left(4.14 \times 10^{-15} \times \frac{6}{2\pi} \times 10^{15} \right) \text{ eV}$$

$$\text{and } \left(4.14 \times 10^{-15} \times \frac{9}{2\pi} \times 10^{15} \right) \text{ eV}$$

$$= 3.95 \text{ eV and } 5.93 \text{ eV}$$

$\Rightarrow$ Energy of maximum energetic electrons

$$= 5.93 - 2.50 = 3.43 \text{ eV}$$

Question78

An electron with speed v and a photon with speed c have the same de-Broglie wavelength. If the kinetic energy and momentum of electron are E_e and p_e and that of photon are E_{ph} and p_{ph} respectively. Which of the following is correct?
[26-Jun-2022-Shift-1]

Options:

A. $\frac{E_e}{E_{ph}} = \frac{2c}{v}$

B. $\frac{E_e}{E_{ph}} = \frac{v}{2c}$

C. $\frac{p_e}{p_{ph}} = \frac{2c}{v}$

$$D. \frac{p_e}{p_{ph}} = \frac{v}{2c}$$

Answer: B

Solution:

$$\lambda = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda}$$

$$\text{Now, A/Q, } \frac{h}{p_e} = \frac{h}{p_{\text{photon}}} \Rightarrow p_e = p_{\text{photon}} \dots\dots\dots (i)$$

$$\text{Now, } K_e = \frac{1}{2} M v^2 = \frac{p v}{2}$$

$$K_{ph} = m c^2 = p c \dots\dots (ii)$$

$$\frac{K_e}{K_{eq}} = \frac{v}{2c}$$

Question 79

The stopping potential for photoelectrons emitted from a surface illuminated by light of wavelength $6630 \text{ \AA}$ is 0.42 V . If the threshold frequency is $x \times 10^{13} / \text{s}$, where x is ____ (nearest integer).

(Given, speed light $= 3 \times 10^8 \text{ m/s}$, Planck's constant $= 6.63 \times 10^{-34} \text{ Js}$)

[26-Jun-2022-Shift-2]

Answer: 35

Solution:

$$\begin{aligned} \frac{hc}{\lambda} - \phi &= K E = e V_0 \\ \Rightarrow \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{6630 \times 10^{-10}} - 6.63 \times 10^{-34} f_{th} &= 1.6 \times 10^{-19} \times 0.4 \\ \Rightarrow f_{th} &\simeq 35.11 \times 10^{13} \text{ Hz} \end{aligned}$$

Question80

An α particle and a carbon 12 atom has same kinetic energy K . The ratio of their de-Broglie wavelengths ($\lambda_\alpha : \lambda_{C12}$) is :

[27-Jun-2022-Shift-1]

Options:

A. $1 : \sqrt{3}$

B. $\sqrt{3} : 1$

C. $3 : 1$

D. $2 : \sqrt{3}$

Answer: B

Solution:

$$K_\alpha = K_C$$

$$\frac{p_\alpha^2}{2m_\alpha} = \frac{p_C^2}{2m_C}$$

$$\frac{p_\alpha}{p_C} = \sqrt{\frac{m_C}{m_\alpha}}$$

$$\text{So } \frac{\lambda_\alpha}{\lambda_C} = \frac{h/p_\alpha}{h/p_C} = \sqrt{\frac{m_C}{m_\alpha}}$$

$$\text{So } \frac{\lambda_\alpha}{\lambda_C} = \sqrt{3}$$

Question81

Given below are two statements :

Statement I : In hydrogen atom, the frequency of radiation emitted when an electron jumps from lower energy orbit (E_1) to higher energy orbit (E_2), is given as $hf = E_2 - E_1$

Statement II : The jumping of electron from higher energy orbit (E_2) to lower energy orbit (E_1) is associated with frequency of radiation

given as $f = (E_2 - E_1)/h$

This condition is Bohr's frequency condition.

In the light of the above statements, choose the correct answer from the options given below :

[27-Jun-2022-Shift-2]

Options:

- A. Both Statement I and Statement II are true.
- B. Both Statement I and Statement II are false.
- C. Statement I is correct but Statement II is false.
- D. Statement I is incorrect but Statement II is true.

Answer: D

Solution:

Radiation is not emitted but absorbed when an electron jumps from low energy to high energy.

Also, $E_2 - E_1$ is the energy of photon

$$\Rightarrow E_2 - E_1 = hf$$

$$\Rightarrow f = \frac{E_2 - E_1}{h}$$

Question82

A metal exposed to light of wavelength 800nm and emits photoelectrons with a certain kinetic energy. The maximum kinetic energy of photo-electron doubles when light of wavelength 500nm is used. The work function of the metal is : (Take $hc = 1230\text{eV} - \text{nm}$).
[25-Jul-2022-Shift-1]

Options:

- A. 1.537eV
- B. 2.46eV
- C. 0.615eV

D. 1.23eV

Answer: C

Solution:

$$\therefore K_m = \frac{hc}{\lambda} - \phi$$

$$\Rightarrow K = \frac{1230}{800} - \phi$$

$$\text{and, } 2K = \frac{1230}{500} - \phi$$

$$\Rightarrow 2 \times \frac{1230}{800} - 2\phi = \frac{1230}{500} - \phi$$

$$\Rightarrow \phi = 0.615 \text{ eV}$$

Question83

The energy band gap of semiconducting material to produce violet (wavelength = $4000\text{\AA}$) LED is eV . (Round off to the nearest integer).

[25-Jul-2022-Shift-1]

Answer: 3

Solution:

Energy corresponding to wavelength $4000\text{\AA}$

$$E = \frac{hc}{\lambda}$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{4000 \times 10^{-10} \times 1.6 \times 10^{-19}} \text{ eV}$$

$$= \frac{12400}{4000}$$

$$= 3.1 \text{ eV}$$

$$\approx 3 \text{ eV}$$

Question84

The ratio of wavelengths of proton and deuteron accelerated by potential V_p and V_d is $1 : \sqrt{2}$. Then the ratio of V_p to V_d will be :
[25-Jul-2022-Shift-2]

Options:

A. $1 : 1$

B. $\sqrt{2} : 1$

C. $2 : 1$

D. $4 : 1$

Answer: D

Solution:

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2meV}}$$
$$\text{so } \frac{\lambda_p}{\lambda_d} = \frac{\sqrt{m_d V_d}}{\sqrt{m_p V_p}} = \frac{1}{\sqrt{2}}$$
$$\frac{2V_d}{V_p} = \frac{1}{2}$$
$$\frac{V_p}{V_d} = \frac{4}{1}$$

Question85

A parallel beam of light of wavelength 900 nm and intensity 100 W m^{-2} is incident on a surface perpendicular to the beam. The number of photons crossing 1 cm^2 area perpendicular to the beam in one second is :
[26-Jul-2022-Shift-1]

Options:

A. 3×10^{16}

B. 4.5×10^{16}

C. 4.5×10^{17}

D. 4.5×10^{20}

Answer: B

Solution:

$$\lambda = 900 \text{ nm}$$

$$I = 100 \text{ W/m}^2$$

$$A = 10^{-4}$$

$$\Rightarrow P = 10^{-2} \text{ W}$$

$\Rightarrow$ Number of photons incident per second

$$= \frac{10^{-2} \lambda}{hc}$$

$$= \frac{9 \times 10^{-11} \times 10^2}{6.63 \times 10^{-34} \times 3 \times 10^8} \simeq 4.5 \times 10^{16}$$

Question 86

With reference to the observations in photo-electric effect, identify the correct statements from below

(A) The square of maximum velocity of photoelectrons varies linearly with frequency of incident light.

(B) The value of saturation current increases on moving the source of light away from the metal surface.

(C) The maximum kinetic energy of photo-electrons decreases on decreasing the power of LED (light emitting diode) source of light.

(D) The immediate emission of photo-electrons out of metal surface can not be explained by particle nature of light/electromagnetic waves.

(E) Existence of threshold wavelength can not be explained by wave nature of light / electromagnetic waves.

Choose the correct answer from the options given below :

[27-Jul-2022-Shift-2]

Options:

A. (A) and (B) only

B. (A) and (E) only

C. (C) and (E) only

D. (D) and (E) only

Answer: B

Solution:

$$\because \frac{1}{2}mv_m^2 = hv - \phi$$

$\Rightarrow v_m^2$ varies linearly with frequency.

And, threshold wavelength can be explained by particle nature of light.

Question87

The equation $\lambda = \frac{1.227}{x} \text{ nm}$ can be used to find the de-Broglie wavelength of an electron. In this equation x stands for:

Where

m = mass of electron

P – momentum of electron

K = Kinetic energy of electron

V = Accelerating potential in volts for electron

[28-Jul-2022-Shift-1]

Options:

A. $\sqrt{mK}$

B. $\sqrt{P}$

C. $\sqrt{K}$

D. $\sqrt{V}$

Answer: D

Solution:

$$\therefore \lambda = \frac{1.227}{\sqrt{V}} \text{ nm}$$

$$\Rightarrow x = \sqrt{V}$$

Question88

Two streams of photons, possessing energies equal to five and ten times the work function of metal are incident on the metal surface successively. The ratio of maximum velocities of the photoelectron emitted, in the two cases respectively, will be
[28-Jul-2022-Shift-2]

Options:

- A. 1 : 2
- B. 1 : 3
- C. 2 : 3
- D. 3 : 2

Answer: C

Solution:

$$\frac{1}{2}mv_1^2 = 5\phi - \phi$$

$$\text{And, } \frac{1}{2}mv_2^2 = 10\phi - \phi$$

$$\Rightarrow \left(\frac{v_1}{v_2} \right)^2 = \frac{4}{9}$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{2}{3}$$

Question89

The kinetic energy of emitted electron is E when the light incident on the metal has wavelength λ . To double the kinetic energy, the

incident light must have wavelength:
[29-Jul-2022-Shift-1]

Options:

A. $\frac{hc}{E\lambda - hc}$

B. $\frac{hc\lambda}{E\lambda + hc}$

C. $\frac{h\lambda}{E\lambda + hc}$

D. $\frac{hc\lambda}{E\lambda - hc}$

Answer: B

Solution:

$$E = \frac{hc}{\lambda} - \phi \quad \text{--- (i)}$$

$$2E = \frac{hc}{\lambda'} - \phi \quad \text{--- (ii)}$$

$$(ii) - (i)$$

$$E = hc \left(\frac{1}{\lambda'} - \frac{1}{\lambda} \right)$$

$$\Rightarrow \lambda' = \frac{hc\lambda}{E\lambda + hc}$$

Question90

An α particle and a proton are accelerated from rest through the same potential difference. The ratio of linear momenta acquired by above two particles will be:

[29-Jul-2022-Shift-2]

Options:

A. $\sqrt{2} : 1$

B. $2\sqrt{2} : 1$

C. $4\sqrt{2} : 1$

D. 8 : 1

Answer: B

Solution:

We know,

$$\text{Momentum}(p) = \sqrt{2mE_k}$$

$$\text{and } E_k = qV_{\text{acc}}$$

$$\therefore p = \sqrt{2mqV_{\text{acc}}}$$

Both α particle and proton are passed through same potential difference.

$$\therefore (V_{\text{acc}})_{\alpha} = (V_{\text{acc}})_p = v$$

$$\therefore p_{\alpha} = \sqrt{2m_{\alpha}q_{\alpha}v}$$

$$p_p = \sqrt{2m_pq_pv}$$

$$\therefore \frac{p_{\alpha}}{p_p} = \sqrt{\frac{m_{\alpha}q_{\alpha}}{m_pq_p}}$$

$$= \sqrt{\frac{4m_p \times 2e}{m_p \times e}}$$

$$= \sqrt{\frac{8}{1}}$$

$$= \frac{2\sqrt{2}}{1}$$

Question91

Two stream of photons, possessing energies equal to twice and ten times the work function of metal are incident on the metal surface successively. The value of ratio of maximum velocities of the photoelectrons emitted in the two respective cases is $x : y$. The value of x is

[26 Feb 2021 Shift 2]

Answer: 1

Solution:

Given, first incident energy, $E_{i_1} = 2$ (work function ϕ_0) Second incident energy, $E_{i_2} = 10\phi_0$

If v_1 and v_2 be the respective maximum velocities in two cases, then by using the concept of photoelectric effect,

$$K E = E_i - \phi_0$$

$$K E = E_i - \phi_0$$

$$\therefore \frac{1}{2}mv_1^2 = 2\phi_0 - \phi_0 = \phi_0 \dots (i)$$

$$\text{and } \frac{1}{2}mv_2^2 = 10\phi_0 - \phi_0 = 9\phi_0 \dots (ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\left(\frac{v_1}{v_2} \right)^2 = \frac{1}{9}$$

$$\frac{v_1}{v_2} = \frac{1}{3} = \frac{x}{y}$$

$$\therefore x = 1$$

Question92

The stopping potential for electrons emitted from a photosensitive surface illuminated by light of wavelength 491nm is 0.710V . When the incident wavelength is changed to a new value, the stopping potential is 1.43V . The new wavelength is
[25 Feb 2021 Shift 2]

Options:

A. 309nm

B. 329nm

C. 382nm

D. 400nm

Answer: C

Solution:

Given, stopping potential (V_1) = 0.710V

Incident wavelength of electrons (λ_1) = 491nm

$$= 491 \times 10^{-9} \text{ m}$$

Let λ_2 will be the new incident wavelength of electrons.

Stopping potential (V_2) = 1.43V

$$\text{As, energy (E)} = \frac{1240}{\lambda_1} = \phi_0 + eV$$

where, ϕ_0 is work-function and V is applied potential

$$\therefore E_1 = \frac{1240}{491} = \phi_0 + 0.71 - 1.43 \dots (i)$$

$$\text{and } E_2 = \frac{1240}{\lambda_2} = \phi_0 + 1.43 \dots (ii)$$

Now, subtracting Eqs. (i) from (ii), we get

$$E_2 - E_1 = 1240 \left(\frac{1}{\lambda_2} - \frac{1}{491} \right) = 0.72$$

$$\Rightarrow \frac{1}{\lambda_2} = 0.00058 + 0.00204 = 0.00262$$

$$\therefore \lambda_2 = 381.7 \text{ nm}$$

$$\lambda_2 = 382 \text{ nm}$$

Question93

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A An electron microscope can achieve better resolving power than an optical microscope.

Reason R The de-Broglie's wavelength of the electrons emitted from an electron gun is much less than wavelength of visible light.

In the light of the above statements, choose the correct answer from the options given below.

[26 Feb 2021 Shift 1]

Options:

A. A is true but R is false.

B. Both A and R are true and R is the correct explanation of A.

C. Both A and R are true but R is not the correct explanation of A.

D. A is false but R is true.

Answer: B

Solution:

As we know that,

$$\text{Resolving power of microscope} = \frac{1}{\Delta\theta} = \frac{2u \sin \theta}{\lambda}$$

$$\text{i.e. resolving power} \propto \frac{1}{\lambda}$$

and since, wavelength of electron emitted (λ_e) < wavelength of visible light (λ_v)

$\therefore$ Resolving power of electron microscope > Resolving power of optical microscope.

Hence, option (b) is the correct.

Question94

The wavelength of an X-ray beam is $10\text{\AA}$. The mass of a fictitious particle having the same energy as that of the X-ray photons is $\frac{x}{3}h\text{kg}$. The value of x is

(h = Planck's constant)

[25 Feb 2021 Shift 2]

Answer: 10

Solution:

Given, wavelength of X-rays,

$$\lambda = 10\text{\AA} = 10 \times 10^{-10}\text{m}$$

Speed of light in free space, $c = 3 \times 10^8\text{m/s}$

$$\text{Since, energy (E)} = \frac{hc}{\lambda} = mc^2 \dots (i)$$

$$\Rightarrow m = \frac{h}{c\lambda} = \frac{h}{3 \times 10^8 \times 10 \times 10^{-10}} = \frac{h}{3 \times 10^{-1}} = \frac{10h}{3}\text{kg}$$

$$\therefore x = 10$$

Question95

An electron of mass m_e and a proton of mass $m_p = 1836m_e$ are moving with the same speed. The ratio of their de-Broglie

wavelength $\frac{\lambda_{\text{electron}}}{\lambda_{\text{proton}}}$ will be
[25 Feb 2021 Shift 2]

Options:

A. 1

B. 1836

C. $\frac{1}{1836}$

D. 918

Answer: B

Solution:

Given, mass of proton (m_p) is 1836 times the mass of electron (m_e) and velocity of proton (v_p) is equal to velocity of electron (v_e).

$$\text{As, wavelength } (\lambda) = \frac{h}{p} = \frac{h}{mv}$$

where, h is Planck's constant and p is momentum. $\therefore \lambda \propto \frac{1}{m}$

$$\Rightarrow \frac{\lambda_e}{\lambda_p} = \frac{m_p}{m_e} = \frac{1836m_e}{m_e} = 1836$$

Question96

A proton, a deuteron and an α -particle are moving with same momentum in a uniform magnetic field. The ratio of magnetic forces acting on them is and their speed is in the ratio.

[25 Feb 2021 Shift 1]

Options:

A. 1 : 2 : 4 and 2 : 1 : 1

B. 2 : 1 : 1 and 4 : 2 : 1

C. 4 : 2 : 1 and 2 : 1 : 1

D. 1 : 2 : 4 and 1 : 1 : 2

Answer: B

Solution:

Let F_p , F_d and F_α be the forces and v_p , v_d and v_α be the velocities of proton, deuteron and α -particle, respectively.

Since, $F = Bqv$

On dividing and multiplying F by m , we get

$$F = Bqv \frac{m}{m}$$

$$\Rightarrow F = B \frac{q}{m} p \quad (\because p = mv)$$

$$\Rightarrow F \propto q/m \quad (\because p \text{ and } B \text{ are same})$$

$$\therefore F_p : F_d : F_\alpha = \frac{+q}{m} : \frac{+q}{2m} : \frac{+2q}{4m}$$
$$= 1 : 1/2 : 1/2 = 2 : 1 : 1$$

$$\text{and } F = Bqv$$
$$\Rightarrow F \propto v$$

$$\therefore v_p : v_d : v_\alpha = \frac{F_p}{q_p} : \frac{F_d}{q_d} : \frac{F_\alpha}{q_\alpha} = \frac{2x}{q} : \frac{x}{q} : \frac{x}{2q}$$
$$= 2 : 1 : 1/2 = 4 : 2 : 1$$

Question97

The de-Broglie wavelength of a proton and α -particle are equal. The ratio of their velocities is
[24 Feb 2021 Shift 2]

Options:

A. 4 : 3

B. 4 : 1

C. 4 : 2

D. 1: 4

Answer: B

Solution:

Let λ_p , λ_α , m_p , m_α , v_p , v_α , p_p and p_α be the wavelength, mass, velocity and momentum of proton and α -particle, respectively.

Given, $\lambda_p = \lambda_\alpha$

As we know that,

$$\lambda = h/p$$

$$\frac{h}{p_p} = \frac{h}{p_\alpha}$$

$$\therefore p_p = p_\alpha$$

$$\Rightarrow m_p v_p = m_\alpha v_\alpha$$

$$\Rightarrow m_p v_p = 4m_p v_\alpha$$

$$\Rightarrow \frac{v_p}{v_\alpha} = \frac{4}{1} \text{ or } 4 : 1$$

Question98

An X-ray tube is operated at 1.24 million volt. The shortest wavelength of the produced photon will be
[24 Feb 2021 Shift 2]

Options:

A. 10^{-3} nm

B. 10^{-1} nm

C. 10^{-2} nm

D. 10^{-4} nm

Answer: A

Solution:

Given, $V = 1.24 \text{ million volt} = 1.24 \times 10^6 \text{ volt}$

Since, energy (E) = eV

where, e is the charge of electron = $1.6 \times 10^{-19} \text{ C}$

$$\therefore E = 1.6 \times 10^{-19} \times 1.24 \times 10^6 \dots \text{(i)}$$

As we know that,

$$\text{Energy of photon, } E = \frac{hc}{\lambda} \dots \text{(ii)}$$

Here, Planck's constant, $h = 6.67 \times 10^{-34} \text{ J-s}$,

C = speed of light in free space, $C = 3 \times 10^8 \text{ ms}^{-1}$

Equating Eqs. (i) and (ii), we get

$$1.6 \times 10^{-19} \times 1.24 \times 10^6 = \frac{6.67 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

$$\Rightarrow \lambda = \frac{20.01 \times 10^{-13}}{1.6 \times 1.24} = 10.09 \times 10^{-13}$$

$$= 1.009 \times 10^{-12} \sim \text{eq} 10^{-3} \times 10^{-9}$$

$$= 10^{-3} \text{ nm}$$

Question99

Given below are two statements :

Statement-I : Two photons having equal linear momenta have equal wavelengths.

Statement-II: If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease.

In the light of the above statements, choose the correct answer from the options given below.

[24feb2021shift1]

Options:

- A. Both Statement I and Statement II are true
- B. Statement I is false but Statement II is true
- C. Both Statement I and Statement II are false
- D. Statement I is true but Statement II is false

Answer: D

Solution:

Momentum of photon,

$$p = \frac{h}{\lambda}$$

Here, h = Plank's constant,

λ = wavelength of light

$$\text{Energy of photon } E = \frac{hc}{\lambda}$$

If linear momentum are equal then wavelength also equal.

On decreasing wavelength, momentum and energy of photon increases.

Question100

Two identical photocathodes receive the light of frequencies f_1 and f_2 , respectively. If the velocities of the photoelectrons coming out are v_1 and v_2 respectively, then
[17 Mar 2021 Shift 2]

Options:

A. $v_1^2 - v_2^2 = \frac{2h}{m}[f_1 - f_2]$

B. $v_1^2 + v_2^2 = \frac{2h}{m}[f_1 + f_2]$

C. $v_1 - v_2 = \left[\frac{2h}{m}(f_1 + f_2) \right]^{\frac{1}{2}}$

D. $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2) \right]^{\frac{1}{2}}$

Answer: A

Solution:

We know that, the photoelectric effect equation,

$$KE = hf - \phi$$

Here, KE = kinetic energy of the electrons

h = Planck's constant

f = frequency of the light and

ϕ = work-function of the photocathodes.

For two identical photocathodes,

$$\frac{1}{2}mv_1^2 = hf_1 - \phi \dots (i)$$

$$\frac{1}{2}mv_2^2 = hf_2 - \phi \dots (ii)$$

Since, the material of photocathodes is same, so the value of the work-function ϕ is same.

Subtracting Eq. (ii) from Eq. (i) we get

$$\frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 = hf_1 - hf_2$$

$$v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$$

Question101

The stopping potential in the context of photoelectric effect depends on the following property of incident electromagnetic radiation
[16 Mar 2021 Shift 1]

Options:

- A. phase
- B. intensity
- C. amplitude
- D. frequency

Answer: D

Solution:

The stopping potential V_0 is related to the frequency of incident radiations by following relation

$$eV_0 = h\nu - \phi \quad (\phi = \text{work-function})$$

The stopping potential in the context of photoelectric effect depends only on the frequency ν of incident electromagnetic radiation.

Question102

If $2.5 \times 10^{-6} \text{ N}$ average force is exerted by a light wave on a non-reflecting surface of 30 cm^2 area during 40 min of time span, the energy flux of light just before it falls on the surface is $\text{W} \sqrt{\text{cm}^2}$. (Round off to the nearest integer. Assume complete absorption and normal incidence conditions are there.)
[17 Mar 2021 Shift 1]

Answer: 25

Solution:

Given, average force, $F = 2.5 \times 10^{-6} \text{N}$

Area, $A = 30 \text{cm}^2$

Time, $t = 40 \text{min}$

We know that, $F = \frac{IA}{c}$

where, I = energy flux of light

and c = speed of light in air.

$$\Rightarrow I = \frac{FC}{A} = \frac{2.5 \times 10^{-6} \times 3 \times 10^8}{30} = 25 \text{W/cm}^2$$

Question103

The speed of electrons in a scanning electron microscope is $1 \times 10^7 \text{ms}^{-1}$. If the protons having the same speed are used instead of electrons, then the resolving power of scanning proton microscope will be changed by a factor of
[18 Mar 2021 Shift 2]

Options:

A. 1837

B. $\frac{1}{1837}$

C. $\sqrt{1837}$

D. $\frac{1}{\sqrt{1837}}$

Answer: A

Solution:

According to de-Broglie wavelength,

$$\lambda = \frac{h}{mv} \text{ Since, resolving power,}$$

$$\text{RP} \propto \frac{1}{\lambda}$$

$$\Rightarrow \text{RP} \propto \frac{mv}{h}$$

$$\Rightarrow \text{RP} \propto m \quad (\text{speed} = \text{constant})$$

$$\Rightarrow \frac{\text{RP}_p}{\text{RP}_e} = \frac{m_p}{m_e}$$

$$\Rightarrow \frac{RP_p}{RP_e} = 1837 \quad (\text{given, } m_p = 1837m_e)$$

Hence, the resolving power of scanning proton microscope will be changed by factor of 1837.

Question104

A particle is travelling 4 times as fast as an electron. Assuming the ratio of de-Broglie wavelength of a particle to that of electron is 2 : 1, the mass of the particle is
[18 Mar 2021 Shift 1]

Options:

- A. $\frac{1}{16}$ times the mass of electron
- B. 8 times the mass of electron
- C. 16 times the mass of electron
- D. $\frac{1}{8}$ times the mass of electron

Answer: D

Solution:

$$\text{Given, } \frac{\lambda_p}{\lambda_e} = \frac{2}{1}, \frac{v_e}{v_p} = \frac{1}{4}$$

We know that, de-Broglie wavelength,

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow \frac{\lambda_p}{\lambda_e} = \frac{m_e v_e}{m_p v_p} \Rightarrow \frac{2}{1} = \frac{m_e}{m_p} \left(\frac{1}{4} \right)$$

$$\Rightarrow m_p = \frac{1}{8} m_e$$

Question105

An electron of mass m and a photon have same energy E . The ratio of wavelength of electron to that of photon is (c being the velocity of

light)

[17 Mar 2021 Shift 1]

Options:

A. $\frac{1}{c} \left(\frac{2m}{E} \right)^{1/2}$

B. $\frac{1}{c} \left(\frac{E}{2m} \right)^{1/2}$

C. $\left(\frac{E}{2m} \right)^{1/2}$

D. $c(2mE)^{1/2}$

Answer: B

Solution:

We know

$$\lambda = \frac{h}{p} \dots (i)$$

where, λ = wavelength,

h = Planck's constant

and p = linear momentum

$$\text{Also, kinetic energy (K E)} = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$\Rightarrow p = \sqrt{2m \text{ K E}} \dots (ii)$$

From Eqs. (i) and (ii), we get

$$\lambda = \frac{h}{\sqrt{2m \text{ K E}}} \text{ or } \lambda = \frac{h}{\sqrt{2mE}} \quad [\because \text{K E} = E]$$

$$\text{Wavelength of electron, } \lambda_e = \frac{h}{\sqrt{2mE}}$$

$$\text{and wavelength of photon, } \lambda_p = \frac{hc}{E}$$

[$\because$ Both electron and photon have same energy.]

$$\Rightarrow \frac{\lambda_e}{\lambda_p} = \frac{\frac{h}{\sqrt{2mE}}}{\frac{hc}{E}} \Rightarrow \frac{\lambda_e}{\lambda_p} = \frac{1}{c} \left(\frac{E}{2m} \right)^{1/2}$$

Question106

The de-Broglie wavelength associated with an electron and a proton were calculated by accelerating them through same potential of 100V . What should nearly be the ratio of their wavelengths?

($m_p = 1.00727u$, $m_e = 0.00055u$)

[16 Mar 2021 Shift 2]

Options:

A. 1860 : 1

B. $(1860)^2$: 1

C. 41.4 : 1

D. 43 : 1

Answer: D

Solution:

Given, mass of proton, $m_p = 1.00727u$

Mass of electron, $m_e = 0.00055u$

Potential, $V = 100V$

As, de-Broglie wavelength can be given as

$$\lambda = \frac{h}{\sqrt{2mqV}} \Rightarrow \lambda \propto \frac{1}{\sqrt{m}}$$

$$\Rightarrow \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{m_2}{m_1}}$$

$$\Rightarrow \frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}} = \sqrt{\frac{1.00727u}{0.00055u}}$$

$$\frac{\lambda_e}{\lambda_p} = \sqrt{1831.4} = 42.79$$

$$\text{or } \frac{\lambda_e}{\lambda_p} \approx \frac{43}{1} \text{ or } \lambda_e : \lambda_p = 43 : 1$$

Question107

A light beam of wavelength 500nm is incident on a metal having work function of 1.25eV , placed in a magnetic field of intensity B. The electrons emitted perpendicular to the magnetic field B, with

maximum kinetic energy are bent into circular arc of radius 30cm.

The value of B is _____ $\times 10^{-7}$ T

Given $hc = 20 \times 10^{-26}$ J – m, mass of electron = 9×10^{-31} kg

[25 Jul 2021 Shift 2]

Answer: 125

Solution:

By photoelectric equation

$$\frac{hc}{\lambda} - \phi = k_{\max}$$

$$k_{\max} = \frac{1240}{500} - 1.25 \approx 1.25$$

$$r = \frac{\sqrt{2mk}}{eB}$$

$$B = \frac{\sqrt{2mk}}{er}$$

$$= 125 \times 10^{-7} \text{ T}$$

Question108

When radiation of wavelength λ is incident on a metallic surface, the stopping potential of ejected photoelectrons is 4.8V . If the same surface is illuminated by radiation of double the previous wavelength, then the stopping potential becomes 1.6V . The threshold wavelength of the metal is :

[25 Jul 2021 Shift 2]

Options:

A. 2λ

B. 4λ

C. 8λ

D. 6λ

Answer: B

Solution:

$$V_s = h\nu - \phi$$

$$4.8 = \frac{hc}{\lambda} - \phi \dots\dots(i)$$

$$1.6 = \frac{hc}{2\lambda} - \phi \dots\dots(ii)$$

Using above equation (i) - (ii)

$$3.2 = \frac{hc}{\lambda} - \frac{hc}{2\lambda}$$

$$3.2 = \frac{hc}{2\lambda} \dots\dots(iii)$$

$$\left[\lambda = \frac{hc}{6.4} \right]$$

Put in equation (ii)

$$\phi = 1.6$$

$$\frac{hc}{\lambda_{th}} = 1.6$$

$$\lambda_{th} = \frac{hc}{1.6}$$

$$= \left(\frac{hc}{6.4} \right) \times 4 = 4\lambda$$

Question109

A certain metallic surface is illuminated by monochromatic radiation of wavelength λ . The stopping potential for photoelectric current for this radiation is $3V_0$. If the same surface is illuminated with a radiation of wavelength 2λ , the stopping potential is V_0 . The threshold wavelength of this surface for photoelectric effect is _____ λ .

[20 Jul 2021 Shift 2]

Answer: 4

Solution:

$$K E = \frac{hc}{\lambda} - \phi$$

$$e(3V_0) = \frac{hc}{\lambda_0} - \phi \dots\dots(i)$$

$$eV_0 = \frac{hc}{2\lambda_0} - \phi \dots\dots(ii)$$

Using (i) & (ii)

$$\phi = \frac{hc}{4\lambda_0} = \frac{hc}{\lambda_t}$$

$$\lambda_t = 4\lambda_0$$

Question110

A particle of mass 9.1×10^{-31} kg travels in a medium with a speed of 10^6 m/s and a photon of aradiation of linear momentum 10^{-27} kgm/s travelsin vacuum. The wavelength of photon is _____ times the wavelength of the particle.

[27 Jul 2021 Shift 1]

Answer: 910

Solution:

$$\text{For photon } \lambda_1 = \frac{h}{p} = \frac{6.6 \times 10^{-34}}{10^{-27}}$$

$$\text{For particle } \lambda_2 = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \times 10^6}$$

$$\therefore \frac{\lambda_1}{\lambda_2} = 910$$

Question111

An electron moving with speed v and a photon moving with speed c, have same D-Broglie wavelength. The ratio of kinetic energy of electron to that of photon is :

[25 Jul 2021 Shift 2]

Options:

A. $\frac{3c}{v}$

B. $\frac{v}{3c}$

C. $\frac{v}{2c}$

D. $\frac{2c}{v}$

Answer: C

Solution:

$$\lambda_e = \lambda_{ph}$$

$$\frac{h}{p_e} = \frac{h}{p_{ph}}$$

$$\sqrt{2mk_e} = \frac{E_{ph}}{c}$$

$$2mk_e = \frac{(E_{ph})^2}{c^2}$$

$$\frac{k_e}{E_{ph}} = \frac{E_{ph}}{c^2} \left(\frac{1}{2m} \right)$$

$$= \frac{p_{ph}}{c} \left(\frac{1}{2m} \right)$$

$$= \frac{p_e}{c} \left(\frac{1}{2m} \right)$$

$$= \frac{mv}{c} \frac{1}{2m}$$

$$= \frac{v}{2c}$$

Question112

An electron of mass m_e and a proton of mass m_p are accelerated through the same potential difference. The ratio of the de-Broglie wavelength associated with the electron to that with the proton is :-
[22 Jul 2021 Shift 2]

Options:

A. $\frac{m_p}{m_e}$

B. 1

C. $\sqrt{\frac{m_p}{m_e}}$

D. $\frac{m_e}{m_p}$

Answer: C

Solution:

$$K E = e\Delta V$$

$$\lambda_e = \frac{h}{\sqrt{2m_e(e\Delta V)}}$$

$$\lambda_p = \frac{h}{\sqrt{2m_p(e\Delta V)}}$$

$$\Rightarrow \frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$$

Question113

An electron having de-Broglie wavelength λ is incident on a target in a X-ray tube. Cut-off wavelength of emitted X-ray is :

[20 Jul 2021 Shift 2]

Options:

A. 0

B. $\frac{2m^2c^2\lambda^2}{h^2}$

C. $\frac{2mc\lambda^2}{h}$

D. $\frac{hc}{mc}$

Answer: C

Solution:

$$\lambda = \frac{h}{mv}$$

$$\text{kinetic energy, } \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2} = \frac{hc}{\lambda_c}$$

$$\lambda_c = \frac{2m\lambda^2 c}{h}$$

Question114

In a photoelectric experiment, increasing the intensity of incident light

[27 Aug 2021 Shift 1]

Options:

- A. increases the number of photons incident and also increases the KE of the ejected electrons.
- B. increases the frequency of photons incident and increases the KE of the ejected electrons.
- C. increases the frequency of photons incident and the KE of the ejected electrons remains unchange
- D. d. increases the number of photons incident and the KE of the ejected electrons remains unchanged.

Answer: D

Solution:

As per Einstein's theory of photoelectric effect when a photon strikes the metal surface it ejects an electron. If we increase the intensity of light, the number of photons incident on surface will increase and thus, the number of electrons ejected will increase proportionally and the photocurrent will also increase. Kinetic energy of ejected electrons is given as

$$\frac{1}{2}m_e v_e^2 = eV_0$$

Here, V_0 is stopping potential.

The stopping potential of surface is constant and it will not increase with increase in intensity of light, so the kinetic energy of ejected electrons will not increase.

Thus, increasing the intensity of light will increase the number of photons incident and the kinetic energy will remain unchanged.

Question115

A monochromatic neon lamp with wavelength of 670.5 nm illuminates a photo-sensitive material which has a stopping voltage of 0.48V. What will be the stopping voltage if the source light is changed with another source of wavelength of 474.6 nm ?
[27 Aug 2021 Shift 2]

Options:

A. 0.96 V

B. 1.25 V

C. 0.24 V

D. 1.5 V

Answer: B

Solution:

In, case I given, incident wavelength,

$$\lambda_1 = 670.5\text{nm}$$

Stopping potential, $V_1 = 0.48\text{V}$

In case II Incident wavelength,

$$\lambda_2 = 474.6\text{nm}$$

Stopping potential is V_2 .

By using Einstein photoelectric

equation, $eV_1 = \frac{hc}{\lambda_1} - \phi_0$

where, h is Planck's constant

$$= 6.63 \times 10^{-34}\text{J s}$$

c is speed of light $= 3 \times 10^8\text{ms}^{-1}$

ϕ_0 is work function

$$\therefore eV_1 = \frac{hc}{\lambda_1} - \phi_0$$

$$\text{and similarly, } eV_2 = \frac{hc}{\lambda_2} - \phi_0$$

$$\Rightarrow \phi_0 = \frac{hc}{\lambda_2} - eV_2$$

From Eqs. (i) and (ii), we get

$$\therefore \frac{hc}{\lambda_1} - eV_1 = \frac{hc}{\lambda_2} - eV_2$$

$$\begin{aligned}
\Rightarrow eV_1 - eV_2 &= hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) \\
\Rightarrow V_1 - V_2 &= \frac{hc}{e} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) \\
\Rightarrow V_2 &= V_1 + \frac{hc}{e} \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right) \\
&= 0.48 + \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19}} \\
&= \left(\frac{1}{474.6} - \frac{1}{670.5} \right) \\
&= 0.48 + 12.4 \times 10^{-7+9} (2.11 \times 10^{-3} - 1.49 \times 10^{-3}) \\
&= 0.48 + 12.4 \times 10^{2-3} \times 0.62 \\
&= 0.48 + 0.77 = 1.25V
\end{aligned}$$

Question116

In a photoelectric experiment ultraviolet light of wavelength 280 nm is used with lithium cathode having work - function $\phi = 2.5 \text{ eV}$. If the wavelength of incident light is switched to 400 nm, find out the change in the stopping potential. ($h = 6.63 \times 10^{-34} \text{ Js.}$, and $c = 3 \times 10^8 \text{ ms}^{-1}$)
[26 Aug 2021 Shift 1]

Options:

- A. 1.3V
- B. 1.1V
- C. 1.9V
- D. 0.6V

Answer: A

Solution:

By Einstein's photoelectric equation, we have

$$\Rightarrow hf = \phi + (KE)_{\max}$$

where, f is the frequency, ϕ is the work-function of metal and $(KE)_{\max}$ is the maximum kinetic energy of emitted photoelectron.

$$\Rightarrow \frac{hc}{\lambda} = \phi + eV_0$$

$$\Rightarrow eV_0 = \frac{hc}{\lambda} - \phi$$

Here, $\lambda = 280 \text{ nm}$ and $\phi = 2.5 \text{ eV}$

$$\Rightarrow e(V_0)_1 = \frac{1240}{280} - 2.5$$

$$\Rightarrow e(V_0)_1 = 1.93 \text{ eV}$$

$$\Rightarrow (V_0)_1 = 1.93 \text{ V}$$

Similarly, stopping potential for the light of wavelength of 400 nm ,

$$e(V_0)_2 = \frac{1240}{400} - 2.5$$

$$\Rightarrow (V_0)_2 = 0.6 \text{ V}$$

$\therefore$ Change in stopping potential,

$$\Delta V = (V_0)_1 - (V_0)_2$$

$$= 1.93 - 0.6 = 1.33 \approx 1.3 \text{ V}$$

Question117

Consider two separate ideal gases of electrons and protons having same number of particles. The temperature of both the gases are same. The ratio of the uncertainty in determining the position of an electron to that of a proton is proportional to
[31 Aug 2021 Shift 2]

Options:

A. $\left(\frac{m_p}{m_e} \right)^{\frac{3}{2}}$

B. $\sqrt{\frac{m_e}{m_p}}$

C. $\sqrt{\frac{m_p}{m_e}}$

D. $\frac{m_p}{m_e}$

Answer: C

Solution:

At same temperature kinetic energy of proton and electron are same

$$KE_p = KE_e$$

$$\Rightarrow \frac{p_p^2}{2m_p} = \frac{p_e^2}{2m_e}$$

$$\frac{p_p^2}{p_e^2} = \frac{m_p}{m_e} \dots (i)$$

Using uncertainty in determining the position, $\Delta x \approx \frac{h}{\Delta p}$

$$\text{Ratio, } \therefore \frac{\Delta x_e}{\Delta x_p} = \frac{\Delta p_p}{\Delta p_e} = \sqrt{\frac{m_p}{m_e}} \text{ [from Eq. (i)]}$$

Question118

A moving proton and electron have the same de-Broglie wavelength. If k and p denote the KE and momentum, respectively. Then, choose the correct option.

[31 Aug 2021 Shift 1]

Options:

A. $K_p < K_e$ and $p_p = p_e$

B. $K_p = K_e$ and $p_p = p_e$

C. $K_p < K_e$ and $p_p < p_e$

D. $K_p > K_e$ and $p_p = p_e$

Answer: A

Solution:

Let de-Broglie wavelength of proton and electron are λ_p and λ_e .

Kinetic energy of electron and proton are K_e and K_p .

Momentum of electron and proton are p_e and p_p .

According to question, $\lambda_p = \lambda_e = \lambda$

By using de-Broglie wavelength,

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

where, m is mass and h is Planck's constant.

$$\therefore p = \frac{h}{\lambda} \dots (i)$$

From Eq. (i)

$$p_e = p_p \text{ as } \lambda_e = \lambda_p$$

$$\text{Again, } K = \frac{h^2}{2m\lambda} \dots (ii)$$

From Eq. (ii)

$$\therefore \frac{K_e}{K_p} = \frac{m_p}{m_e} \times \frac{\lambda_p}{\lambda_e}$$

$$\Rightarrow \frac{K_e}{K_p} = \frac{m_p}{m_e} [\because \lambda_e = \lambda_p]$$

$$\because m_p > m_e$$

$$K_e > K_p$$

Question119

The de-Broglie wavelength of a particle having kinetic energy E is λ . How much extra energy must be given to this particle, so that the de-Broglie wavelength reduces to 75% of the initial value ?
[26 Aug 2021 Shift 2]

Options:

A. $\frac{1}{9}E$

B. $\frac{7}{9}E$

C. E

D. $\frac{16}{9}E$

Answer: B

Solution:

Given, initial kinetic energy, $E_1 = E$

Initial de-Broglie wavelength, $\lambda_1 = \lambda$

Consider the wavelength of particle changes to 75% of λ after providing energy ΔE to the particle.

Hence, final wavelength of particle, $\lambda_2 = 0.75\lambda$

Final energy, $E_2 = E + \Delta E$

Relationship between de-Broglie wavelength and energy of particle is given as

$$\lambda = \frac{h}{\sqrt{2mE}}$$

Here, h is Planck's constant and m is mass of particle which is also constant term. Therefore, we get

$$\lambda \propto \frac{1}{\sqrt{E}}$$

Thus, we can write the relationship as

$$\frac{\lambda_1}{\lambda_2} = \frac{\sqrt{E_2}}{\sqrt{E_1}}$$

$$\Rightarrow \frac{\lambda}{0.75\lambda} = \frac{\sqrt{E + \Delta E}}{\sqrt{E}}$$

$$\Rightarrow \frac{4}{3} = \frac{\sqrt{E + \Delta E}}{\sqrt{E}}$$

Squaring both sides of above equation, we get

$$\Rightarrow \frac{16}{9} = \frac{E + \Delta E}{E}$$

$$\Rightarrow 16E = 9E + 9\Delta E$$

$$\Delta E = \frac{7}{9}E$$

Question 120

The temperature of an ideal gas in 3 - dimensions is 300K. The corresponding de-Broglie wavelength of the electron approximately at 300K, is

$$[m_e = \text{mass of electron} = 9 \times 10^{-31} \text{ kg}]$$

$$h = \text{Planck's constant} = 6.6 \times 10^{-34} \text{ Js}$$

$$K_B = \text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}]$$

[1 Sep 2021 Shift 2]

Options:

A. 6.26 nm

B. 8.46 nm

C. 2.26 nm

D. 3.25 nm

Answer: A

Solution:

Given, Planck's constant, $h = 6.6 \times 10^{-34} \text{ Js}$

Boltzmann constant, $k_B = 1.38 \times 10^{-23} \text{ J/K}$

Mass of an electron, $m_e = 9 \times 10^{-31} \text{ kg}$

Temperature of an ideal gas, $T = 300 \text{ K}$

As we know that, de-Broglie wavelength,

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} \dots (i)$$

Here, E is the kinetic energy.

$$E = \frac{3K_B T}{2}$$

Substituting value of E in Eq. (i), we get,

$$\lambda = \frac{h}{\sqrt{3mK_B T}}$$

Substituting the given values in the above equation, we get

$$\lambda = \frac{6.6 \times 10^{-34}}{\sqrt{3 \times 9 \times 10^{-31} \times 138 \times 10^{-23} \times 300}}$$

$$= 6.26 \text{ nm}$$

$\therefore$ The corresponding de-Broglie wavelength of an electron approximately at 300 K is 6.26 nm.

Question 121

A particle moving with kinetic energy E has de Broglie wavelength λ . If energy ΔE is added to its energy, the wavelength become $\frac{\lambda}{2}$.

Value of ΔE , is:

[9 Jan. 2020 I]

Options:

A. E

B. $4E$

C. $3E$

D. $2E$

Answer: C

Solution:

As per question, when K E of particle E , wavelength λ and when K E becomes $E + \Delta E$ wavelength becomes $\lambda/2$

$$\begin{aligned}
 \text{Using, } \lambda &= \frac{h}{\sqrt{2mK E}} \\
 \frac{\lambda}{2} &= \frac{h}{\sqrt{2m(K E + \Delta E)}} \\
 \Rightarrow \frac{\lambda}{\lambda/2} &= \sqrt{\frac{K E + \Delta E}{K E}} \\
 \Rightarrow 4 &= \frac{K E + \Delta E}{K E} \\
 \Rightarrow 4K E - K E &= \Delta E \\
 \therefore \Delta E &= 3K E = 3E
 \end{aligned}$$

Question122

An electron of mass m and magnitude of charge $|e|$ initially at rest gets accelerated by a constant electric field E . The rate of change of de-Broglie wavelength of this electron at time t ignoring relativistic effects is:

[9 Jan. 2020 II]

Options:

A. $-\frac{h}{|e|E \sqrt{t}}$

B. $\frac{|e|E t}{h}$

C. $-\frac{h}{|e|E t}$

D. $\frac{-h}{|e|E t^2}$

Answer: D

Solution:

Acceleration of electron in electric field, $a = \frac{eE}{m}$

Using equation

$$v = u + at$$

$$\Rightarrow v = 0 + \frac{eE}{m}t$$

$$\Rightarrow v = \frac{eE t}{m} \dots\dots(i)$$

De-broglie wavelength λ is given by

$$\lambda = \frac{h}{mv} = \frac{h}{m \left(\frac{eEt}{m} \right)} \text{ [using (i)]}$$

$$\Rightarrow \lambda = \frac{h}{eEt}$$

Differentiating w.r.t.

$$\frac{d\lambda}{dt} = \frac{d \left(\frac{h}{eEt} \right)}{dt} \Rightarrow \frac{d\lambda}{dt} = \frac{-h}{eEt^2}$$

Question123

An electron (mass m) with initial velocity $\vec{v} = v_0 \hat{i} + v_0 \hat{j}$ is in an electric field $\vec{E} = -E_0 \hat{k}$. If λ_0 is initial de-Broglie wavelength of electron, its de-Broglie wave length at time t is given by:
[8 Jan. 2020 II]

Options:

A. $\frac{\lambda_0 \sqrt{2}}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$

B. $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$

C. $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{2m^2 v_0^2}}}$

D. $\frac{\lambda_0}{\sqrt{2 + \frac{e^2 E_0^2 t^2}{m^2 v_0^2}}}$

Answer: C

Solution:

Given, Initial velocity, $u = v_0 \hat{i} + v_0 \hat{j}$

$$\text{Acceleration, } a = \frac{qE_0}{m} = \frac{eE_0}{m}$$

Using $v = u + at$

$$v = v_0 \hat{i} + v_0 \hat{j} + \frac{eE_0}{m} t \hat{k}$$

$$\therefore |\vec{v}| = \sqrt{2v_0^2 + \left(\frac{eE_0 t}{m}\right)^2}$$

$$\text{de-Broglie wavelength, } \lambda = \frac{h}{p}$$

$$\Rightarrow \lambda = \frac{h}{mv} \quad (\because p = mv)$$

$$\text{Initial wavelength, } \lambda_0 = \frac{h}{mv_0 \sqrt{2}}$$

Final wavelength,

$$\lambda = \frac{h}{m \sqrt{2v_0^2 + \left(\frac{eE_0 t}{m}\right)^2}}$$

$$\frac{\lambda}{\lambda_0} = \frac{1}{\sqrt{1 + \left(\frac{eE_0 t}{\sqrt{2}mv_0}\right)^2}}$$

$$\Rightarrow \lambda = \frac{\lambda_0}{\sqrt{1 + \frac{e^2 E_0^2 t^2}{2m^2 v_0^2}}}$$

Question 124

A beam of electrons of energy E scatters from a target having atomic spacing of $1\text{\AA}$. The first maximum intensity occurs at $\theta = 60^\circ$. Then E (in eV) is _____.

(Plank constant $h = 6.64 \times 10^{-34} \text{ J s}$, $1\text{eV} = 1.6 \times 10^{-19} \text{ J}$,

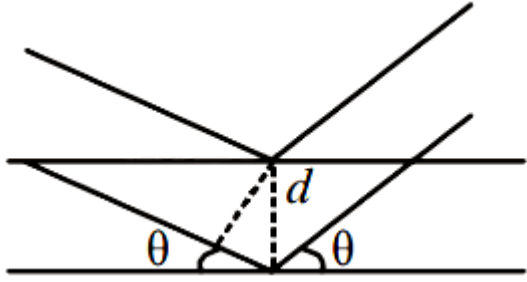
electron mass $m = 9.1 \times 10^{-31} \text{ kg}$)

[NA Sep. 05, 2020 (I)]

Answer: 50

Solution:

From Bragg's equation $2d \sin \theta = \lambda$ and de-Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$



$$2d \sin \theta = \lambda = \frac{h}{\sqrt{2mE}}$$

$$\Rightarrow 2 \times 10^{-10} \times \frac{\sqrt{3}}{2} = \frac{6.6 \times 10^{-34}}{\sqrt{2mE}}$$

$$[\because \theta = 60^\circ \text{ and } d = 1 \text{ \AA} = 1 \times 10^{-10} \text{ m}]$$

$$\therefore E = \frac{1}{2} \times 6.64^2 \times \frac{10^{-48}}{9.1 \times 10^{-31} \times 3 \times 1.6 \times 10^{-19}} \approx 50 \text{ eV}$$

Question125

Radiation, with wavelength $6561 \text{ \AA}$ falls on a metal surface to produce photoelectrons. The electrons are made to enter a uniform magnetic field of $3 \times 10^{-4} \text{ T}$. If the radius of the largest circular path followed by the electrons is 10 mm , the work function of the metal is close to:

[9 Jan. 2020 I]

Options:

A. 1.1 eV

B. 0.8 eV

C. 1.6 eV

D. 1.8 eV

Answer: A

Solution:

Using Einstein's photoelectric equation,

$$E = \omega_0 + K E_{\max}$$

$$\Rightarrow \omega_0 = K E_{\max} - E$$

$$p = \sqrt{2mK E} \Rightarrow K E = \frac{p^2}{2m}$$

$$r = \frac{p}{eB} \Rightarrow p = reB$$

$$K_{\max} = \frac{r^2 e^2 B^2}{2m} \quad K E_{\max} = \frac{12420}{\lambda} - \omega_0$$

$$\begin{aligned} \Rightarrow \omega_0 &= \frac{12420}{6561} - \frac{r^2 e^2 B^2}{2m} \text{ (I neV)} \\ &= 1.89 \text{ (eV)} - \frac{(10^{-4})(1.6 \times 10^{-19})9 \times 10^5}{2 \times 9.07 \times 10^{-31}} \\ &= 1.89 \text{ (eV)} - \frac{(10^{-4})(1.6 \times 10^{-19})9 \times 10^5}{2 \times 9.07 \times 10^{-31}} \\ &= (1.89 - 0.79) \text{ eV} = 1.1 \text{ eV} \end{aligned}$$

Question126

When photon of energy 4.0 eV strikes the surface of a metal A, the ejected photoelectrons have maximum kinetic energy $T_A \text{ eV}$ and de-Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photon of energy 4.50 eV is $T_B = (T_A - 1.5) \text{ eV}$. If the de-Broglie wavelength of these photoelectrons $\lambda_B = 2\lambda_A$, then the work function of metal B is:
[8 Jan. 2020 I]

Options:

- A. 4 eV
- B. 2 eV
- C. 1.5 eV
- D. 3 eV

Answer: A

Solution:

de-Broglie wavelength (λ),

$$\text{Momentum, } mv = \frac{h}{\lambda} = p = \sqrt{2m(K E)}$$

$$\therefore \lambda = \frac{h}{\sqrt{2mKE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{KE}}$$

$$\therefore \frac{\lambda_A}{\lambda_B} = \sqrt{\frac{K_B}{K_A}} = \sqrt{\frac{T_A - 1.5}{T_A}} \text{ (as given)}$$

$$\text{Also, } \frac{\lambda_A}{\lambda_B} = \frac{1}{2}$$

On solving we get, $T_A = 2\text{eV}$

$$\therefore KE_B = T_A - 1.5 = 2 - 1.5 = 0.5\text{eV}$$

$\therefore$ Work function of metal B is

$$\phi_B = E_B - KE_B = 4.5 - 0.5 = 4\text{eV}$$

Question127

A beam of electromagnetic radiation of intensity $6.4 \times 10^{-5} \text{W} / \text{cm}^2$ is comprised of wavelength, $\lambda = 310 \text{ nm}$. It falls normally on a metal (work function $\phi = 2\text{eV}$) of surface area of 1cm^2 . If one in 10^3 photons ejects an electron, total number of electrons ejected in 1s is 10^x . ($hc = 1240\text{eV nm}$, $1\text{eV} = 1.6 \times 10^{-19}\text{J}$), then x is _____.
[NA 7 Jan. 2020 I]

Answer: 11

Solution:

Energy of photon

$$E = \frac{hc}{\lambda} = \frac{1240}{310} = 4\text{eV} > 2\text{eV} [\phi]$$

(so emission of photoelectron will take place)

$$= 4 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-19} \text{ joule}$$

$$N = \frac{6.4 \times 10^{-5} \times 1}{4 \times 6.4 \times 10^{-19}} = 10^{14}$$

$$\text{No. of photoelectrons emitted per second} = \frac{10^{14}}{10^3} = 10^{11} (\because 1 \text{ in } 10^3 \text{ photons ejects an electron})$$

$$\therefore \text{Value of } X = 11.00$$

Question128

An electron, a doubly ionized helium ion (He^{++}) and a proton are having the same kinetic energy. The relation between their respective de-Broglie wavelengths λ_e , $\lambda_{\text{He}^{++}}$ and λ_p is

[Sep. 06, 2020 (I)]

Options:

A. $\lambda_e > \lambda_{\text{He}^{++}} > \lambda_p$

B. $\lambda_e < \lambda_{\text{He}^{++}} = \lambda_p$

C. $\lambda_e > \lambda_p > \lambda_{\text{He}^{++}}$

D. $\lambda_e < \lambda_p < \lambda_{\text{He}^{++}}$

Answer: C

Solution:

$$\text{de-Broglie wavelength, } \lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(\text{K.E.})}}$$

$$\therefore \lambda \propto \frac{1}{\sqrt{m}}$$

$$\text{As } m_{\text{He}^{++}} > m_p > m_e$$

$$\lambda_{\text{He}^{++}} > \lambda_p > \lambda_e \text{ or } \lambda_e > \lambda_p > \lambda_{\text{He}^{++}}$$

Question129

Assuming the nitrogen molecule is moving with r.m.s. velocity at 400 K, the de-Broglie wavelength of nitrogen molecule is close to :

(Given : nitrogen molecule weight : 4.64×10^{-26} kg, Boltzman constant : 1.38×10^{-23} J/K, Planck constant : 6.63×10^{-34} J.s)

[Sep. 06, 2020 (II)]

Options:

A. $0.24 \text{ \AA}$

B. $0.20 \text{ \AA}$

C. $0.34 \text{ \AA}$

D. $0.44 \text{ \AA}$

Answer: A

Solution:

Rms speed of gas molecule, $V_{\text{rms}} = \sqrt{\frac{3kT}{m}}$

de Broglie wavelength, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mk}}$

$$\therefore \lambda = \frac{h}{\sqrt{2m \times \frac{1}{2} m V_{\text{rms}}^2}} = \frac{h}{\sqrt{m \times \frac{3}{2} kT}} = \frac{h}{\sqrt{3mkT}}$$

Substituting the respective values we get

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{3 \times 4.64 \times 10^{-26} \times 1.38 \times 10^{-13} \times 400}} = 0.24 \text{ \AA}$$

Question 130

Particle A of mass $m_A = \frac{m}{2}$ moving along the x -axis with velocity v_0 collides elastically with another particle B at rest having mass $m_B = \frac{m}{3}$. If both particles move along the x -axis after the collision, the change $\Delta\lambda$ in de-Broglie wavelength of particle A, in terms of its de-Broglie wavelength (λ_0) before collision is :

[Sep. 04, 2020 (I)]

Options:

A. $\Delta\lambda = \frac{3}{2}\lambda_0$

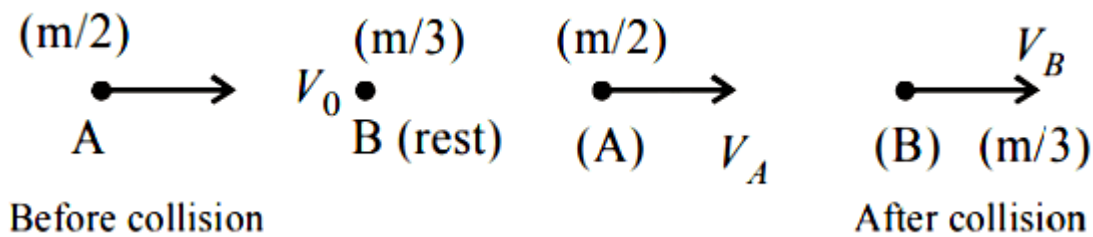
B. $\Delta\lambda = \frac{5}{2}\lambda_0$

C. $\Delta\lambda = 2\lambda_0$

D. $\Delta\lambda = 4\lambda_0$

Answer: D

Solution:



Applying momentum conservation

$$\frac{m}{2} \times V_0 + \frac{m}{3} \times (0) = \frac{m}{2} V_A + \frac{m}{3} V_B$$
$$= \frac{V_0}{2} = \frac{V_A}{2} + \frac{V_B}{3} \dots\dots(i)$$

Since, collision is elastic

$$e = 1 = \frac{V_B - V_A}{V_0} \Rightarrow V_0 = V_B - V_A \dots\dots(ii)$$

On solving equations (i) and (ii) : $V_A = \frac{V_0}{5}$

Now, de-Broglie wavelength of A before collision :

$$\lambda_0 = \frac{h}{m_A V_0} = \frac{h}{\left(\frac{m}{2}\right) V_0} \Rightarrow \lambda_0 = \frac{2h}{m V_0}$$

Final de-Broglie wavelength :

$$\lambda_f = \frac{h}{m_A V_0} = \frac{h}{\frac{m}{2} \times \frac{V_0}{5}} \Rightarrow \lambda_f = \frac{10h}{m V_0}$$

$$\therefore \Delta\lambda = \lambda_f - \lambda_0 = \frac{10h}{m V_0} - \frac{2h}{m V_0}$$

$$\Rightarrow \Delta\lambda = \frac{8h}{m V_0} \Rightarrow \Delta\lambda = 4 \times \frac{2h}{m V_0}$$

$$\therefore \Delta\lambda = 4\lambda_0$$

Question131

A particle is moving 5 times as fast as an electron. The ratio of the de-Broglie wavelength of the particle to that of the electron is 1.878×10^{-4} . The mass of the particle is close to :
[Sep. 02, 2020 (II)]

Options:

A. $4.8 \times 10^{-27} \text{ kg}$

B. $9.1 \times 10^{-31} \text{ kg}$

C. $1.2 \times 10^{-28} \text{kg}$

D. $9.7 \times 10^{-28} \text{kg}$

Answer: D

Solution:

de Broglie wavelength

$$\lambda = \frac{h}{mv} \Rightarrow m = \frac{h}{\lambda v}$$

Clearly, $m \propto \frac{1}{\lambda v}$

If λ and v be the wavelength and velocity of electron and λ' and v' be the wavelength and velocity of the particle then

$$\Rightarrow \frac{m'}{m} = \frac{v\lambda}{v'\lambda'} = \frac{1}{5} \times \frac{1}{1.878} \times 10^{-4}$$

$$\Rightarrow m = 9.7 \times 10^{-28} \text{kg}$$

Question132

The surface of a metal is illuminated alternately with photons of energies $E_1 = 4 \text{ eV}$ and $E_2 = 2.5 \text{ eV}$ respectively. The ratio of maximum speeds of the photoelectrons emitted in the two cases is 2. The work function of the metal in (eV) is _____. [NA Sep. 05, 2020 (II)]

Answer: 2

Solution:

From the Einstein's photoelectric equation

Energy of photon = Kinetic energy of photoelectrons + Work function

$\Rightarrow$ Kinetic energy = Energy of Photon – Work Function

Let ϕ_0 be the work function of metal and v_1 and v_2 be the velocity of photoelectrons. Using

Einstein's photoelectric equation we have

$$\frac{1}{2}mv_1^2 = 4 - \phi_0 \dots\dots(i)$$

$$\frac{1}{2}mv_2^2 = 2.5 - \phi_0 \dots (ii)$$

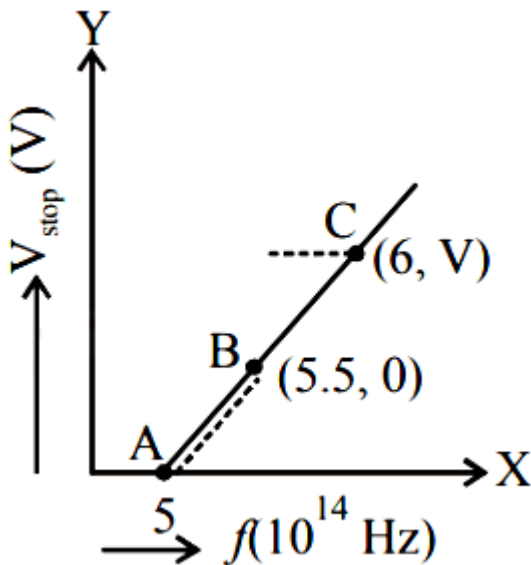
$$\Rightarrow \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2} = \frac{4 - \phi_0}{2.5 - \phi_0}$$

$$\Rightarrow (2)^2 = \frac{4 - \phi_0}{2.5 - \phi_0} \Rightarrow 10 - 4\phi_0 = 4 - \phi_0$$

$$\phi_0 = 2\text{eV}$$

Question133

Given figure shows few data points in a photo electric effect experiment for a certain metal. The minimum energy for ejection of electron from its surface is : (Plancks constant $h = 6.62 \times 10^{-34} \text{ J.s}$)



[Sep. 04, 2020 (I)]

Options:

A. 2.27 eV

B. 2.59 eV

C. 1.93 eV

D. 2.10 eV

Answer: A

Solution:

Graph of V and f given at B(5.5, 0)

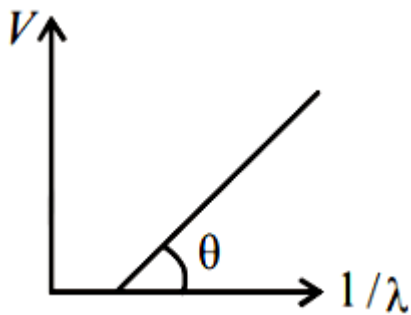
Minimum energy for ejection of electron = Work function (ϕ).

$$\phi = h\nu \text{ joule or } \phi = \frac{h\nu}{e} \text{ eV (for } \nu = 0)$$

$$\therefore \phi = \frac{6.62 \times 10^{-34} \times 5.5 \times 10^{14}}{1.6 \times 10^{-19}} \text{ eV} = 2.27 \text{ eV}$$

Question 134

In a photoelectric effect experiment, the graph of stopping potential V versus reciprocal of wavelength obtained is shown in the figure. As the intensity of incident radiation is increased :



[Sep. 04, 2020 (II)]

Options:

- A. Straight line shifts to right
- B. Slope of the straight line get more steep
- C. Straight line shifts to left
- D. Graph does not change

Answer: D

Solution:

According to Einstein's photoelectric equation

$$K_{\max} = h\nu - \phi_0$$

$$\Rightarrow eV_s = \frac{hc}{\lambda} - \phi_0$$

$$\Rightarrow V_s = \frac{hc}{\lambda e} - \frac{\phi_0}{e}$$

where λ = wavelength of incident light

ϕ_0 = work function

V_s = stopping potential

Comparing the above equation with $y = mx + c$, we get slope = $\frac{hc}{e}$

Increasing the frequency of incident radiation has no effect on work function and frequency. So, graph will not change.

Question 135

When the wavelength of radiation falling on a metal is changed from 500 nm to 200 nm, the maximum kinetic energy of the photoelectrons becomes three times larger. The work function of the metal is close to :

[Sep. 03, 2020 (I)]

Options:

A. 0.81 eV

B. 1.02 eV

C. 0.52 eV

D. 0.61 eV

Answer: D

Solution:

Using equation, $= \frac{hc}{\lambda} - \phi$

$$K E_{\max} = \frac{hc}{\lambda} - \phi = \frac{hc}{500} - \phi \dots (1)$$

$$\text{Again, } 3K E_{\max} = \frac{hc}{200} - \phi \dots (2)$$

Dividing equation (2) by (1),

$$\frac{3K E_{\max}}{K E_{\max}} = \frac{3}{1} = \frac{\frac{hc}{200} - \phi}{\frac{hc}{500} - \phi}$$

Putting the value of $hc = 1237.5$ and solving we get, work function, $\phi = 0.61 \text{ eV}$.

Question 136

Two sources of light emit X-rays of wavelength 1 nm and visible light of wavelength 500 nm, respectively. Both the sources emit light of the same power 200 W. The ratio of the number density of photons of X-rays to the number density of photons of the visible light of the given wavelengths is :

[Sep. 03, 2020 (II)]

Options:

A. $\frac{1}{500}$

B. 250

C. $\frac{1}{250}$

D. 500

Answer: A

Solution:

Given,

Wavelength of X -rays, $\lambda_1 = 1\text{nm} = 1 \times 10^{-9}\text{m}$

Wavelength of visible light, $\lambda_2 = 500 \times 10^{-9}\text{m}$

The number of photons emitted per second from a source of monochromatic radiation of wavelength λ and power P is given as

$$n = \frac{P}{E} = \frac{P}{h\nu} = \frac{P\lambda}{hc} \quad (\because E = h\nu \text{ and } \nu = \frac{c}{\lambda})$$

$\Rightarrow$ Clearly $n \propto \lambda$

$$\Rightarrow \frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2} = \frac{1}{500}$$

Question137

When radiation of wavelength λ is used to illuminate a metallic surface, the stopping potential is V. When the same surface is illuminated with radiation of wavelength 3λ , the stopping potential is $\frac{V}{4}$. If the threshold wavelength for the metallic surface is $n\lambda$ then value of $n\lambda$ will be _____.

[NA Sep. 02, 2020 (I)]

Answer: 9

Solution:

When radiation of wavelength A, λ_A is used to illuminate, stopping potential $V_A = V$

$$\frac{hc}{\lambda} = \phi + eV \dots\dots(i)$$

When radiation of wavelength B, λ_B is used to illuminate, stopping potential, $V_B = \frac{V}{4}$

$$\frac{hc}{3\lambda} = \phi + \frac{eV}{4} \dots\dots(ii)$$

From eq. (i) - (ii)

$$\frac{hc}{\lambda} \left(1 - \frac{1}{3} \right) = \frac{3}{4}eV$$

$$\Rightarrow \frac{hc}{\lambda} \frac{2}{3} = \frac{3}{4}eV \Rightarrow eV = \frac{8}{9\lambda}$$

$$\frac{hc}{\lambda} = \phi + \frac{8}{9} \frac{hc}{\lambda}$$

$$\therefore \phi = \frac{hc}{9\lambda} = \frac{hc}{n\lambda}, \text{ so, } n = 9.$$

Question138

A particle A of mass 'm' and charge 'q' is accelerated by a potential difference of 50v Another particle B of mass 4m ' and charge'q is accelerated by a potential differnce of2500V . The ratio of de-Broglie wavelength $\frac{\lambda_A}{\lambda_B}$ is

[12 Jan. 2019 I]

Options:

A. 10.00

B. 0.07

C. 14.14

D. 4.47

Answer: C

Solution:

de Broglie wavelength (λ) is given by $K = qV$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} (\because p = \sqrt{2mK})$$

Substituting the values we get

$$\therefore \frac{\lambda_A}{\lambda_B} = \frac{\sqrt{2m_B q_B V_B}}{\sqrt{2m_A q_A V_A}} = \sqrt{\frac{4m \cdot q \cdot 2500}{m \cdot q \cdot 50}}$$
$$= 2\sqrt{50} = 2 \times 7.07 = 14.14$$

Question139

If the deBroglie wavelength of an electron is equal to 10^{-3} times the wavelength of a photon of frequency 6×10^{14} Hz, then the speed of electron is equal to :

(Speed of light $= 3 \times 10^8$ m/s)

Planck's constant $= 6.63 \times 10^{-34}$ J.s

Mass of electron $= 9.1 \times 10^{-31}$ kg)

[11 Jan. 2019 I]

Options:

A. 1.1×10^6 m/s

B. 1.7×10^6 m/s

C. 1.8×10^6 m/s

D. 1.45×10^6 m/s

Answer: D

Solution:

de-Broglie wavelength,

$$\lambda = \frac{h}{mv} = 10^{-3} \left(\frac{3 \times 10^8}{6 \times 10^{14}} \right) \left[\because \lambda = \frac{c}{\nu} \right]$$

$$v = \frac{6.63 \times 10^{-34} \times 6 \times 10^{14}}{9.1 \times 10^{-31} \times 3 \times 10^5}$$

$$v = 1.45 \times 10^6 \text{ m/s}$$

Question140

In an electron microscope, the resolution that can be achieved is of the order of the wavelength of electrons used. To resolve a width of 7.5×10^{-12} m, the minimum electron energy required is close to: [10 Jan. 2019 I]

Options:

- A. 500 keV
- B. 100 keV
- C. 1 keV
- D. 25 keV

Answer: D

Solution:

Using, $\lambda = \frac{h}{p}$ { given: $\lambda = 7.5 \times 10^{-12}$ }

$$\Rightarrow P = \frac{h}{\lambda}$$

Minimum energy required,

$$K E = \frac{P^2}{2m} = \frac{(h/\lambda)^2}{2m} = \frac{\left\{ \frac{6.6 \times 10^{-34}}{7.5 \times 10^{-12}} \right\}^2}{2 \times 9.1 \times 10^{-31}} \text{ J} = 25 \text{ keV}$$

Question141

When a certain photosensitive surface is illuminated with monochromatic light of frequency ν , the stopping potential for the photo current is $-V_0/2$. When the surface is illuminated by monochromatic light of frequency $\nu/2$, the stopping potential is

– V_0 . The threshold frequency for photoelectric emission is:

[12 Jan. 2019 II]

Options:

A. $\frac{5v}{3}$

B. $\frac{4}{3}v$

C. $2v$

D. (BONUS)

Answer: D

Solution:

(BONUS)

Question142

In a Frank-Hertz experiment, an electron of energy 5.6 eV passes through mercury vapour and emerges with an energy 0.7 eV. The minimum wavelength of photons emitted by mercury atoms is close to :

[12 Jan. 2019 II]

Options:

A. 1700 nm

B. 2020 nm

C. 220 nm

D. 250 nm

Answer: D

Solution:

Using, wavelength, $\lambda = \frac{12375}{\Delta E}$

$$\text{or, } \lambda = \frac{12375}{4.9} \approx 250\text{nm}$$

Question143

In a photoelectric experiment, the wavelength of the light incident on a metal is changed from 300 nm to 400 nm. The decrease in the stopping potential is close to :

$$\left(\frac{hc}{e} = 1240\text{nm} - V \right)$$

[11 Jan. 2019 II]

Options:

A. 0.5 V

B. 1.5 V

C. 1.0 V

D. 2.0 V

Answer: C

Solution:

Let ϕ = work function of the metal,

$$\frac{hc}{\lambda_1} = \phi + eV_1 \dots\dots(i)$$

$$\frac{hc}{\lambda_2} = \phi + eV_2 \dots\dots(ii)$$

Subtracting (ii) from (i) we get

$$hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) = e(V_1 - V_2)$$

$$\Rightarrow V_1 - V_2 = \frac{hc}{e} \left(\frac{\lambda_2 - \lambda_1}{\lambda_1 \cdot \lambda_2} \right)$$

$$\left[\lambda_1 = 300 \text{ nm}, \lambda_2 = 400 \text{ nm}, \frac{hc}{e} = 1240 \text{ nm} - V \right]$$

$$= (1240\text{nm} - v) \left(\frac{100\text{nm}}{300\text{nm} \times 400\text{nm}} \right)$$

$$= 1.03\text{V} \approx 1\text{V}$$

Question144

A metal plate of area $1 \times 10^{-4} \text{ m}^2$ is illuminated by a radiation of intensity $16 \text{ mW} / \text{m}^2$. The work function of the metal is 5 eV . The energy of the incident photons is 10 eV and only 10% of it produces photo electrons. The number of emitted photo electrons per second and their maximum energy, respectively, will be:

$$[1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}]$$

[10 Jan. 2019 II]

Options:

A. 10^{14} and 10 eV

B. 10^{12} and 5 eV

C. 10^{11} and 5 eV

D. 10^{10} and 5 eV

Answer: C

Solution:

using, intensity $I = \frac{nE}{At}$

n = no. of photoelectrons

$$\Rightarrow 16 \times 10^{-3} = \left(\frac{n}{t} \right) \times \frac{10 \times 1.6 \times 10^{-19}}{10^{-4}} \text{ or, } \frac{n}{t} = 10^{12}$$

So, effective number of photoelectrons ejected per unit time $= 10^{12} \times 10/100 = 10^{11}$

Question145

Surface of certain metal is first illuminated with light of wavelength $\lambda_1 = 350 \text{ nm}$ and then, by light of wavelength $\lambda_2 = 540 \text{ nm}$. It is found that the maximum speed of the photo electrons in the two cases differ by a factor of (2) The work function of the metal (in eV) is

close to:

$$(\text{Energy of photon} = \frac{1240}{\lambda(\text{ in nm})} \text{eV})$$

[9 Jan. 2019 I]

Options:

A. 1.8

B. 2.5

C. 5.6

D. 1.4

Answer: A

Solution:

From Einstein's photoelectric equation,

$$\frac{hc}{\lambda_1} = \phi + \frac{1}{2}m(2v)^2 \dots\dots(i)$$

$$\text{and } \frac{hc}{\lambda_2} = \phi + \frac{1}{2}mv^2 \dots\dots(ii)$$

As per question, maximum speed of photoelectrons in two cases differ by a factor 2

From eqn. (i) & (ii)

$$\Rightarrow \frac{hc}{\lambda_1} - \frac{\phi}{hc\lambda_2 - \phi} = 4 \Rightarrow \frac{hc}{\lambda_1} - \phi = \frac{4hc}{\lambda_2} - 4\phi$$

$$\Rightarrow \frac{4hc}{\lambda_2} - \frac{hc}{\lambda_1} = 3\phi \Rightarrow \phi = \frac{1}{3}hc \left(\frac{4}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

$$= \frac{1}{3} \times 1240 \left(\frac{4 \times 350 - 540}{350 \times 540} \right) = 1.8 \text{eV}$$

Question146

The magnetic field associated with a light wave is given at the origin by

$$\mathbf{B} = B_0[\sin(3.14 \times 10^7) ct + \sin(6.28 \times 10^7) ct]$$

If this light falls on a silver plate having a work function of 4.7 eV, what will be the maximum kinetic energy of thephotoelectrons?

$$(c = 3 \times 10^8 \text{ms}^{-1}, h = 6.6 \times 10^{-34} \text{J - s})$$

[9 Jan. 2019 II]

Options:

A. 6.82 eV

B. 12.5 eV

C. 8.52 eV

D. 7.72 eV

Answer: D**Solution:**

According to question, there are two EM waves with different frequency,

$$B_1 = B_0 \sin(\pi \times 10^7 c) t$$

$$\text{and } B_2 = B_0 \sin(2\pi \times 10^7 c) t$$

To get maximum kinetic energy we take the photon with higher frequency using, $B = B_0 \sin \omega t$ and $\omega = 2\pi\nu \Rightarrow \nu = \omega/2\pi$

$$B_1 = B_0 \sin(\pi \times 10^7 c) t \Rightarrow \nu_1 = \frac{10^7}{2} \times c$$

$$B_2 = B_0 \sin(2\pi \times 10^7 c) t \Rightarrow \nu_2 = 10^7 c$$

where c is speed of light $c = 3 \times 10^8 \text{ m/s}$

Clearly, $\nu_2 > \nu_1$

so K E of photoelectron will be maximum for photon of higher energy.

$$\nu_2 = 10^7 \text{ cHz}$$

$$h\nu = \phi + K E_{\text{max}}$$

energy of photon

$$E_{\text{ph}} = h\nu = 6.6 \times 10^{-34} \times 10^7 \times 3 \times 10^9$$

$$E_{\text{ph}} = 6.6 \times 3 \times 10^{-19} \text{ J}$$

$$= \frac{6.6 \times 3 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV} = 12.375 \text{ eV}$$

$$K E_{\text{max}} = E_{\text{ph}} - \phi$$

$$= 12.375 - 4.7 = 7.675 \text{ eV} \approx 7.7 \text{ eV}$$

Question147

A particle ' P ' is formed due to a completely inelastic collision of particles ' x ' and ' y ' having de-Broglie wavelengths ' λ_x ' and ' λ_y ' respectively. If x and y were moving in opposite directions, then the

de-Broglie wavelength of ' P ' is:
[9 Apr. 2019 II]

Options:

A. $\frac{\lambda_x \lambda_y}{\lambda_x + \lambda_y}$

B. $\frac{\lambda_x \lambda_y}{|\lambda_x - \lambda_y|}$

C. $\lambda_x - \lambda_y$

D. $\lambda_x + \lambda_y$

Answer: B

Solution:

$$P_1 - P_2 = (P_1 + P_2) = P$$

$$\text{As } P \propto \frac{1}{\lambda}$$

$$\text{or } \frac{1}{\lambda_x} - \frac{1}{\lambda_y} = \frac{1}{\lambda}$$

$$\text{or } \frac{\lambda_y - \lambda_x}{\lambda_x \lambda_y} = \frac{1}{\lambda}$$

Question148

Two particles move at right angle to each other. Their de Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de Broglie wavelength λ , of the final particle, is given by:

[8 April 2019 I]

Options:

A. $\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

B. $\lambda = \sqrt{\lambda_1 \lambda_2}$

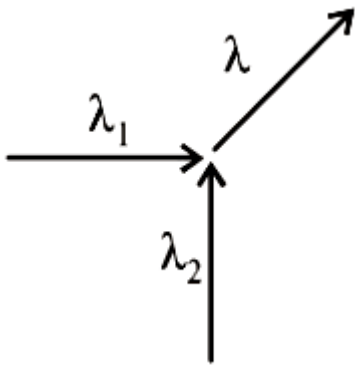
$$C. \lambda = \frac{\lambda_1 + \lambda_2}{2}$$

$$D. \frac{2}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$$

Answer: A

Solution:

From the de-Broglie relation,



$$p_1 = \frac{h}{\lambda_1}$$

$$p_2 = \frac{h}{\lambda_2}$$

Momentum of the final particle (p_f) is given by

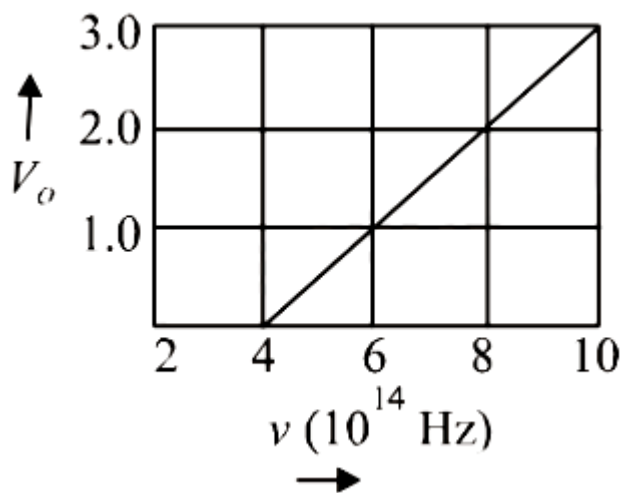
$$\therefore p_f = \sqrt{p_1^2 + p_2^2}$$

$$\Rightarrow \frac{h}{\lambda} = \sqrt{\frac{h^2}{\lambda_1^2} + \frac{h^2}{\lambda_2^2}}$$

$$\Rightarrow \frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$$

Question 149

The stopping potential V_0 (in volt) as a function of frequency (ν) for a sodium emitter, is shown in the figure. The work function of sodium, from the data plotted in the figure, will be: (Given : Planck's constant (h) = 6.63×10^{-34} J s, electron charge $e = 1.6 \times 10^{-19}$ C)



[12 Apr. 2019 I]

Options:

A. 1.82 eV

B. 1.66 eV

C. 1.95 eV

D. 2.12 eV

Answer: B

Solution:

$$f_0 = 4 \times 10^{14} \text{ Hz}$$

$$W_0 = hf_0 = 6.63 \times 10^{-34} \times (4 \times 10^{14}) \text{ J}$$

$$= \frac{(6.63 \times 10^{-34}) \times (4 \times 10^{14})}{1.6 \times 10^{-19}}$$

$$= 1.66 \text{ eV}$$

Question150

In a photoelectric effect experiment the threshold wavelength of light is 380 nm. If the wavelength of incident light is 260 nm, the maximum kinetic energy of emitted electrons will be:

Given E (in eV) =

[10 Apr. 2019 I]

Options:

- A. 1.5 eV
- B. 3.0 eV
- C. 4.5 eV
- D. 15.1 eV

Answer: A

Solution:

$$\begin{aligned}
 K E_{\max} &= E - \phi_0 \\
 (\text{where } E &= \text{energy of incident light } \phi_0 = \text{work function}) \\
 &= \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \\
 &= 1237 \left[\frac{1}{260} - \frac{1}{380} \right] \\
 &= \frac{1237 \times 120}{380 \times 260} = 1.5 \text{ eV}
 \end{aligned}$$

Question151

A 2mW laser operates at a wavelength of 500nm. The number of photons that will be emitted per second is :

[Given Planck's constant $h = 6.6 \times 10^{-34} \text{ J s}$, speed of light $c = 3.0 \times 10^8 \text{ m/s}$]
[10 Apr. 2019 II]

Options:

- A. 5×10^{15}
- B. 1.5×10^{16}
- C. 2×10^{16}
- D. 1×10^{16}

Answer: A

Solution:

Energy of photon (E) is given by

$$E = \frac{hc}{\lambda}$$

Number of photons of wavelength λ emitted in t second from laser of power P is given by

$$n = \frac{Pt\lambda}{hc}$$

$$\Rightarrow n = \frac{2 \times \lambda}{hc} = \frac{2 \times 10^{-3} \times 5 \times 10^{-7}}{2 \times 10^{-25}} (\because t = 1\text{S})$$

$$\Rightarrow n = 5 \times 10^{15}$$

Question 152

The electric field of light wave is given as

$$\vec{E} = 10^3 \cos \left(\frac{2\pi x}{5 \times 10^{-7}} - 2\pi \times 6 \times 10^{14} t \right) \frac{\text{N}}{\text{C}}$$

This light falls on a metal plate of work function 2eV . The stopping potential of the photo-electrons is:

$$\text{Given, } E (\text{ in eV }) = \frac{12375}{\lambda (\text{ in } \text{\AA})}$$

[9 April 2019 I]

Options:

A. 2.0 V

B. 0.72 V

C. 0.48 V

D. 2.48 V

Answer: C

Solution:

$$\text{Here } w = 2\pi \times 6 \times 10^{14} \text{ or } f = 6 \times 10^{14} \text{ Hz}$$

$$\text{Wavelength } \lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^{14}} = 0.5 \times 10^{-6} \text{ m} = 5000 \text{\AA}$$

$$\text{Now } E = \frac{12374}{5000} = 2.48 \text{ eV}$$

$$\text{Using } E = w + eV$$

$$2.48 = 2 + eV_s \text{ or } V_s = 0.48 \text{ V}$$

Question153

Two electrons are moving with non-relativistic speeds perpendicular to each other. If corresponding de Broglie wavelengths are λ_1 and λ_2 , their de Broglie wavelength in the frame of reference attached to their centre of mass is:

[Online April 15, 2018]

Options:

A. $\lambda_{\text{CM}} = \lambda_1 = \lambda_2$

B. $\frac{1}{\lambda_1} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

C. $\lambda_{\text{CM}} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$

D. $\lambda_{\text{CM}} = \left(\frac{\lambda_1 + \lambda_2}{2} \right)$

Answer: C

Solution:

Momentum (p) of each electron $\frac{h}{\lambda_1} \hat{i}$ and $\frac{h}{\lambda_2} \hat{j}$

Velocity of centre of mass

$$V_{\text{cm}} = \frac{h}{2m\lambda_1} \hat{i} + \frac{h}{2m\lambda_2} \hat{j} (\because p = mv)$$

Velocity of 1 st particle about centre of mass

$$V_{1\text{cm}} = \frac{h}{2m\lambda_1} \hat{i} - \frac{h}{2m\lambda_2} \hat{j}$$

$$\lambda_{\text{cm}} = \frac{h}{\sqrt{\frac{h^2}{4\lambda_1^2} + \frac{h^2}{4\lambda_2^2}}} = \frac{2\lambda_1\lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}} \left(\because \lambda = \frac{h}{p} \right)$$

Question154

If the de Broglie wavelengths associated with a proton and an α -particle are equal, then the ratio of velocities of the proton and the α -particle will be:

[Online April 15, 2018]

Options:

A. 1 : 4

B. 1 : 2

C. 4 : 1

D. 2 : 1

Answer: C

Solution:

According to question, $\lambda_p = \lambda_\alpha$

Using, $\lambda = \frac{h}{p} = \frac{h}{mv}$

So, $\frac{h}{m_p \times v_p} = \frac{h}{m_\alpha \times v_\alpha}$

$$\Rightarrow \frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p} = \frac{4m_p}{m_p}$$

($\because$ mass of α -particle is 4 times of mass of proton)

$$\text{So, } \frac{v_p}{v_\alpha} = \frac{4}{1}; \text{ i.e., } 4 : 1$$

Question155

A particle A of mass m and initial velocity v collides with a particle B of mass $\frac{m}{2}$ which is at rest. The collision is head on, and elastic. The ratio of the de-Broglie wavelengths λ_A to λ_B after the collision is [2017]

Options:

A. $\frac{\lambda_A}{\lambda_B} = \frac{2}{3}$

$$\text{B. } \frac{\lambda_A}{\lambda_B} = \frac{1}{2}$$

$$\text{C. } \frac{\lambda_A}{\lambda_B} = \frac{1}{3}$$

$$\text{D. } \frac{\lambda_A}{\lambda_B} = 2$$

Answer: D

Solution:

From question, $m_A = M$; $m_B = \frac{m}{2}$

$$u_A = V \quad u_B = 0$$

Let after collision velocity of A = V_1 and velocity of B = V_2

Applying law of conservation of momentum,

$$mu = mv_1 + \left(\frac{m}{2}\right)v_2$$

$$\text{or, } 24 = 2v_1 + v_2 \dots\dots(i)$$

By law of collision

$$e = \frac{v_2 - v_1}{u - 0}$$

$$\text{or, } u = v_2 - v_1 \dots\dots(ii)$$

[$\because$ collision is elastic, $e = 1$]

using eqns (i) and (ii)

$$v_1 = \frac{4}{3} \text{ and } v_2 = \frac{4}{3}u$$

$$\text{de-Broglie wavelength } \lambda = \frac{h}{p}$$

$$\therefore \frac{\lambda_A}{\lambda_B} = \frac{P_B}{P_A} = \frac{\frac{m}{2} \times \frac{4}{3}u}{m \times \frac{4}{3}} = 2$$

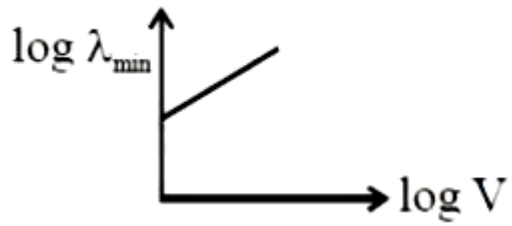
Question156

An electron beam is accelerated by a potential difference V to hit a metallic target to produce X-rays. It produces continuous as well as characteristic X-rays. If $\lambda_{\min}$ is the smallest possible wavelength of X-ray in the spectrum, the variation of $\log \lambda_{\min}$ with $\log V$ is correctly represented in :

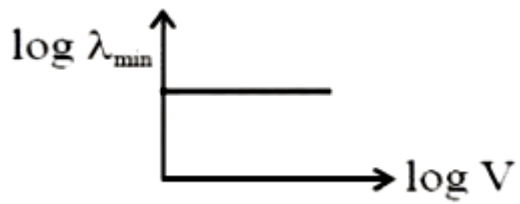
[2017]

Options:

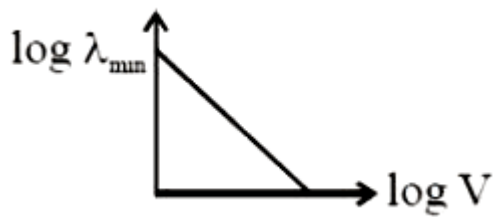
A.



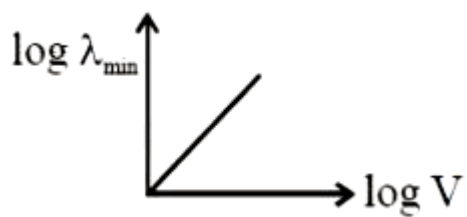
B.



C.



D.



Answer: C

Solution:

In X-ray tube, $\lambda_{\min} = \frac{hc}{eV}$

$$\ln \lambda_{\min} = \ln \left(\frac{hc}{e} \right) - \ln V$$

Clearly, $\log \lambda_{\min}$ versus $\log V$ graph

slope is negative hence option (c) correctly depicts.

Question157

A Laser light of wavelength 660nm is used to weld Retina detachment. If a Laser pulse of width 60ms and power 0.5 kW is used the approximate number of photons in the pulse are : [Take Planck's constant $h = 6.62 \times 10^{-34} \text{ J s}$]
[Online April 9, 2017]

Options:

A. 10^{20}

B. 10^{18}

C. 10^{22}

D. 10^{19}

Answer: A

Solution:

Given, $\lambda = 660\text{nm}$, Power = 0.5kW , t = 60ms

$$\begin{aligned}\text{Power } P &= \frac{nhc}{\lambda t} \Rightarrow n = \frac{P\lambda}{hc} \\ &= 0.5 \times 10^3 \times \frac{660 \times 10^{-9} \times 60 \times 10^{-3}}{6.6 \times 10^{-34} \times 3 \times 10^8} \\ &= 100 \times 10^{18} = 10^{20}\end{aligned}$$

Question158

The maximum velocity of the photoelectrons emitted from the surface is v when light of frequency n falls on a metal surface. If the incident frequency is increased to 3n, the maximum velocity of the ejected photoelectrons will be :
[Online April 8, 2017]

Options:

A. less than $\sqrt{3}v$

B. v

C. more than $\sqrt{3}v$

D. equal to $\sqrt{3}v$

Answer: C

Solution:

As the metal surface is same, work function (ϕ) is same for both the case.

Initially $K E_{\max} = nh - \phi \dots\dots(i)$

After increase $K E_{\max} = 3nh - \phi \dots\dots(ii)$

For work function ϕ – not to be – ve or zero, $v' > \sqrt{3}v$

Question159

Radiation of wavelength λ , is incident on a photocell. The fastest emitted electron has speed v . If the wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will be:

[2016]

Options:

A. $= v \left(\frac{4}{3} \right)^{\frac{1}{2}}$

B. $= v \left(\frac{3}{4} \right)^{\frac{1}{2}}$

C. $> v \left(\frac{4}{3} \right)^{\frac{1}{2}}$

D. $< v \left(\frac{4}{3} \right)^{\frac{1}{2}}$

Answer: C

Solution:

$$\begin{aligned}h\frac{c}{\lambda} - hv_0 &= \frac{1}{2}mv^2 \\ \therefore \frac{4hc}{3\lambda} - hv_0 &= \frac{1}{2}mv'^2 \\ \therefore \frac{v'^2}{v^2} &= \frac{\frac{4}{3}v - v_0}{v - v_0} \therefore v' = v \sqrt{\frac{\frac{4}{3}v - v_0}{v - v_0}} \\ \therefore v' &> v \sqrt{\frac{4}{3}}\end{aligned}$$

Question160

A photoelectric surface is illuminated successively by monochromatic light of wavelengths λ and $\frac{\lambda}{2}$. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface is :
[Online April 10,2016]

Options:

- A. $\frac{hc}{2\lambda}$
- B. $\frac{hc}{\lambda}$
- C. $\frac{hc}{3\lambda}$
- D. $\frac{3hc}{\lambda}$

Answer: A

Solution:

From Einstein's photoelectric equation

$$K.E._{\lambda} = \frac{hc}{\lambda} - \phi \dots\dots(i)$$

(for monochromatic light of wavelength λ) where ϕ is work function

$$K.E. \frac{\lambda}{2} = \frac{hc}{\lambda/2} - \phi \dots\dots(ii)$$

(for monochromatic light of wavelength $\lambda/2$)

From question,

$$K.E. \frac{\lambda}{2} = 3(K.E. \lambda) \Rightarrow \frac{hc}{\lambda/2} - \phi = 3 \left(\frac{hc}{\lambda} - \phi \right)$$

$$\frac{2hc}{\lambda} - \phi = 3 \frac{hc}{\lambda} - 3\phi$$

$$\Rightarrow 2\phi = \frac{hc}{\lambda} \therefore \phi = \frac{hc}{2\lambda}$$

Question161

When photons of wavelength λ_1 are incident on an isolated sphere, the corresponding stopping potential is found to be V . When photons of wavelength λ_2 are used, the corresponding stopping potential was thrice that of the above value. If light of wavelength λ_3 is used then find the stopping potential for this case :
[Online April 9,2016]

Options:

A. $\frac{hc}{e} \left[\frac{1}{\lambda_3} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right]$

B. $\frac{hc}{e} \left[\frac{1}{\lambda_3} + \frac{1}{2\lambda_2} - \frac{1}{\lambda_1} \right]$

C. $\frac{hc}{e} \left[\frac{1}{\lambda_3} - \frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right]$

D. (None)

Answer: D

Solution:

From Einstein's photoelectric equation, we have

$$\frac{hc}{\lambda_1} = \frac{hc}{\lambda_0} + eV \dots\dots(i)$$

$$\frac{hc}{\lambda_2} = \frac{hc}{\lambda_0} + eV \dots\dots(ii)$$

$$\frac{hc}{\lambda_3} = \frac{hc}{\lambda_0} + 3eV' \dots\dots(iii)$$

From equation (1)&(2)

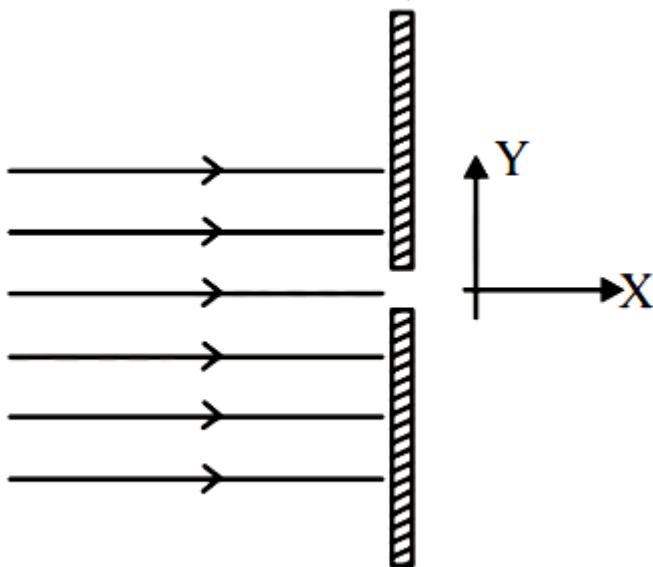
$$\frac{3}{2\lambda_1} - \frac{2}{2\lambda_2} = \frac{1}{\lambda_0}$$

$$\frac{hc}{\lambda_1} - hc \left[\frac{3}{2\lambda_1} - \frac{1}{2\lambda_2} \right] = eV'$$

$$\frac{hc}{e} \left[\frac{1}{\lambda_3} - \frac{3}{2\lambda_1} + \frac{1}{2\lambda_2} \right] = V'$$

Question162

A parallel beam of electrons travelling in x-direction falls on a slit of width d (see figure). If after passing the slit, an electron acquires momentum p_y in the y-direction then for a majority of electrons passing through the slit (h is Planck's constant) :



[Online April 10, 2015]

Options:

A. $|P_y|d > h$

B. $|P_y|d < h$

C. $|P_y|d \simeq h$

D. $|P_y|d >> h$

Answer: A

Solution:

From Bragg's equation

$$d \sin \theta = \lambda$$

$$\sin \theta = \frac{\lambda}{d} < 1 \therefore \lambda < d$$

$$\frac{h}{|p_y|} < d \left[\because \lambda = \frac{h}{|p_y|} \right]$$

$$\therefore h < |p_y| d$$

Question163

de-Broglie wavelength of an electron accelerated by a voltage of 50V is close to ($|e| = 1.6 \times 10^{-19} \text{C}$, $m_e = 9.1 \times 10^{-31} \text{kg}$ $h = 6.6 \times 10^{-34} \text{J s}$):
[Online April 10, 2015]

Options:

A. 2.4 Å

B. 0.5 Å

C. 1.7 Å

D. 1.2 Å

Answer: C

Solution:

de-Broglie wavelength,

$$\lambda = \frac{h}{P} = \frac{h}{mv} = \frac{h}{\sqrt{2mqV}}$$

$$\begin{aligned} \text{or, } \lambda &= \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 50}} \\ &= 1.7 \text{Å} \end{aligned}$$

Question164

For which of the following particles will it be most difficult to experimentally verify the de-Broglie relationship?

[Online April 9, 2015]

Options:

- A. an electron
- B. a proton
- C. an α -particle
- D. a dust particle

Answer: D

Solution:

Among the given particles most difficult to experimentally verify the de-broglie relationship is for a dust particle.

Question165

Match List - I (Fundamental Experiment) with List - II (its conclusion) and select the correct option from the choices given below the list:

List - I	List - II
A . Franck-Hertz Experiment	(i) Particle nature of light
B. Photo-electric experiment	(ii) Discrete energy levels of atom
C. Davis on-Germer experiment	(iii) Wave nature of electron
	(iv) Structure of atom

[2015]

Options:

- A. (A)-(ii); (B)-(i); (C)-(iii)
- B. (A)-(iv); (B)-(iii); (C)-(ii)

C. (A)-(i); (B)-(iv); (C)-(iii)

D. (A)-(ii); (B)-(iv); (C)-(iii)

Answer: A

Solution:

Frank-Hertz experiment - Discrete energy levels of atom, Photoelectric effect - Particle nature of light.

Davison - Germer experiment - wave nature of electron.

Question166

A beam of light has two wavelengths of $4972\text{\AA}$ and $6216\text{\AA}$ with a total intensity of $3.6 \times 10^{-3} \text{W m}^{-2}$ equally distributed among the two wavelengths. The beam falls normally on an area of 1cm^2 of a clean metallic surface of work function 2.3eV . Assume that there is no loss of light by reflection and that each capable photon ejects one electron. The number of photoelectrons liberated in 2 s is approximately:

[Online April 12,2014]

Options:

A. 6×10^{11}

B. 9×10^{11}

C. 11×10^{11}

D. 15×10^{11}

Answer: B

Solution:

Given, $\lambda_1 = 4972\text{\AA}$

and $\lambda_2 = 6216\text{\AA}$

and $I = 3.6 \times 10^{-3} \text{W m}^{-2}$

Intensity associated with each wavelength

$$= \frac{3.6 \times 10^{-3}}{2}$$

$$= 1.8 \times 10^{-3} \text{ W m}^{-2}$$

work function $\phi = h\nu$

$$= \frac{hc}{\lambda}$$

$$= \frac{(6.62 \times 10^{-34})(3 \times 10^8)}{\lambda}$$

$$= \frac{12.4 \times 10^3}{\lambda} \text{ eV}$$

for different wavelengths

$$\phi_1 = \frac{12.4 \times 10^3}{\lambda_1} = \frac{12.4 \times 10^3}{4972} = 2.493 \text{ eV} = 3.984 \times 10^{-19} \text{ J}$$

$$\phi_2 = \frac{12.4 \times 10^3}{\lambda_2} = \frac{12.4 \times 10^3}{6216} = 1.994 \text{ eV} = 3.184 \times 10^{-19} \text{ J}$$

Work function for metallic surface $\phi = 2.3 \text{ eV}$ (given)

$$\phi_2 < \phi$$

Therefore, ϕ_2 will not contribute in this process.

Now, no. of electrons per $\text{m}^2 - \text{s} = \text{no. of photons per } \text{m}^2 - \text{s}$

$$\text{no. of electrons per } \text{m}^2 - \text{s} = \frac{1.8 \times 10^{-3}}{3.984 \times 10^{-19}} \times 10^{-4}$$

$$(\because 1 \text{ cm}^2 = 10^{-4} \text{ m}^2) = 0.45 \times 10^{12}$$

So, the number of photo electrons liberated in 2 sec.

$$= 0.45 \times 10^{12} \times 2$$

$$= 9 \times 10^{11}$$

Question 167

A photon of wavelength λ is scattered from an electron, which was at rest. The wavelength shift $\Delta\lambda$ is three times of λ and the angle of scattering θ is 60° . The angle at which the electron recoiled is ϕ . The value of $\tan \phi$ is : (electron speed is much smaller than the speed of light)

[Online April 11, 2014]

Options:

A. 0.16

B. 0.22

C. 0.25

D. 0.28

Answer: B

Question168

Electrons are accelerated through a potential difference V and protons are accelerated through a potential difference $4V$. The de-Broglie wavelengths are λ_e and λ_p for electrons and protons respectively. The ratio of $\frac{\lambda_e}{\lambda_p}$ is given by : (given m_e is mass of electron and m_p is mass of proton).

[Online April 23, 2013]

Options:

A. $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$

B. $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_e}{m_p}}$

C. $\frac{\lambda_e}{\lambda_p} = \frac{1}{2} \sqrt{\frac{m_e}{m_p}}$

D. $\frac{\lambda_e}{\lambda_p} = 2 \sqrt{\frac{m_p}{m_e}}$

Answer: D

Solution:

Energy in joule (E) = charge $\times$ potential diff. in volt

$$E_{\text{electron}} = q_e V \text{ and } E_{\text{proton}} = q_p 4V$$

$$\text{de-Broglie wavelength } \lambda = \frac{h}{P} = \frac{h}{\sqrt{2mE}}$$

$$\lambda_e = \frac{h}{\sqrt{2m_e eV}} \text{ and } \lambda_p = \frac{h}{\sqrt{2m_p e4V}} (\because q_e = q_p)$$

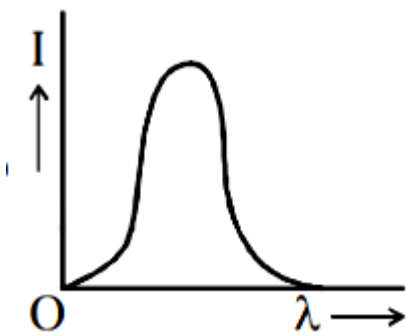
$$\therefore \frac{\lambda_e}{\lambda_p} = \frac{\frac{h}{\sqrt{2m_e eV}}}{\frac{h}{\sqrt{2m_p e4V}}} = \sqrt{\frac{2m_p e4V}{2m_e eV}} = 2 \sqrt{\frac{m_p}{m_e}}$$

Question169

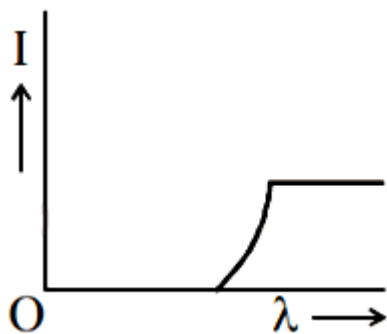
The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows :
[2013]

Options:

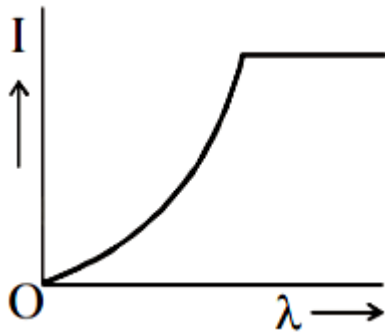
A.



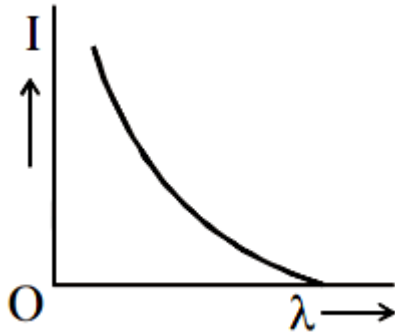
B.



C.



D.



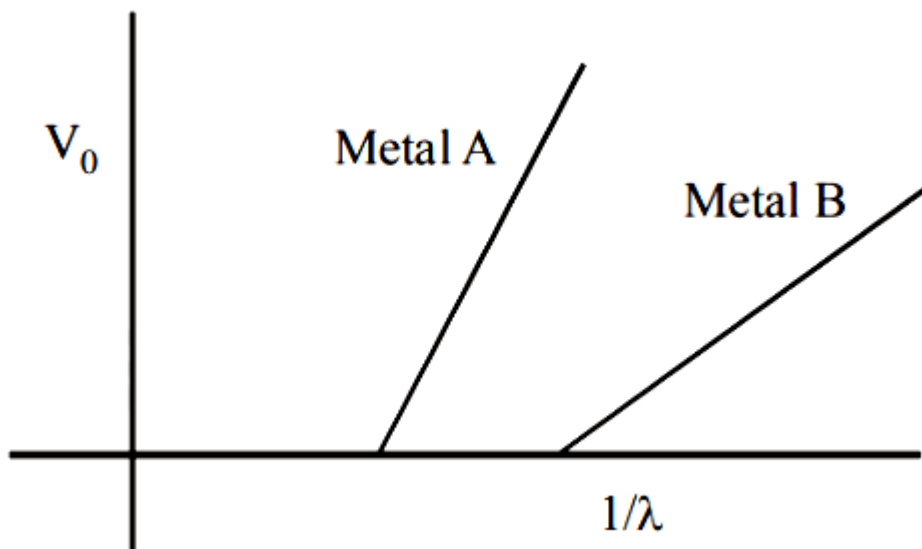
Answer: D

Solution:

As λ is increased, there will be a value of λ above which photoelectrons will cease to come out so photocurrent will become zero. Hence (d) is correct answer.

Question170

In an experiment on photoelectric effect, a student plots stopping potential V_0 against reciprocal of the wavelength λ of the incident light for two different metals A and B These are shown in the figure.



Looking at the graphs, you can most appropriately say that:
[Online April 25, 2013]

Options:

- A. Work function of metal B is greater than that of metal A
- B. For light of certain wavelength falling on both metal, maximum kinetic energy of electrons emitted from A will be greater than those emitted from B.
- C. Work function of metal A is greater than that of metal B
- D. Students data is not correct

Answer: D

Solution:

$$\frac{hc}{\lambda} - \phi = eV_0$$

$$V_0 = \frac{hc}{e\lambda} - \frac{\phi}{e}$$

For metal A

$$\frac{\phi_A}{hc} = \frac{1}{\lambda}$$

For metal B

$$\frac{\phi_B}{hc} = \frac{1}{\lambda}$$

As the value of $\frac{1}{\lambda}$ (increasing and decreasing) is not specified hence we cannot say that which metal has comparatively greater or lesser work function (ϕ).

A copper ball of radius 1 cm and work function 4.47eV is irradiated with ultraviolet radiation of wavelength 2500 Å. The effect of irradiation results in the emission of electrons from the ball. Further the ball will acquire charge and due to this there will be a finite value of the potential on the ball. The charge acquired by the ball is : [Online April 25, 2013]

Options:

A. $5.5 \times 10^{-13}\text{C}$

B. $7.5 \times 10^{-13}\text{C}$

C. $4.5 \times 10^{-12}\text{C}$

D. $2.5 \times 10^{-11}\text{C}$

Answer: A

Question172

This equation has statement 1 and statement 2. Of the four choices given after the statements, choose the one that describes the two statements.

Statement 1: Davisson-Germer experiment established the wave nature of electrons.

Statement 2 : If electrons have wave nature, they can interfere and show diffraction.

[2012]

Options:

A. Statement 1 is false, Statement 2 is true.

B. Statement 1 is true, Statement 2 is false

C. Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of statement 1

D. Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1

Answer: A

Solution:

Davisson Germer experiment showed that electron beams can undergo diffraction when passed through atomic crystal. This established wave nature of electron as waves can exhibit interference and diffraction.

Question173

Photoelectrons are ejected from a metal when light of frequency ν falls on it. Pick out the wrong statement from the following.
[Online May 26, 2012]

Options:

A. No electrons are emitted if ν is less than W/h , where W is the work function of the metal

B. The ejection of the photoelectrons is instantaneous.

C. The maximum energy of the photoelectrons is $h\nu$.

D. The maximum energy of the photoelectrons is independent of the intensity of the light.

Answer: C

Solution:

According to photo-electric equation:

$$K.E_{\max} = h\nu - h\nu_0 \text{ (Work function)}$$

Some sort of energy is used in ejecting the photoelectrons.

Question174

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement 1: A metallic surface is irradiated by a monochromatic light of frequency $\nu > \nu_0$ (the threshold frequency). If the incident frequency is now doubled, the photocurrent and the maximum kinetic energy are also doubled.

Statement 2: The maximum kinetic energy of photoelectrons emitted from a surface is linearly dependent on the frequency of the incident light. The photocurrent depends only on the intensity of the incident light.

[Online May 19, 2012]

Options:

- A. Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
- B. Statement 1 is false, Statement 2 is true.
- C. Statement 1 is true, Statement 2 is false.
- D. Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.

Answer: B

Solution:

The maximum kinetic energy of photoelectrons depends upon frequency on incident light and photo current depends upon intensity of incident light.

Question175

This question has Statement -1 and Statement -2. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement -1 : A metallic surface is irradiated by a monochromatic light of frequency $\nu > \nu_0$ (the threshold frequency). The maximum

kinetic energy and the stopping potential are $K_{\max}$ and V_0 respectively. If the frequency incident on the surface is doubled, both the $K_{\max}$ and V_0 are also doubled.

Statement -2 : The maximum kinetic energy and the stopping potential of photoelectrons emitted from a surface are linearly dependent on the frequency of incident light.

[2011]

Options:

- A. Statement-1 is true, Statement-2 is true, Statement - 2 is the correct explanation of Statement - 1.
- B. Statement-1 is true, Statement-2 is true, Statement - 2 is not the correct explanation of Statement - 1.
- C. Statement - 1 is false, Statement - 2 is true.
- D. Statement - 1 is true, Statement - 2 is false

Answer: C

Solution:

By Einstein photoelectric equation,

$$K_{\max} = eV_0 = h\nu - h\nu_0$$

When ν is doubled, $K_{\max}$ and V_0 become more than double.

Question176

Statement -1 : When ultraviolet light is incident on a photocell, its stopping potential is V_0 and the maximum kinetic energy of the photoelectrons is $K_{\max}$. When the ultraviolet light is replaced by X - rays, both V_0 and K_m increase.

Statement -2 : Photoelectrons are emitted with speeds ranging from zero to a maximum value because of the range of frequencies present in the incident light.

[2010]

Options:

- A. Statement -1 is true, Statement -2 is true ; Statement -2 is the correct explanation of Statement -1
- B. Statement -1 is true, Statement -2 is true; Statement -2 is not the correct explanation of Statement -1
- C. Statement -1 is false, Statement -2 is true.
- D. Statement -1 is true, Statement -2 is false.

Answer: D

Solution:

We know that

$$eV_0 = K_{\max} = h\nu - \phi$$

where, ϕ is the work function.

X-rays have higher frequency (ν) than ultraviolet rays. Therefore as ν increases $K_{\max}$, E and V_0 both increases.

The kinetic energy ranges from zero to maximum because of loss of energy due to subsequent collisions before getting ejected.

Question177

The surface of a metal is illuminted with the light of 400 nm. The kinetic energy of the ejected photoelectrons was found to be 1.68 eV. The work function of the metal is :

($hc = 1240 \text{ eV.nm}$)

[2009]

Options:

- A. 1.41 eV
- B. 1.51 eV
- C. 1.68 eV
- D. 3.09 eV

Answer: A

Solution:

Wavelength of incident light, $\lambda = 400\text{nm}$ $hc = 1240 \text{ eV.nm}$

$$K.E = 1.68\text{eV}$$

Using Einstein's photoelectric equation

$$\frac{hc}{\lambda} - W = K.E$$

$$\Rightarrow W = \frac{hc}{\lambda} - K.E$$

$$\Rightarrow W = \frac{1240}{400} - 1.68$$

$$= 3.1 - 1.68$$

$$= 1.41\text{eV}$$

Question178

Electrons accelerated by potential V are diffracted from a crystal. If $d = 1\text{\AA}$ and $i = 30^\circ$, V should be about

($h = 6.6 \times 10^{-34} \text{ J s}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$)

[2008]

Options:

A. 2000 V

B. 50 V

C. 500 V

D. 1000 V

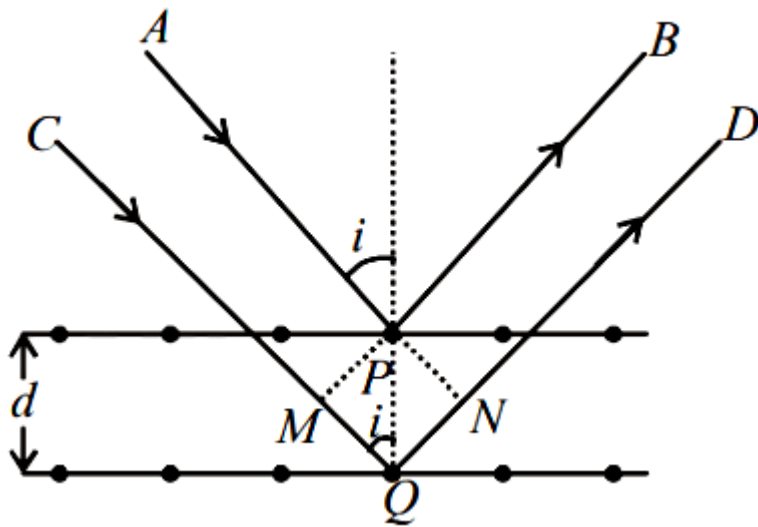
Answer: B

Solution:

The path difference between the rays APB and CQD is

$$\Delta x = MQ + QN = d \cos i + d \cos i$$

$$\Delta x = 2d \cos i$$



For constructive interference the path difference is integral multiple of wavelength

$$\therefore n\lambda = 2d \cos i$$

From de-broglie concept

Wavelength,

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK.E}} = \frac{h}{\sqrt{2meV}}$$

$$\therefore \frac{nh}{\sqrt{2meV}} = 2d \cos i$$

Squaring both side

$$\frac{n^2 h^2}{2meV} = 4d^2 \cos^2 i$$

For first order interference $n = 1$

$$\begin{aligned} \therefore V &= \frac{h^2}{8me d^2 \cos^2 i} \\ &= \frac{(6.6 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times (10^{-10})^2 \times \cos^2 30} \\ &= 50V \end{aligned}$$

Question179

If a strong diffraction peak is observed when electrons are incident at an angle i' from the normal to the crystal planes with distance ' d ' between them (see figure), de Broglie wavelength λ_{dB} of electrons can be calculated by the relationship (n is an integer) [2008]

Options:

A. $d \sin i = n\lambda_{dB}$

B. $2d \cos i = n\lambda_{dB}$

C. $2d \sin i = n\lambda_{dB}$

D. $d \cos i = n\lambda_{dB}$

Answer: B

Solution:

For constructive interference,
 $2d \cos i = n\lambda_{dB}$

Question180

Photon of frequency ν has a momentum associated with it. If c is the velocity of light, the momentum is [2007]

Options:

A. $h\nu/c$

B. ν/c

C. $h\nu c$

D. $h\nu/c^2$

Answer: A

Solution:

Energy of a photon of frequency ν is given by $E = h\nu$

Also, $E = mc^2$, $mc^2 = h\nu$

$$\Rightarrow mc = \frac{h\nu}{c} \Rightarrow p = \frac{h\nu}{c}$$

Question181

The threshold frequency for a metallic surface corresponds to an energy of 6.2 eV and the stopping potential for a radiation incident on this surface is 5 V. The incident radiation lies in [2006]

Options:

- A. ultra-violet region
- B. infra-red region
- C. visible region
- D. X-ray region

Answer: A

Solution:

Work function, $\phi = 6.2\text{eV} = 6.2 \times 1.6 \times 10^{-19}\text{J}$

Stopping potential, $V = 5\text{ volt}$

From the Einstein's photoelectric equation

$$\frac{hc}{\lambda} - \phi = eV_0$$

$$\Rightarrow \lambda = \frac{hc}{\phi + eV_0}$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19}(6.2 + 5)} \approx 10^{-7}\text{m}$$

This range lies in ultra violet range.

Question182

The time taken by a photoelectron to come out after the photon strikes is approximately [2006]

Options:

- A. 10^{-4}s
- B. 10^{-10}s
- C. 10^{-16}s

D. 10^{-1} s

Answer: B

Solution:

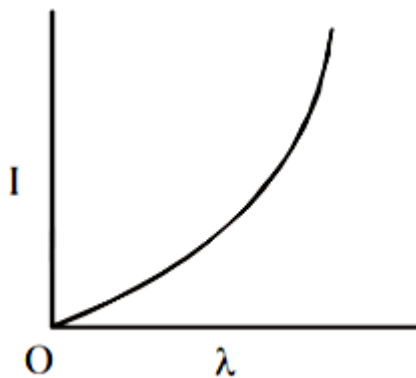
The photoelectric emission is an instantaneous process without any apparent time lag. It is known that emission starts in the time of the order of 10^{-9} second. So, the approximate time taken by a photoelectron to come out after the photon strikes is 10^{-10} second.

Question183

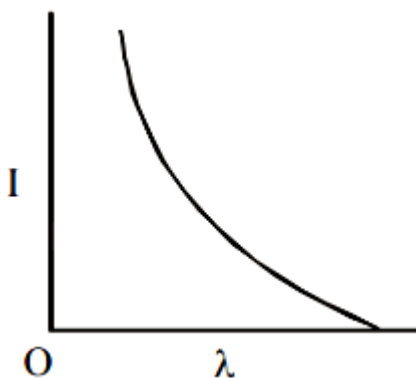
The anode voltage of a photocell is kept fixed. The wavelength λ of the light falling on the cathode is gradually changed. The plate current I of the photocell varies as follows [2006]

Options:

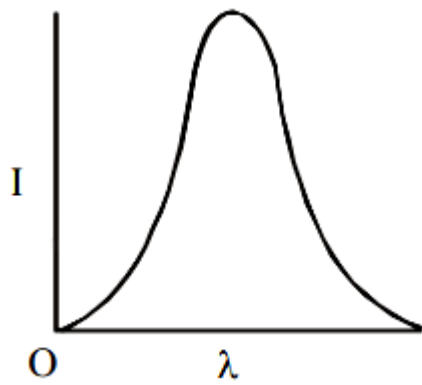
A.



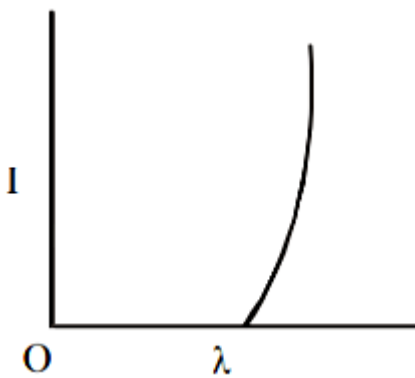
B.



C.



D.



Answer: B

Solution:

As λ decreases, y increases and hence the speed of photoelectron increases. The chances of photo electron to meet the anode increases and hence photo electric current increases.

Question184

If the kinetic energy of a free electron doubles, it's de-Broglie wavelength changes by the factor [2005]

Options:

A. 2

B. $\frac{1}{2}$

C. $\sqrt{2}$

D. $\frac{1}{\sqrt{2}}$

Answer: D

Solution:

de-Broglie wavelength,

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\text{but } K.E = \frac{1}{2}mv^2$$

$$\Rightarrow K.E = \frac{(mv)^2}{2m}$$

$$\Rightarrow mv = \sqrt{2mK.E}$$

$$\lambda = \frac{h}{\sqrt{2mK.E}}$$

$$\therefore \lambda \propto \frac{1}{\sqrt{K.E}}$$

So, if $K.E$ is doubled, wavelength becomes $\frac{\lambda}{\sqrt{2}}$

Question 185

A photocell is illuminated by a small bright source placed 1m away. When the same source of light is placed $\frac{1}{2}$ m away, the number of electrons emitted by photocathode would [2005]

Options:

A. increase by a factor of 4

B. decrease by a factor of 4

C. increase by a factor of 2

D. decrease by a factor of 2

Answer: A

Solution:

$$I \propto \frac{I_1}{r^2}; \frac{I_1}{I_2} = \left(\frac{r_2}{r_1} \right)^2 = \frac{1}{4}$$

$$I_2 \rightarrow 4 \times I_1$$

When intensity becomes 4 times, no. of photoelectrons emitted would increase by 4 times, since number of electrons emitted per second is directly proportional to intensity.

Question186

A radiation of energy E falls normally on a perfectly reflecting surface. The momentum transferred to the surface is [2004]

Options:

A. Ec

B. $2E / c$

C. E / c

D. E / c^2

Answer: B

Solution:

Momentum of photon of energy E is $= \frac{E}{c}$

When a photon hits a perfectly reflecting surface, it reflects back in opposite direction with same energy and momentum.

$$\therefore \text{Change in momentum} = \frac{E}{C} - \left(\frac{-E}{C} \right) = \frac{2E}{C}$$

This is equal to momentum transferred to the surface.

Question187

According to Einstein's photoelectric equation, the plot of the kinetic energy of the emitted photo electrons from a metal Versus the frequency, of the incident radiation gives a straight line whose

slope
[2004]

Options:

- A. depends both on the intensity of the radiation and the metal used
- B. depends on the intensity of the radiation
- C. depends on the nature of the metal used
- D. is the same for the all metals and independent of the intensity of the radiation

Answer: D

Solution:

From the Einstein photoelectric equation $K.E. = h\nu - \phi$

Here, ϕ = work function of metal

h = Planck's constant

slope of graph of $K.E.$ & ν is h (Planck's constant) which is same for all metals.

Question188

The work function of a substance is 4.0 eV. The longest wavelength of light that can cause photoelectron emission from this substance is approximately
[2004]

Options:

- A. 310 nm
- B. 400 nm
- C. 540 nm
- D. 220 nm

Answer: A

Solution:

Work function of metal (ϕ) is given by

$$\phi = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda = \frac{hc}{\phi}$$

$$\Rightarrow \lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{4 \times 1.6 \times 10^{-19}} = 310 \text{nm}$$

Question 189

Two identical photocathodes receive light of frequencies f_1 and f_2 . If the velocities of the photo electrons (of mass m) coming out are respectively v_1 and v_2 , then

[2003]

Options:

A. $v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

B. $v_1 + v_2 = \left[\frac{2h}{m}(f_1 + f_2) \right]^{1/2}$

C. $v_1^2 + v_2^2 = \frac{2h}{m}(f_1 + f_2)$

D. $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2) \right]^{1/2}$

Answer: A

Solution:

Let work function be W and v_1 and v_2 be the velocity of electrons for frequencies f_1 and f_2 .

Using Einstein's photo electric equation for one photodiode, we get

$$hf_1 - W = \frac{1}{2}mv_1^2 \dots\dots(i)$$

Using Einstein's photo electric equation for another photodiode we get

$$hf_2 - W = \frac{1}{2}mv_2^2 \dots\dots(ii)$$

Subtracting (ii) from (i) we get

$$(hf_1 - W) - (hf_2 - W) = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2$$

$$\therefore h(f_1 - f_2) = \frac{m}{2}(v_1^2 - v_2^2)$$

$$\therefore v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$$

Question190

**Formation of covalent bonds in compounds exhibits
[2002]**

Options:

- A. wave nature of electron
- B. particle nature of electron
- C. both wave and particle nature of electron
- D. none of these

Answer: A

Solution:

Covalent bonds are formed by sharing of electrons with different compounds. Formation of covalent bond is best explained by molecular orbital theory.

Question191

**Sodium and copper have work functions 2.3 eV and 4.5 eV respectively. Then the ratio of the wavelengths is nearest to
[2002]**

Options:

- A. 1 : 2
- B. 4 : 1
- C. 2 : 1
- D. 1 : 4

Answer: C

Solution:

We know that work function,

$$E = h\nu = \frac{hC}{\lambda}$$

where

h = Planck's constant

C = velocity of light

λ = wavelength of light

$$\therefore \frac{E_{Na}}{E_{Cu}} = \frac{\lambda_{Cu}}{\lambda_{Na}}$$

$$\Rightarrow \frac{\lambda_{Na}}{\lambda_{Cu}} = \frac{E_{Cu}}{E_{Na}} = \frac{4.5}{2.3} \approx \frac{2}{1}$$
